

# CLINICAL SKILLS & EXAMINATIONS

## NOTES

**FOURTH EDITION**

PRE-SUMMARIZED  
READY-TO-STUDY  
HIGH-YIELD NOTES

FOR THE TIME-POOR  
MEDICAL, PRE-MED,  
USMLE OR PA STUDENT



MED  
STUDENT  
NOTES  
.COM

168 PAGES





## A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

**If you are new to us, here are a few things to help get the most out of your notes:**

1. **Once saved, the notes are yours for life!** However, we strongly advise that you download and save the files immediately upon purchasing for permanent offline access.
2. **Sharing notes is prohibited.** All files are share-protected and our system will automatically revoke access to and lock files if it detects a customer attempting to share or distribute our notes.
3. **Your license permits you to do the following:**
  - a. You may download/save/view files on up to 2 simultaneous devices.
  - b. You may save the files to an external hard drive for backup purposes only.
  - c. You may print your notes to hard copy on any home printer/photocopier.
4. **Once saved, you do not need to download the notes again.** You can simply transfer your file/s to your second personal device (eg: your iPad/tablet) without the need to re-download your files. If you wish to retire an old device, simply transfer your files to your new device, then delete the files from your old device.
5. **Each order has an allocated download allowance.** Exceeding your download quota will require you to request an extension by emailing us at [admin@medstudentnotes.com](mailto:admin@medstudentnotes.com) and quoting your order number.
6. **If you have any further queries,** feel free to check out our FAQ page: <https://www.medstudentnotes.com/pages/faqs> or email us at [admin@medstudentnotes.com](mailto:admin@medstudentnotes.com)
7. **Please ensure you use the SAME Email address for any future purchases.** That way, all of your files will be visible in the one place on your account page.
8. **Feel free to use the discount code "TAKE20OFF"** for 20% off any future purchases. (Valid for a limited time)
9. **If you find these notes valuable but you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit this link for instructions: <https://www.medstudentnotes.com/pages/bundle-upgrade-process> .**



## Table Of Contents:

**What's included:** Ready-to-study clinical examination guidelines of each of the major bodily systems presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

### CONTENTS: (CLICKABLE HYPERLINKS)

- [THE CARDIOVASCULAR EXAMINATION](#)
- [THE ENDOCRINE EXAMINATION](#)
- [THE GASTROINTESTINAL EXAMINATION](#)
- [THE HAEMATOLOGICAL \(HEMATOLOGICAL\) EXAMINATION](#)
- [THE MUSCULOSKELETAL EXAMINATION](#)
- [THE NEUROLOGICAL EXAMINATIONS](#)
- [THE RENAL SYSTEM EXAMINATION](#)
- [THE RESPIRATORY EXAMINATION](#)
- [THE RHEUMATOLOGICAL EXAMINATION](#)
- [THE PREOPERATIVE ASSESSMENT:](#)
- [THE EAR HISTORY & HEARING TESTS:](#)
- [THE EYE EXAMINATION:](#)
- [THE WELL-WOMAN'S HEALTH CHECK](#)
- [MALE GENITOURINARY EXAM](#)

## **THE CARDIOVASCULAR EXAMINATION**



## THE CARDIOVASCULAR EXAMINATION

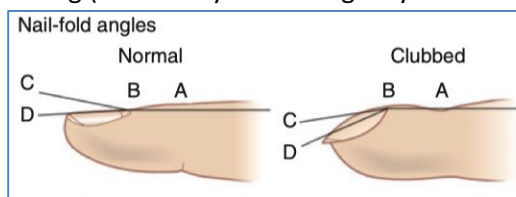
### THE FULL CARDIO EXAM:

- **Introduction + Consent + Wash Hands**
- **General Inspection:**
  - Body Habitus:
    - Cardiac Cachexia (Portal Hypertension/RHF)
    - Obesity (Diabetes/Dyslipidaemia/Poor Diet)
  - Alert & Orientated?
  - Dyspnoea/Respiratory Distress (CCF, Pulmonary Hypertension, Cor-Pulmonale, MI)
  - Discomfort/Pain (Angina, MI, Pericarditis, Tamponade)
  - Diaphoresis (Angina, MI, Pain)
  - Chest Deformities/Surgical Scars (CABG, Valve Repairs)



- Congenital Facies (Marfan's, Down's, Turner's Syndromes)
- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (Shock, MI, Pain, Anaemia)
      - Irregular (AF – MI, Alcohol, Mitral Regurgitation/Stenosis)
      - Low Volume (Shock, MI, Tamponade)
  - **Respiratory Rate:**
    - Tachypnoea (Shock, MI, Pain, Anaemia)
  - **Blood Pressures:**
    - Hypertension (Pain, Essential Hypertension)
    - Hypotension (Shock, Heart Failure)
  - **Temperature:**
    - Fever (Infective Endocarditis, Pericarditis, Myocarditis)

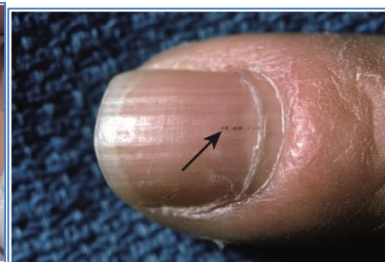
- **Hands:**
  - Perfusion + CRT
  - Pale Nails & Palmar Crease Pallor (Anaemia)
  - Palmar Erythema (Polycythaemia)
  - Peripheral Cyanosis (Heart Failure, Pulmonary Oedema)
  - Clubbing (Chronic Cyanosis – Eg: "Cyanotic Heart")



- Splinter Haemorrhages/Osler's Nodes (Painful Fingertips)/Janeway Lesions (Palms Infective Endocarditis)



**FIGURE 3.55**  
Osler's nodes in infective endocarditis

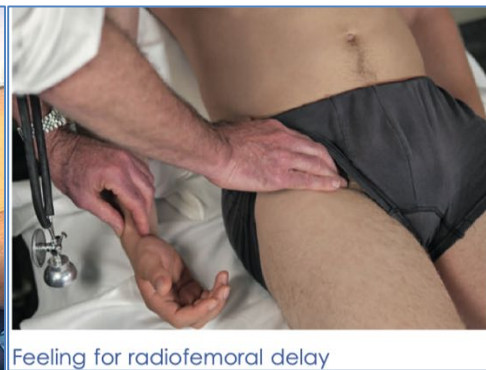


**FIGURE 3.75**  
Splinter haemorrhages

- Xanthomata (Cholesterol in tendons - Dyslipidaemia)
  - Dupuytren's Contracture (Alcohol – Dilated Cardiomyopathy)

- **Arms:**

- Radio-Radial-Delay (Coarctation of the Aorta)
- Radio-Femoral-Delay (Coarctation of the Aorta)
- Track marks (IVDU/Infective Endocarditis)



- **Face:**

○ **Eyes:**

- Conjunctival Pallor (Anaemia)



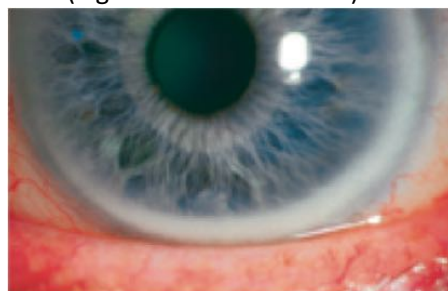
- Scleral Icterus (Jaundice)



- Xanthelasma (periorbital cholesterol)



- +Fundoscopy for Roth's Spots (Infective Endocarditis)
- Arcus Senilis (Sign of CVD risk factors)



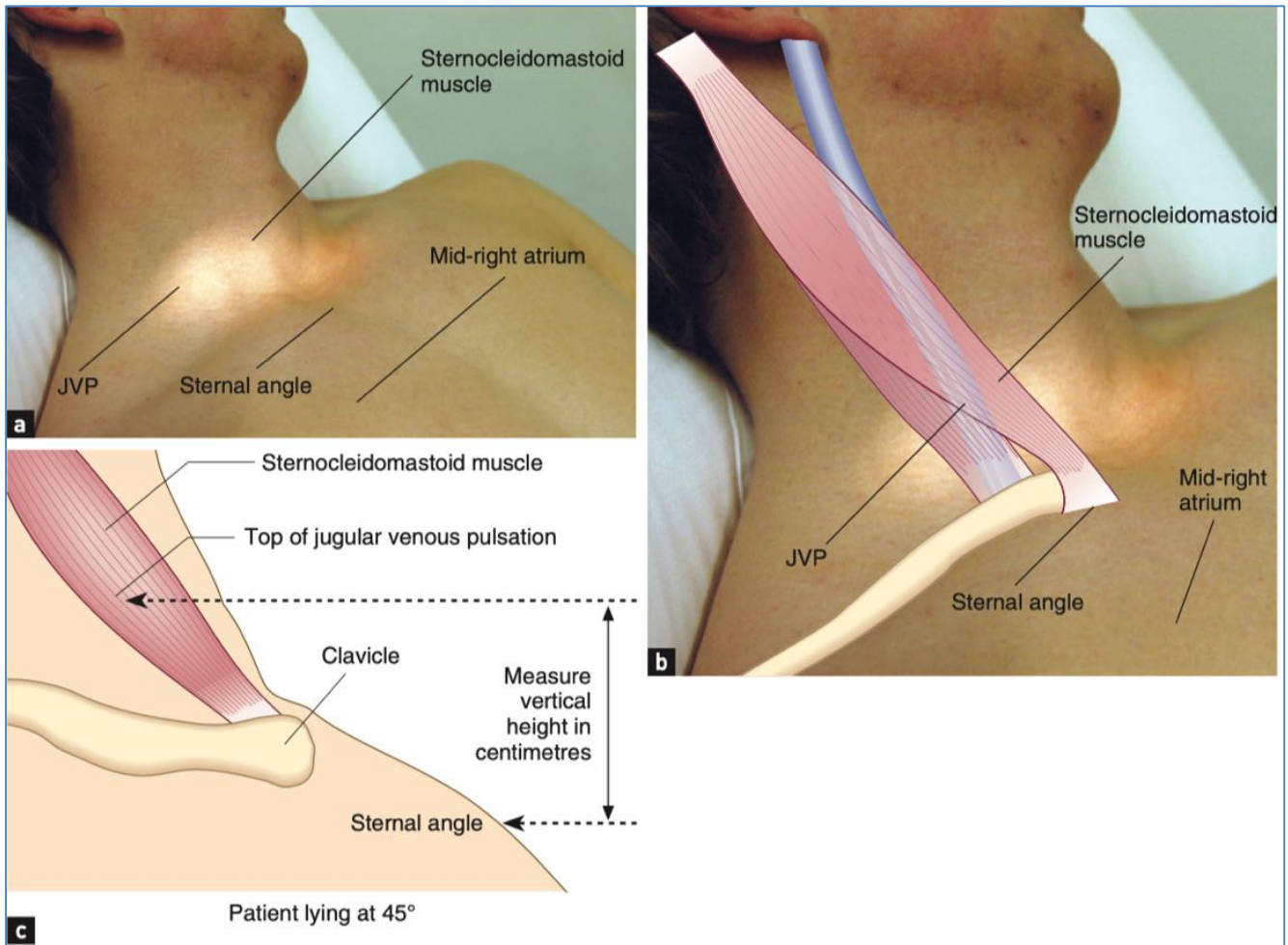
- **Mitral Facies/Malar Rash (Mitral Stenosis)**

○ **Mouth:**

- Hydration
- Central Cyanosis/Peripheral Cyanosis (CCF - Pulmonary Oedema)
- Gum Pallor (Anaemia)
- Poor Dental Hygiene (Infective Endocarditis)
- High Arched Palate (Marfan's Syndrome)

- **Neck:**

- ↑JVP (RVF, Pulmonary Hypertension) + Hepatojugular Reflex
- Jugular Venous Pulsations (Tricuspid Regurgitation)
- Carotid Pulses (Character/Volume)
- Carotid Bruits (Carotid Stenosis – Atheroma)

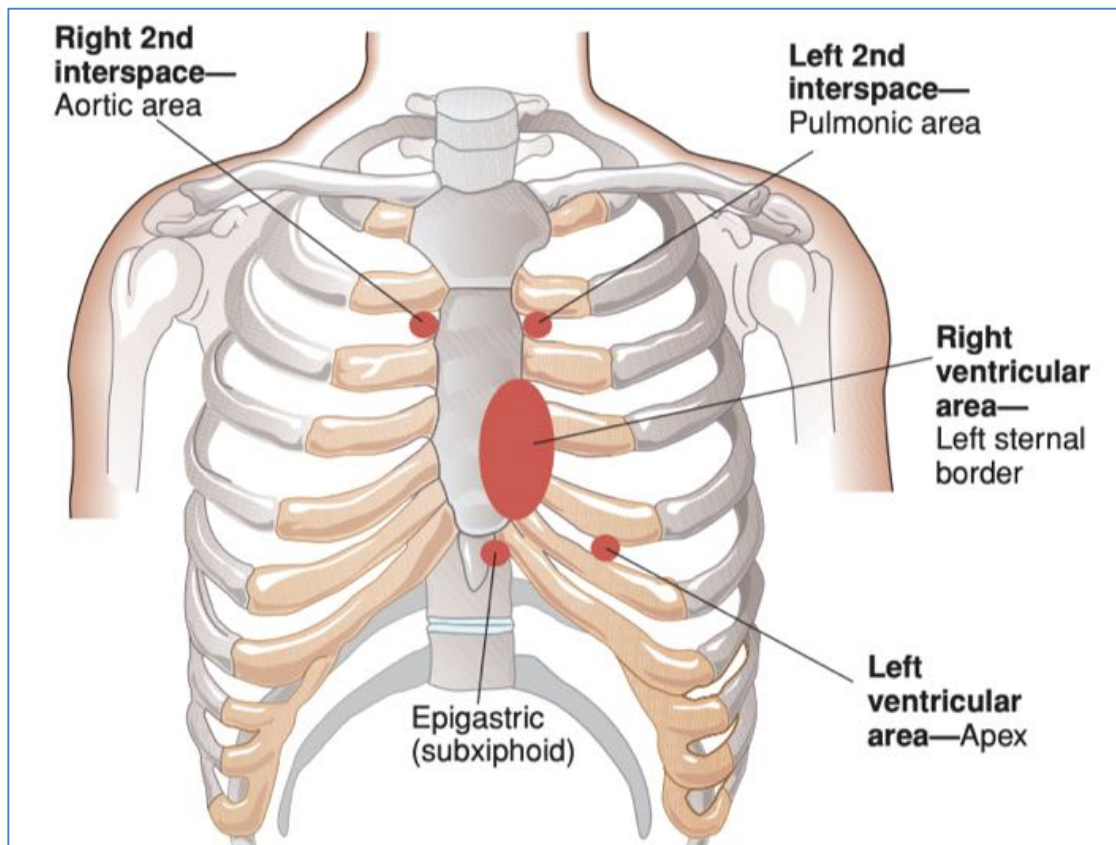


Unattributable

- **Chest:**

- **Inspection:**
  - Scars (CABG, Sternotomy)
  - Chest Deformities (Pectus Excavatum, Pectus Carinatum, Barrel Chest, Kyphosis, Lordosis)
  - Bruising
  - Visible Apex Beat
  - Pacemaker
- **Palpation:**
  - Apex Beat (Normally 5ICSMCL) – Displaced in Cardiomegaly & Hypertrophic Cardiomyopathy.
  - Heaves (Mitral/Tricuspid Regurgitation)
  - Thrills (Palpable Systolic Murmurs)
- **Percussion (NOT NECESSARY):**
  - Heart Borders
- **Auscultation:**
  - Muffled Heart Sounds (COPD, Tamponade)
  - Murmurs (Mitral/Tricuspid/Aortic/Pulmonary Valves) (Bell – Diastolic)(Diaphragm – Systolic)
    - +/- Axillary/Carotid Radiation
  - Pericardial Friction Rub (Pericarditis)





Unattributable

- **Back:**
  - (+ Respiratory if CCF – Basal Inspiratory Crackles)
  - Sacral Oedema (RHF)
- **Abdomen – LYING FLAT:**
  - Visible Pulsatile Masses (Aneurysm)
  - Scars
  - Tenderness
  - Hepatomegaly (Portal Hypertension/RHF)
    - + Pulsatile (If Tricuspid Regurgitation)
  - Splenomegaly (Infective Endocarditis)
  - Ascites/Shifting Dullness/Fluid Thrill (RHF)
  - Aortic Width (Aneurysm)
  - Renal Bruits
- **Legs & Feet:**
  - Peripheral Oedema (RHF/fluid overload/Valvular failure)



- Venous Stasis (Shiny Skin, Hair Loss, & Venous Ulcers)



- 
- Calf Tenderness (PVD, PE)
- Arterial Ulcers
- Pulses (Popliteal/Dorsalis Pedis/Posterior Tibial)
- Cap Refill/Warm/Well Perfused
- Clubbing
- Splinters/Janeways/Osler's



**FIGURE 3.27**

Janeway lesions

-

## LIKELY FOCUSED CARDIO EXAMS FOR OSCE

| Causes (differential diagnosis) of chest pain and typical features |                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pain                                                               | Causes                                                                                                                                                                                                                                 | Typical features                                                                                                                                                                                                                                                                                                                              |
| Cardiac pain                                                       | Myocardial ischaemia or infarction                                                                                                                                                                                                     | Central, tight or heavy; may radiate to the jaw or left arm                                                                                                                                                                                                                                                                                   |
| Vascular pain                                                      | Aortic dissection                                                                                                                                                                                                                      | Very sudden onset, radiates to the back                                                                                                                                                                                                                                                                                                       |
| Pleuropericardial pain                                             | Pericarditis ± myocarditis<br>Infective pleurisy<br>Pneumothorax<br>Pneumonia<br>Autoimmune disease<br>Mesothelioma<br>Metastatic tumour                                                                                               | Pleuritic pain, worse when patient lies down<br>Pleuritic pain<br>Sudden onset, sharp, associated with dyspnoea<br>Often pleuritic, associated with fever and dyspnoea<br>Pleuritic pain<br>Severe and constant<br>Severe and constant, localised                                                                                             |
| Chest wall pain                                                    | Persistent cough<br>Muscular strains<br>Intercostal myositis<br>Thoracic zoster<br>Coxsackie B virus infection<br>Thoracic nerve compression or infiltration<br>Rib fracture<br>Rib tumour, primary or metastatic<br>Tietze's syndrome | Worse with movement, chest wall tender<br>Worse with movement, chest wall tender<br>Sharp, localised, worse with movement<br>Severe, follows nerve root distribution, precedes rash<br>Pleuritic pain<br>Follows nerve root distribution<br>History of trauma, localised tenderness<br>Constant, severe, localised<br>Costal cartilage tender |
| Gastrointestinal pain                                              | Gastro-oesophageal reflux<br>Diffuse oesophageal spasm                                                                                                                                                                                 | Not related to exertion, may be worse when patient lies down—common<br>Relieved by swallowing e.g. of warm water                                                                                                                                                                                                                              |
| Airway pain                                                        | Tracheitis<br>Central bronchial carcinoma<br>Inhaled foreign body                                                                                                                                                                      | Pain in throat, breathing painful                                                                                                                                                                                                                                                                                                             |
| Central pain                                                       | Panic attacks                                                                                                                                                                                                                          | Often preceded by anxiety, associated with breathlessness and hyperventilation symptoms (dizziness, perioral paraesthesia)                                                                                                                                                                                                                    |
| Mediastinal pain                                                   | Mediastinitis<br>Sarcoid adenopathy, lymphoma                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                               |

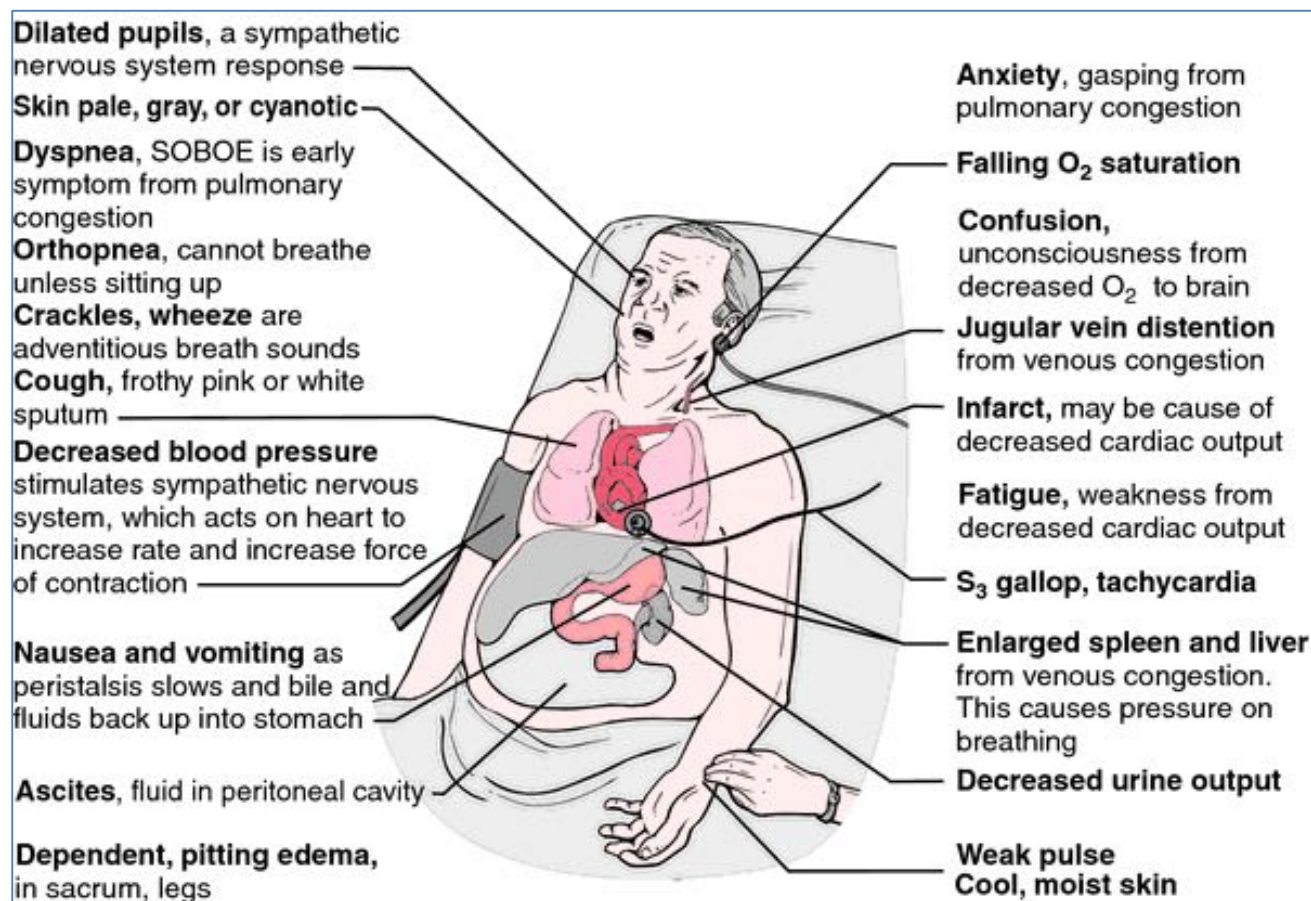
### Left Heart Failure (LVF):

- **TYPICAL Symptoms/Presentation:**
  - Exertional Dyspnoea
  - Orthopnea
  - Paroxysmal Nocturnal Dyspnoea
  - Wheezing Cough ("Cardiac Asthma")
  - (+ Syncope + Angina = Aortic Stenosis)
  - (+ High Arch Palate of Marfan's = Mitral Prolapse)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia (Low Volume), Hypotension
  - **Other:**
    - ↓CO → Cold peripheries, ↑CRT, Peripheral Cyanosis, Cerebral Hypoperfusion (Inattention, Confusion)
    - **Pulmonary Congestion** → Central Cyanosis, Basal Lung Crepitations, Diffuse Wheezes.
    - Abdo-Jugular Reflux Positive.
    - Laterally Displaced Apex Beat (LV Dilation)
  - **Signs of Causes:**
    - Mitral Facies & Mitral Regurgitation (Pan-Systolic Murmur over Mitral/Apex)
    - Mitral Facies, Mitral Stenosis (Pan-Diastolic Crescendo Murmur over Mitral/Apex)
    - OR Aortic Stenosis (Systolic-Ejection Murmur over Right-Sternal Border)
    - OR Aortic Regurgitation (Decrescendo Diastolic Murmur over Right Sternal Border)
    - (Cardiac Cachexia if → Right Heart Failure)



### Right Heart Failure (RVF):

- **TYPICAL Symptoms/Presentation:**
  - Anorexia/Cardiac Cachexia (Portal HTN)
  - Swollen Ankles, Sacrum & Abdomen,
  - Weight Gain (Fluid)
  - **(+ Pulsatile Liver = Tricuspid Regurgitation)**
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Low-Volume Pulse
  - **Other:**
    - **↓CO** → Cold peripheries, ↑CRT, Peripheral Cyanosis, Cerebral Hypoperfusion (Inattention, Confusion)
    - RV-Heave
    - **Peripheral Congestion** → Peripheral Oedema (Sacral/Tibial), Ascites, Small Pleural Effusions.
    - ↑JVP, Kussmaul's Sign (↑JVP on Inspiration)
    - **Portal Hypertension** → "Cardiac Cachexia", Caput Medusa, Tender Hepatomegaly, Pulsatile Liver (TR),
  - **Signs of Causes:**
    - **COPD & Cor-Pulmonale** (RVF due to Pulmonary Hypertension)
    - **LVF** (Pulmonary Congestion)
    - **Tricuspid Regurgitation** (Pansystolic Murmur over Tricuspid + Pulsatile Liver)



## **IHD & Acute Myocardial Infarction:**

### **TYPICAL Symptoms/Presentation:**

#### ○ IHD:

- Angina (Severe Central Crushing Chest Pain):
  - Stable = Transient Exertional
  - Variant/Prinzmetal = Transient @ Rest
  - Unstable = Resting Angina of ↑Severity & Frequency
- (+/- Diaphoresis, Dyspnoea, Anxiety)

#### ○ AMI:

- Angina (Severe Central Crushing Chest Pain)
  - >20mins
  - Radiating to L-Arm, Neck & Jaw
- Dyspnoea
- Diaphoresis
- Anxiety

### **TYPICAL Clinical Signs:**

#### ○ Vitals:

- Tachycardia/Bradycardia
  - +/-Arrhythmias (AF, VT, VF, Heart Block)
- Tachypnoea
- Hypotension
- Afebrile

#### ○ Other:

- Clammy, Sweaty Hands
- **If LV Infarct → LVF →** Pulmonary Congestion, ↓CO, Cool Peripheries, Central & Peripheral Cyanosis, (Cardiogenic Shock)
  - **If Papillary Muscle Dysfunction →** Mitral Regurgitation (Midsystolic Murmur)
- **If RV Infarct → RVF →** ↑JVP, Kussmaul's Sign.
- **If Transmural →** Pericardial Friction Rub

#### ○ Signs of Causes:

- PVD, Hypertension, Diabetes, Hypercholesterolaemia, Obesity, Smoker.

## **Pulmonary Embolism:**

### **TYPICAL Symptoms/Presentation:**

- Sudden, SEVERE Dyspnoea
- Pleuritic Chest Pain
- (+/- Haemoptysis)
- (+/- Syncope)

### **TYPICAL Clinical Signs:**

#### ○ Vitals:

- Tachycardia, Tachypnoea, Hypotension (LVF), (+/-Fever)

#### ○ Other:

- **RVF →** Cool Peripheries, ↑CRT, Peripheral Cyanosis, ↑JVP, RV-Heave, Tricuspid Regurgitation Murmur
- **↓Respiratory Function →** Central Cyanosis
- Pleural Friction Rub,

#### ○ Signs of Causes:

- **DVT –** Calf Pain, Calf Tenderness, Calf Swelling/Erythema, Pedal Oedema.
  - (B/G of Pregnancy, Air Travel, Recent Surgery, Clotting Disorders)



## Rheumatic Fever/ Rheumatic Heart Disease:

### - TYPICAL Symptoms/Presentation:

- **(Preceding Strep Throat):**
  - Fever, Pharyngitis, Tonsillitis, Lymphadenopathy
- **Rheumatic Fever (>2wks Post GABH Strep Pharyngitis):**
  - Fever
  - **Joints** - Migratory Polyarthritits, Ataxia
  - **Heart** - Pleuritic Chest Pain (Fibrinous Pericarditis)
  - **Nodules** – Subcutaneous Nodules
  - **Erythema Marginatum** – Red rings on trunk & limbs
  - **Sydenham Chorea** – Rapid, Involuntary Movements.
  - (↑in Indigenous)
- **Rheumatic Heart Disease:**
  - Palpitations (AF)
  - LVF → Exertional Dyspnoea, Orthopnoea, PND
    - Mitral Facies

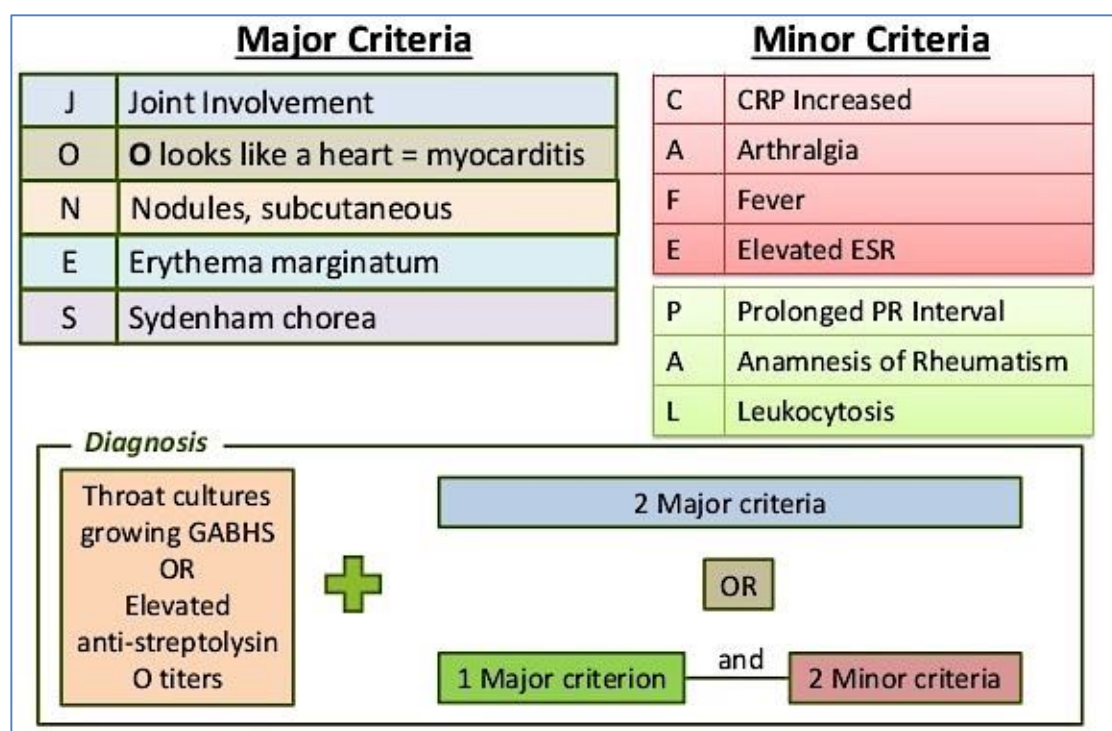
### - TYPICAL Clinical Signs:

#### **Acute Rheumatic Fever:**

- **Vitals:**
  - Fever, Tachycardia, Tachypnoea
- **Other:**
  - Ataxia (due to polyarthritits)
  - Pericardial Friction Rub (Fibrinous Pericarditis)
  - Subcutaneous Nodules
  - Erythema Marginatum (Red rings on trunk and limbs)
  - Sydenham's Chorea (Rapid, Involuntary, Purposeless Movements)

#### **Rheumatic Heart Disease:**

- **Vitals:**
  - Afebrile, Tachycardia (+/- AF), Tachypnoea (CCF), Hypotensive
- **Other:**
  - Mitral Facies & Mitral Stenosis (+/- Mitral Regurgitation)
  - CCF → Inspiratory Creps, Peripheral & Central Cyanosis, ↑JVP, Dyspnoea
- **Signs of Cause:**
  - Indigenous, Low SES, Poor Hygiene

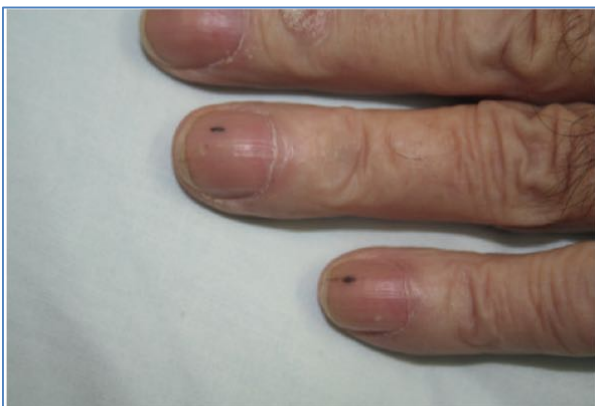


### Acute Aortic Dissection:

- **TYPICAL Symptoms/Presentation:**
  - Sudden, Severe Tearing Chest/Abdo Pain
    - Radiating to Back
  - (+/- Stroke)
  - (+/- ALOC – due to Tamponade)
  - (+/- Sudden Death)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Unequal Radial Pulse Pressures,
    - Asymmetrical Blood Pressure
  - **Other:**
    - Pulsatile, Expansile Abdominal Masses, Renal Bruits
    - Signs of Tamponade (↑JVP, ↓Heart Sounds, Low-Volume Pulse, etc)
    - Aortic Regurgitation (Due to disruption of annulus) (Decrescendo Diastolic Murmur)
  - **Signs of Causes:**
    - Marfan's Habitus
  - **Complications:**
    - Myocardial Infarction (Coronary Occlusion), Mesenteric Ischaemia (Mesenteric Artery Occlusion), Pre-Renal Failure (Renal Artery Occlusion), Limb Ischaemia

### Infective Endocarditis:

- **TYPICAL Symptoms/Presentation:**
  - Acute PUO – (+ Chills, Night-Sweats/Weight Loss)
  - Dyspnoea
  - Mild Arthralgia ("licks joints, bites the heart")
  - Palpitations
  - Haematuria
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea
  - **Other:**
    - Clubbing, Palmar Crease Pallor, **Splinter Haemorrhages** (Septic Emboli), **Janeway Lesions** (Painless Maculopapular Bacterial Colonies on Palms), **Painful Osler's Nodes** (Tender, Red, Raised Nodules on Pulp of Fingers)
    - Track marks (IVDU),
    - Conjunctival Pallor, **Retinal Roth's Spots** (Retinal Haemorrhages), Poor Dentition,
    - **New or Changing Murmur** (Either side of heart) (Often Tricuspid), Prosthetic Valve Click,
    - Splenomegaly
  - **Signs of Causes:**
    - IVDU, Poor Dentition, Open Heart Surgery.



Splinter haemorrhages in the fingernails of a patient with staphylococcal aortic valve endocarditis



Janeway lesion

### **Acute Pericarditis:**

- **TYPICAL Symptoms/Presentation:**
  - Fever
  - Dyspnoea (+/- Dry Cough)
  - Pleuritic Chest Pain (Worse on Supine)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Tachypnoea (Fast, Shallow), Febrile,
  - **Other:**
    - Pericardial Friction Rub (Best heard when sitting forward + hold breath)
    - IF Tamponade → ↑JVP, ↓Heart Sounds, Impalpable apex, & ↓CO.
  - **Signs of Causes:**
    - URTI, Post MI, Post Cardiac Surgery, Uraemia, SLE/Rheumatoid.

### **Chronic Constrictive Pericarditis:**

- **TYPICAL Symptoms/Presentation:**
  - Acute Symptoms for >3mths
  - Cachexia, Fatigue
  - RVF → Oedema, Abdo Distension
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia
    - +\*Pulsus Paradoxus (>10mmHg fall in Arterial Pulse Pressure on Inspiration)
    - Hypotension
  - **Other:**
    - Cachexia
    - ↑JVP + Kussmaul's Sign
    - Impalpable Apex, ↓Heart Sounds, Pericardial 'Knock'
    - Tender Hepatomegaly, Tender Splenomegaly, Ascites & Peripheral Oedema.
  - **Signs of Causes:**
    - Post MI, Post Cardiac Surgery, Uraemia, SLE/Rheumatoid

### **Acute Cardiac Tamponade:**

- **TYPICAL Symptoms/Presentation:**
  - Dyspnoea
  - Anxiety
  - Syncope
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia (Low Volume & Pulsus Paradoxus), Hypotension, Afebrile
  - **Other:**
    - ↑JVP,
    - Impalpable Apex, ↓Heart Sounds,
    - Left Lung: Dull & Bronchial Breathing @ Base (Compressed by Heart)

### Acute Infective Myocarditis:

- **TYPICAL Symptoms/Presentation:**
  - (Viral Infection (Flu, Coxsackie, CMV, HIV, Parvo))
  - **Acute Heart Failure**– Dyspnoea/Orthopnoea, ALOC, Cardiogenic Shock.
  - Palpitations
  - Sudden Death
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Febrile, Tachycardia, Tachypnoea, Hypotension (if HF)
  - **Other:**
    - **Acute Heart Failure** – Peripheral Shutdown, ↑CRT, Low Pulse Pressure, ALOC, ↑JVP, Pulmonary Congestion, Dyspnoea, Basal Creps.
    - Palpitations
  - **Signs of Causes:**
    - URTI, Immunocompromise (HIV).

### Acute Toxic Myocarditis:

- **TYPICAL Symptoms/Presentation:**
  - (Cocaine, Prescription Meds, Arsenic, Alcohol, Snake Venom, Carbon Monoxide)
  - **Acute Heart Failure**– Dyspnoea/Orthopnoea, ALOC, Cardiogenic Shock.
  - Palpitations
  - Sudden Death
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Afebrile, Tachycardia, Tachypnoea, Hypotension (if HF)
  - **Other:**
    - **Acute Heart Failure** – Peripheral Shutdown, ↑CRT, Low Pulse Pressure, ALOC, ↑JVP, Pulmonary Congestion, Dyspnoea, Basal Creps.
    - Palpitations
    - CO-Poisoning → Headache, Dizziness, Nausea, Convulsions, ALOC
  - **Signs of Causes:**
    - Nasal Septal Necrosis (Cocaine), Snake Bite Marks, Confusion/Ataxia/Delirium.

### Dilated Cardiomyopathy: (Alcoholism/Genetic)

- **TYPICAL Symptoms/Presentation:**
  - Any Age
  - **Biventricular Heart Failure** – Dyspnoea, Orthopnoea, PND, Fatigue, Cyanosis, Peripheral Oedema, Ascites, Fatigue, Anorexia.
  - +/- Palpitations
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (+/- Ventricular Arrhythmias), Tachypnoea, Hypotension, Afebrile
  - **Other:**
    - Cool Peripheries, ↑CRT, Peripheral & Central Cyanosis, Low-Volume Pulse,
    - Dyspnoea, Orthopnoea, PND,
    - Peripheral Oedema, Ascites, Tender Hepatomegaly, Pulsatile Liver(TR),
    - ↓Heart Sounds, Basal Inspiratory Creps,
  - **Signs of Causes:**
    - Dupuytren's Contractures (Chronic Alcoholism)



### Hypertrophic Cardiomyopathy: (Genetic)

- **TYPICAL Symptoms/Presentation:**
  - Exertional Dyspnoea (LVF → Pulmonary Congestion)
  - Angina (↑ Demand on Coronary Supply)
  - Palpitations (AF/Ventricular Arrhythmias)
  - Syncope
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Tachypnoea, Hypotension, Afebrile
  - **Other:**
    - Cool Peripheries, ↑ CRT, Peripheral & Central Cyanosis,
    - Sharp “Tapping” Pulse, Low Volume Pulse,
    - Prominent Pulsatile JVP (↓ RV-Compliance)
    - Lateral Apex Displacement, Ejection Systolic Murmur (LV-Outflow Obstruction – ↑ on Valsalva & ↓ on Squatting) & Pansystolic Murmur (Mitral Regurgitation)

### Restrictive Cardiomyopathy: (Amyloid/Sarcoid-osis)

- **TYPICAL Symptoms/Presentation:**
  - Cachexia/Fatigue
  - RVF → Oedema, Abdo Distension, ↑ JVP
  - LVF - Cough/Dyspnoea/PND/Orthopnea
  - Palpitations
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (+/- Arrhythmias), Low-Volume Pulse,
    - Hypotension
    - +\*Pulsus Paradoxus (>10mmHg fall in Arterial Pulse Pressure on Inspiration)
  - **Other:**
    - Cool Peripheries, ↑ CRT, Peripheral & Central Cyanosis,
    - Cachexia
    - ↑ JVP + Kussmaul’s Sign
    - Impalpable Apex, ↓ Heart Sounds,
    - Tender Hepatomegaly, Tender Splenomegaly, Ascites & Peripheral Oedema.
  - **Signs of Causes:**
    - Post MI, Post Cardiac Surgery, Uraemia, SLE/Rheumatoid

### Stress Cardiomyopathy: (Broken Heart Syndrome/Takotsubo)

- **TYPICAL Symptoms/Presentation:**
  - Sudden Onset CCF:
    - Chest Pain
    - Dyspnoea
    - Palpitations (Vent. Arrhythmias)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (+/- Arrhythmias), Low-Volume Pulse,
    - Hypotension
  - **Other:**
    - Cool Peripheries, ↑ CRT, Peripheral & Central Cyanosis,

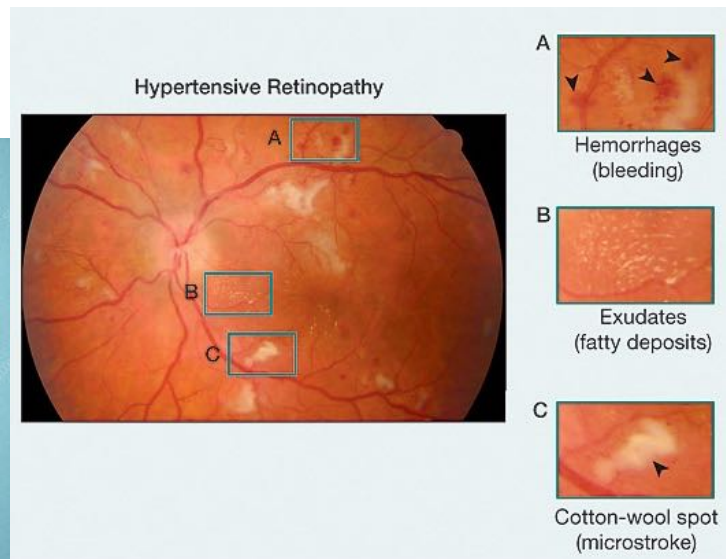
## Systemic Hypertension:

### - TYPICAL Symptoms/Presentation:

- Headaches/Dizziness
- Blurred Vision/Tinnitus
- Hypertensive Encephalopathy (N/V/ALOC)
- (Epistaxis)

### - TYPICAL Clinical Signs:

- **Inspect for Signs of Causes:**
  - Cushing's Syndrome, Conn's Syndrome,  $\uparrow$ PTH, Hypercalcaemia, Hyperthyroidism, Hypothyroidism, Pregnancy (Eclampsia), Acromegaly, Polycythaemia, CKD, Phaeo,
- **Vitals:**
  - Hypertension (Note: Inverse Postural Variation in *Essential Hypertension*, but Normal Postural Variation in *Secondary Hypertension*),
  - RF-Delay (coarctation)
- **Other:**
  - Palmar Erythema,  $\downarrow$ CRT, Facial Flushing
  - Fundoscopy (Hypertensive Retinopathy +/- Papilloedema), Full Neurological Exam
  - Carotid Palpation, Carotid Bruits
  - S4 on Auscultation (4<sup>th</sup> Heart Sound = Splitting of S1 due to late Mitral Closure)
  - Pulsatile Abdominal Masses (Aneurysm), Renal Masses (Tumours), Renal Bruits,
  - **+Urinalysis**



## Pulmonary Hypertension & Cor-Pulmonale: (LVF, COPD)

### - TYPICAL Symptoms/Presentation:

- **RVF Secondary to Pulmonary Hypertension:**
  - **COPD:** Dyspnoea, Cough, Wheeze
  - **Pulmonary HTN:** Cough/Dyspnoea/PND/Orthopnea
  - **RVF:** Swelling (Legs, Abdo), Chest Pain

### - TYPICAL Clinical Signs:

- **Vitals:**
  - Tachycardia (if LVF), Tachypnoea (if COPD/LVF), Hypotension (If LVF), Afebrile
- **Other:**
  - **If LVF:** Cool Peripheries,  $\uparrow$ CRT, Peripheral & Central Cyanosis, Low Volume Pulse
  - **If COPD:** Clubbing, Tar Staining, Peripheral & Central Cyanosis
  - $\uparrow$ JVP + a-Wave, RV-Heave, Loud S2 (closure of Pulmonic Valve) Abdojugular Reflux Positive, Portal Hypertension (Tender Hepatomegaly), Ascites, Sacral/Pedal Oedema,
- **Signs of Causes:**
  - LVF, Smoking, COPD, IPF

### **Carcinoid Heart Disease:**

- **TYPICAL Symptoms/Presentation:**
  - Episodic Flushing of Skin
  - Cramps
  - Diarrhoea/Nausea/Vomiting
  - (+/- Valvular Heart Failure)
  - (+/- Bronchoconstriction → Dyspnoea)
- **TYPICAL Clinical Signs:**
  - Flushing of Skin
  - **RVF:** Tricuspid Regurgitation (Pansystolic Murmurs), Pulmonary Stenosis (Systolic Murmur), Tender Hepatomegaly, Portal Hypertension

### **Mitral Stenosis:**

- **TYPICAL Symptoms/Presentation:**
  - LVF – Exertional Dyspnoea/Orthopnea/PND/ Pink Frothy Sputum
  - (If RVF → Ascites/Peripheral Oedema/Fatigue)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Low-Vol Pulse, +/- AF), Tachypnoea, Hypotensive
  - **Other:**
    - **LVF:** Cool Peripheries, ↑CRT, Peripheral & Central Cyanosis, Low Volume Pulse,
    - Mitral Facies, ↑JVP if Pulmonary HTN, RV-Heave, Inspiratory Creps,
    - Mitral Stenosis (Pan-Diastolic Rumbling Murmur with Opening Snap & late Crescendo)
  - **Signs of Causes:**
    - Rheumatic Heart Disease, Congenital

### **Mitral Regurgitation:**

- **TYPICAL Symptoms/Presentation:**
  - **Acute (Eg: Papillary Rupture, MI, Infective Endocarditis):**
    - Acute Pulmonary Oedema → Exertional Dyspnoea/Orthopnea/PND/ Pink Frothy Sputum
    - +/- Shock
  - **Chronic (Eg: MVP, Ageing, Rheumatic Heart Disease, Cardiomyopathy) :**
    - Compensated Pulmonary Oedema → Exertional Dyspnoea/Fatigue
- **TYPICAL Clinical Signs:**
  - Acute:**
  - **Vitals:**
    - Tachycardia (Low Vol. +/- AF), Tachypnoea, Hypotension
  - **Other:**
    - **LVF:** Cool Peripheries, ↑CRT, Peripheral & Central Cyanosis, Low Volume Pulse,
    - Mitral Facies, ↑JVP if Pulmonary HTN, RV-Heave, Inspiratory Creps,
    - Mitral Regurgitation (Loud Systolic Murmur + Thrill @ Apex)
  - **Signs of Causes:**
    - Papillary Rupture, MI, Infective Endocarditis
  - Chronic:**
  - **Vitals:**
    - Normal Pulse Rate & Volume (+/- AF), Tachypnoea, Normotensive
  - **Other:**
    - Displaced Apex, RV-Heave, Mitral Regurgitation (Loud Systolic Murmur → Axilla + Thrill @ Apex)
    - If Severe: Signs of LVF (Cool Peripheries, ↑CRT, P&C-Cyanosis, Low Volume Pulse)
  - **Signs of Causes:**
    - MVP, Ageing, Rheumatic Heart Disease, Cardiomyopathy

### **Aortic Stenosis:**

- **TYPICAL Symptoms/Presentation:**
  - **Aortic Stenosis Triad:**
    - Exertional Chest Pain
    - Exertional Dyspnoea
    - Exertional Syncope
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal/Tachycardia
  - **Other:**
    - General Inspection Normal
    - *Slow-Rising* (“Anacrotic”) Carotid Pulse, Displaced Apex (LV-Hypertrophy),
    - Systolic R-Parasternal Thrill (if Severe)
    - Reversed Splitting of S2 (Delayed LV Ejection), Harsh Crescendo-Decrescendo Systolic Murmur Over Aorta + Opening Snap (Radiating to Carotids)
  - **Signs of Causes:**
    - Age-Related (Elderly), Rheumatic

### **Aortic Insufficiency:**

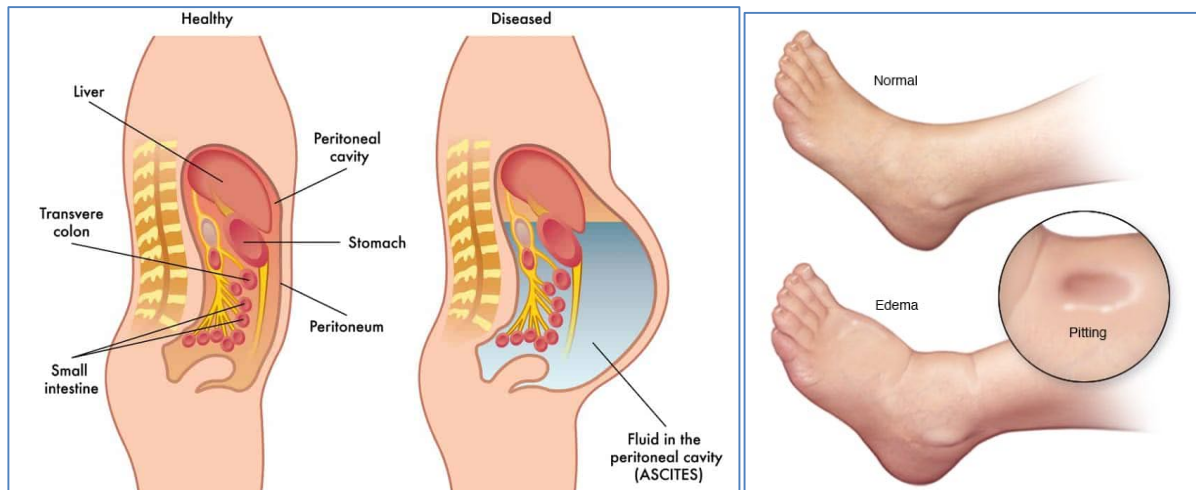
- **TYPICAL Symptoms/Presentation:**
  - **Same as Aortic Stenosis:**
    - Exertional Chest Pain
    - Exertional Dyspnoea
    - Exertional Syncope
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, BP: Wide Pulse Pressure, R: Normal
  - **Other:**
    - Inspect for Marfan’s
    - Collapsing “Water-Hammer” Pulse (↑ by Taking Radial Pulse Above Patient’s Head),
    - Displaced Apex (LV Hypertrophy), Diastolic Thrill @ L-Sternal,
    - Decrescendo Diastolic Murmur @ Aortic Area
    - (Signs of LVF if Severe)
  - **Signs of Causes:**
    - Marfan’s Syndrome
    - Ankylosing Spondylitis,
    - Rheumatoid Arthritis,
    - Tertiary Syphilis,
    - Infective Endo.

### **Tricuspid Stenosis (Rare):**

- **TYPICAL Symptoms/Presentation:**
  - RHF – Peripheral Oedema/Ascites/Portal Hypertension
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia
  - **Other:**
    - ↑JVP with Slow Descent, No Pulmonary Signs
    - Pan-Diastolic Rumbling Tricuspid Murmur + Late Accentuation (Atrial Contraction)
    - Tender Pulsatile Hepatomegaly, Splenomegaly, Ascites, Peripheral Oedema
  - **Signs of Causes:**
    - Rheumatic Heart Disease

### **Tricuspid Insufficiency:**

- **TYPICAL Symptoms/Presentation:**
  - RHF – Peripheral Oedema/Ascites/Portal Hypertension
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia
  - **Other:**
    - ↑JVP with Large Pulsations, RV Heave,
    - Pan-Systolic High-Pitched Tricuspid Murmur
    - Tender Pulsatile Hepatomegaly (+/- Dancing R-Nipple), Splenomegaly, Ascites, Peripheral Oedema, Pulsatile Leg Veins.
  - **Signs of Causes:**
    - \*\*\*IVDU, Infective Endocarditis.



<https://www.alfapump.com/ascites/>

[https://sci.washington.edu/info/newsletters/articles/15\\_spr\\_edema.asp](https://sci.washington.edu/info/newsletters/articles/15_spr_edema.asp)

### **Pulmonary Stenosis:**

- **TYPICAL Symptoms/Presentation:**
  - RHF - Peripheral Oedema/Ascites
  - Chest Pain
  - Dyspnoea
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Low Volume Pulse (If Severe), Hypotension (if Severe)
  - **Other:**
    - (↓CO) → Peripheral Cyanosis, Cool Peripheries, ↑CRT, Low Volume Pulse
    - ↑JVP with Large Pulsations (RA-Hypertrophy), RV Heave (RV-Hypertrophy), Pulmonary Thrill,
    - Ejection Systolic Murmur @ Pulmonary Area (Max on Inspiration)
    - Pulsatile Tender Hepatomegaly
  - **Signs of Causes:**
    - Congenital, Carcinoid.

### **Pulmonary Insufficiency:**

- **TYPICAL Symptoms/Presentation:**
  - RHF - Peripheral Oedema/Ascites
  - Chest Pain
  - Dyspnoea
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Low Volume Pulse (If Severe), Hypotension (if Severe)
  - **Other:**
    - ↓CO, ↑JVP, Decrescendo Diastolic Pulmonary Murmur
  - **Signs of Causes:**
    - Pulmonary HTN, Infective Endocarditis, Congenital



### Pulmonary Hypertension & Cor-Pulmonale: (LVF, COPD)

- **TYPICAL Symptoms/Presentation:**
  - **RVF Secondary to Pulmonary Hypertension:**
    - **COPD:** Dyspnoea, Cough, Wheeze
    - **Pulmonary HTN:** Cough/Dyspnoea/PND/Orthopnea
    - **RVF:** Swelling (Legs, Abdo), Chest Pain
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (if LVF), Tachypnoea (if COPD/LVF), Hypotension (If LVF), Afebrile
  - **Other:**
    - **If LVF:** Cool Peripheries, ↑CRT, Peripheral & Central Cyanosis, Low Volume Pulse
    - **If COPD:** Clubbing, Tar Staining, Peripheral & Central Cyanosis
    - ↑JVP + a-Wave, RV-Heave, Loud S2 (closure of Pulmonic Valve) Abdojugular Reflux Positive, Portal Hypertension (Tender Hepatomegaly), Ascites, Sacral/Pedal Oedema,
  - **Signs of Causes:**
    - LVF, Smoking, COPD, IPF



### Atrial Septal Defect (L-R):

- **TYPICAL Symptoms/Presentation:**
  - Asymptomatic until ≈30yrs.
  - Eventual RVF (Peripheral Oedema/Ascites) due to Pulmonary HTN (Dyspnoea/Orthopnea/PND)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - RV-Hypertrophy, Fixed Splitting of S2 (Late Tricuspid Closure)
    - (If Shunt Reverses → Cyanosis, Clubbing, Tachycardia, Tachypnoea)

### Ventricular Septal Defect (L-R):

- **TYPICAL Symptoms/Presentation:**
  - **\*\* (Infants)** Failure to Thrive
  - LVF & Pulm HTN (Dyspnoea/Orthopnea/PND)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Tachypnoea, Hypotension,
  - **Other:**
    - Signs of ↓CO
    - Pansystolic Murmur @ Lower L-Sternal border, + Systolic Thrill @ L-Sternal Edge
  - **Signs of Causes:**
    - Congenital, (Or Septal MI)

### Patent Ductus Arteriosus (Aorta → Pulmonary Artery):

- **TYPICAL Symptoms/Presentation:**
  - Pulmonary HTN → Pulmonary Congestion/Dyspnoea/Cough
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Collapsing Pulse), Hypotension (with ↓ Diastolic)
  - **Other:**
    - Hyperkinetic Apex Beat, Reversed Splitting of S2 (Early Aortic Valve Closure)
    - “machinery Murmur” max @ Pulmonary
    - **\*\*Note: IF THE SHUNT HAS REVERSED, the Child may become Dyspnoeic/Cyanotic which is relieved by Squatting (which ↑ Aortic Pressure & Reverts shunt back to L→R)**

### Coarctation of Aorta: (Typically distal to L-Subclavian)

- **TYPICAL Symptoms/Presentation:**
  - Poor Lower-Body Development
  - Claudication
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Upper-Body HTN, Lower Body Hypotension,
  - **Other:**
    - Radio-Femoral Delay, Weak Femoral Pulses, Midsystolic Murmur @ Praecordium
  - **Signs of Causes:**
    - Congenital, Mostly Males (Although also associated with Turner’s Syndrome)

### Tetralogy of Fallot: (VSD + RVH + Overriding Aorta + Subpulmonic Stenosis)

- **TYPICAL Symptoms/Presentation:**
  - (Infants) – Poor Feeding, Cyanotic Spells, Failure to Thrive, Syncope
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Tachypnoea, Hypotension
  - **Other:**
    - Central & Peripheral Cyanosis (Blue Baby), Clubbing, Polycythaemia, Parasternal Heave (RVH), Systolic Thrill, Pulmonary Ejection Systolic Murmur.

### Transposition of Great Vessels:

- **TYPICAL Symptoms/Presentation:**
  - (Fatal Without Concurrent ASD/VSD Exists)
  - With ASD/VSD:
    - Cyanosis, Palpitations, Dizziness, Dyspnoea
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Systemic Hypotension, Pulmonary Hypertension, Tachypnoea,
  - **Other:**
    - Cyanosis, Clubbing, Polycythaemia
    - (Adults – Scars, Facial Flushing, Cyanosis, Oedema)

### Peripheral Vascular (Arterial) Disease:

- **TYPICAL Symptoms/Presentation:**
  - Claudication
  - Gangrene
  - Arterial Ulcers
  - \*\*Impotence
  - (Acute Occlusion → Critical Limb Ischaemia)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - ↑CRT, Peripheral Cyanosis, Gangrene, \*\*Raynaud's Phenomenon, Arterial Ulcers (Painful, Dry, Superficial), Atrophic Skin/Hair Loss, Focal Weaknesses in Peripheral Pulses, Absent Pulses, Arterial bruits (Carotid/Renal/ Femoral), Neurological Signs (Dizziness, Syncope, CVA),
    - (If Critical Limb Ischaemia → Acute Limb Pain, Absent Distal Pulses, Pale Limb)
  - **Signs of Causes:**
    - Hypercholesterolaemia, Smoking, Hypertension, Obesity, Diabetes



James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons

### Deep Vein Thrombosis (Phlebothrombosis/Thrombophlebitis):

- **TYPICAL Symptoms/Presentation:**
  - Calf Tenderness
  - Distal Oedema
  - Heat/Redness
  - (Can → PE → Acute Dyspnoea & Collapse)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Asymmetrical Calf Tenderness, Asymmetrical Calf Oedema, Heat/Redness, Dilated superficial veins,
  - **Signs of Causes:**
    - Immobility, Trauma, Cardiac Failure, DIC, Pregnancy.



James Heilman, MD, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

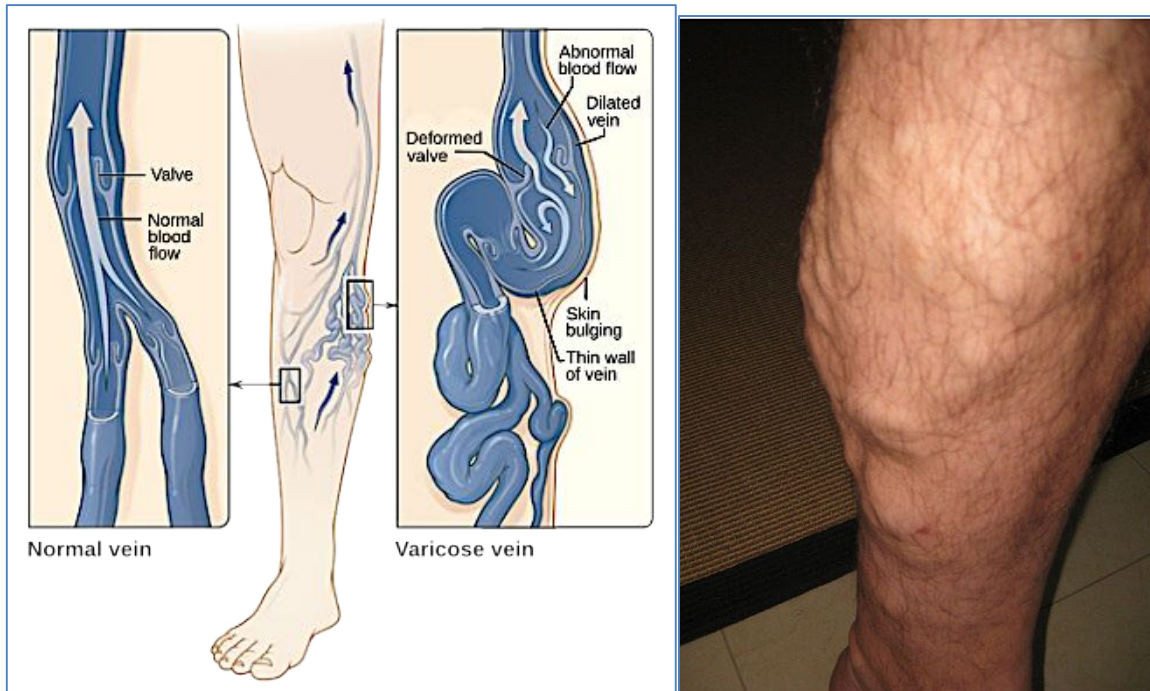
### Varicose Veins:

#### - TYPICAL Symptoms/Presentation:

- Varicosities
- Persistent Oedema
- Venous Ulcerations
- Poor wound healing

#### - TYPICAL Clinical Signs:

- **Vitals:**
  - Normal
- **Other:**
  - Varicosities, Pedal Oedema, Venous Stasis Ulcers, Eczema, Haemosiderin Pigmentation,
  - Palpation (Hard veins = thrombosis; Tenderness = Thrombophlebitis)
  - Special tests (Trendelenburg's & Perthe's Tests)



National Heart Lung and Blood Institute (NIH), CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

### Lymphangitis:

#### - TYPICAL Symptoms/Presentation:

- Fever
- Painful, Red, Subcutaneous Streaks
- Painful Lymphadenopathy

#### - TYPICAL Clinical Signs:

- **Vitals:**
  - Fever, Tachycardia, Tachypnoea
- **Other:**
  - Painful red subcutaneous streaks (inflamed lymphatics), Can → cellulitis



James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

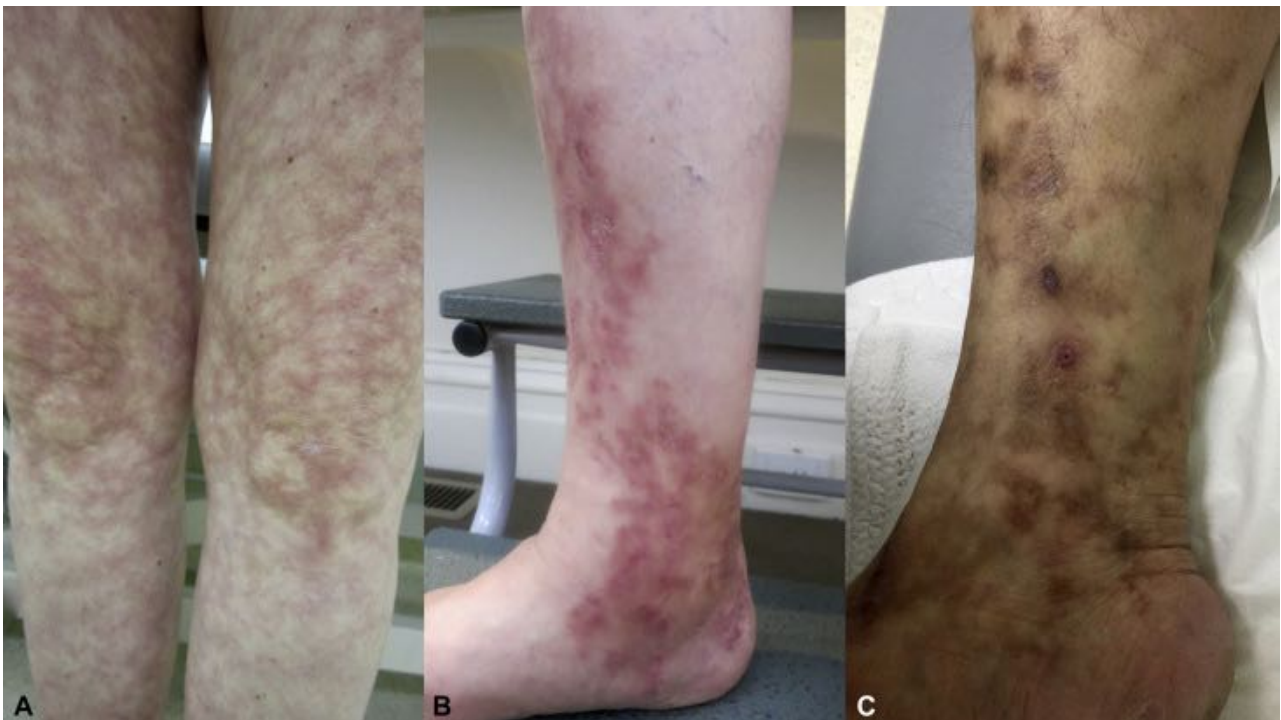


### Giant Cell (Temporal) Arteritis:

- **TYPICAL Symptoms/Presentation:**
  - **Fever/Fatigue/Weight Loss**
  - **Headache** (Pain on wearing a Hat)
  - **Jaw Claudication**
  - Visual Disturbances
  - (+/- *Polymyalgia Rheumatica* - Pain/Stiffness in Neck/Shoulders/Hips)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever
  - **Other:**
    - Temporal Artery Tenderness
    - Cachexia

### Polyarteritis Nodosa:

- **TYPICAL Symptoms/Presentation:**
  - Fever/Rash/Malaise/Weight Loss
  - Diffuse Myalgia, Peripheral Neuropathy
  - Abdo Pain, Melena
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Hypertension (due to *Kidney Failure*)
  - **Other:**
    - Rash
    - Digital Gangrene
    - Peripheral Neuropathy

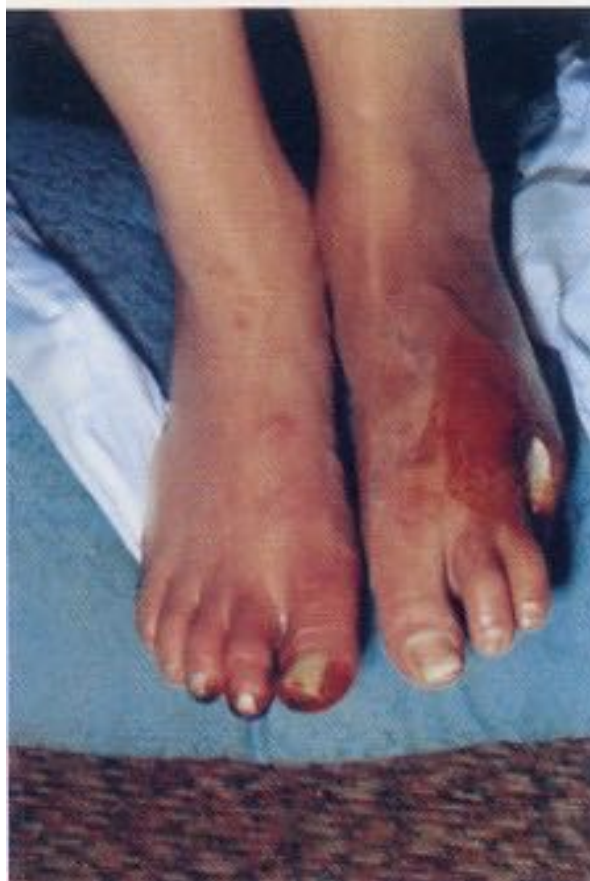


[https://www.jaad.org/article/S0190-9622\(19\)32992-5/fulltext](https://www.jaad.org/article/S0190-9622(19)32992-5/fulltext)



**Buerger's Disease (Thromboangiitis Obliterans/ "Smoker's Gangrene"):**

- **TYPICAL Symptoms/Presentation:**
  - Peripheral Vascular Insufficiency – Gangrenous Fingertips & Toes/Claudication
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal (Tachy if recent cigarette), (Tachypnoea if COPD)
  - **Other:**
    - Claudication
    - Chronic Arterial Ulcers
    - Gangrenous Toes/Feet/Fingers
  - **Signs of Causes:**
    - \*\*\*TAR STAINING FROM SMOKING



Onthelist, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons

## THE ENDOCRINE EXAMINATION

## THE ENDOCRINE EXAMINATION

### THE FULL ENDOCRINE EXAM:

- Introduction, Wash Hands, Consent,
- General Inspection:
  - Facies:

- Moon Facies (Cushings)



- Frightened Stare (Graves - Hyperthyroidism)



- Anhedonic + Puffy Eyes (Hashimotos – Hypothyroidism)
- Frontal Bossing, Enlarged Jaw/Tongue/Nose/Ear/Brow (Acromegaly)



- Hyperpigmented + Wasting (Addison's)



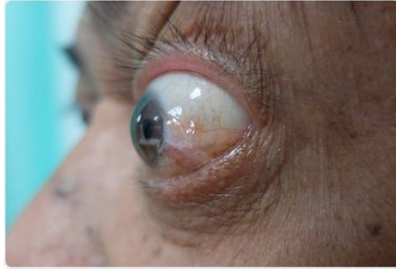
- Hirsutism + Receding Hairline (PCOS)



- **Body Habitus:**
    - Obesity (D2M, Cushing's, Hypothyroidism)
    - Wasting (D1M, Addison's, Hyperthyroidism)
    - Central Adiposity (Cushing's Syndrome)
  - **Hair Distribution:**
    - Hair Loss (Hypothyroidism)
    - Hirsutism (Cushing's & PCOS)
  - **Pigmentation:**
    - Generalised Hyperpigmentation (Addison's, Haemochromatosis)
    - Acanthosis Nigricans (Diabetes, Cushing's, Acromegaly, PCOS)
  - **Mental Changes:**
    - Slowness (Hypothyroidism, DKA, HONC, Addison's)
    - ↓Affect (Hypothyroidism)
  - **Vital Signs:**
    - **Pulse:**
      - Tachycardia (Hyperthyroidism, Pheochromocytoma, DKA)
      - Bradycardia (Hypotension)
    - **Blood Pressure:**
      - Hypertension (Hyper & Hypothyroidism/Phaeo/Cushing's/Conn's/PCOS/Acromegaly)
      - Hypotension (Addison's)
    - **Respiratory Rate:**
      - Typically Normal
      - Tachypnoea (DKA)
    - **Temperature:**
      - Hyperthermia (Hyperthyroidism)
      - Hypothermia (Hypothyroidism)
  - **Hands:**
    - Warm, Well Perfused, ↓CRT, Erythema, Sweaty (Hyperthyroidism/Phaeochromocytoma/Acromegaly)
    - Cool, Poorly Perfused, ↑CRT, Peripheral Cyanosis, Dry (Hypothyroidism)
    - Tremor (Hyperthyroidism, Pheochromocytoma)
    - Clubbing/Thyroid Acropathy (Grave's Disease [Hyperthyroidism])
    - Plummer's Nails/Onycholysis (Grave's Disease [Hyperthyroidism])
- 
- Brittle Nails (Hypothyroidism)
  - Palmar Crease Pigmentation (Addison's Disease)
- 
- Large, Spade-like Hands (Acromegaly)
  - Osteoarthritic Heberdon's & Bouchard's Nodules (Acromegaly)
  - Xanthomata (Hypothyroidism, Diabetes, Cushing's, Acromegaly, PCOS)
- **Arms:**
  - Proximal Myopathy (Hyperthyroidism, Hypothyroidism, Cushing's, Acromegaly)
  - Hyperreflexia (Hyperthyroidism); Hung Reflexes (Hypothyroidism)
  - Muscle Wasting (Diabetes, Addison's)

- **Face:**

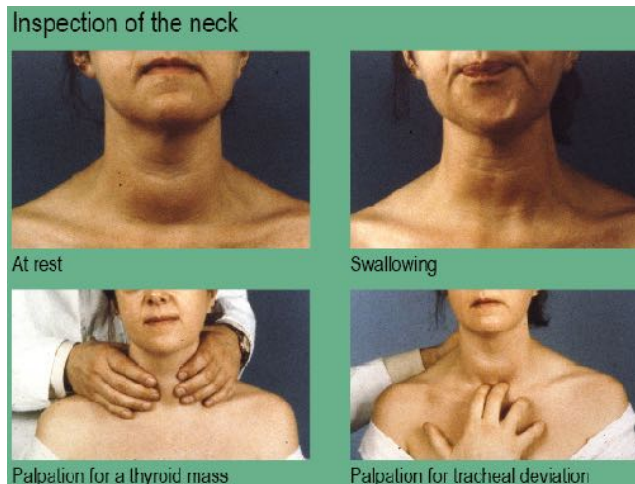
- (Conjunctival Pallor, Central/Peripheral Cyanosis)
- Receding Hairline + Loss of Lateral 1/3 Eyebrow + Periorbital Myxoedema (Hypothyroidism)
- Exophthalmos + Lid Lag (Hyperthyroidism)



- 
- Xanthelasma (Hypothyroidism, Diabetes, Cushing's, Acromegaly, PCOS)
- Bitemporal Hemianopsia (Pituitary Adenomas → Cushing's, Hyperthyroidism, Acromegaly, Prolactinoma, Conn's)
- Fundoscopy (Hypertensive & Diabetic Retinopathy)
- Buccal Hyperpigmentation (Addison's)
- Mouth Infections (Diabetes, Cushing's)
- **Facies:**
  - Moon Facies (Cushing's),
  - Frightened Stare (Graves - Hyperthyroidism)
  - Anhedonic + Puffy Eyes (Hashimoto's – Hypothyroidism)
  - Frontal Bossing, Enlarged Jaw/Tongue/Nose/Ear/Brow (Acromegaly)
  - Hyperpigmented + Wasting (Addison's)
  - Hirsutism + Receding Hairline (PCOS)

- **Neck:**

- **Thyroid Gland Examination:**
  - **General Inspection:**
    - Thyroidectomy Scars
    - Goitre (Hyperthyroidism/Hypothyroidism)
    - Swallow Test (Is it mobile?)
  - **Palpation From Behind:**
    - Locate Thyroid Gland (2cm below laryngeal prominence)
    - Goitre (Hyper/Hypo) – Size, Consistency, Tenderness, Mobility, Thrill
    - Swallow Test (Mobility)
    - Cervical Lymphadenopathy (Grave's & Hashimoto's)
  - **Auscultate:**
    - Thyroid Bruits (Hyperthyroidism)
  - **Retrosternal Goitre?:**
    - Pemberton's Sign
    - Abdominojugular Reflux
    - ↑JVP
    - Percuss Across Sternum





- Acanthosis Nigricans (Diabetes, Cushing's, Acromegaly, PCOS)



- 
- Buffalo Hump & Supraclavicular Fat Pads (Cushing's)



- 
- Carotid Bruits (Diabetes, Cushing's, Acromegaly, PCOS)

- **Chest:**

- Hair Distribution (PCOS)
- Gynaecomastia (Hyperthyroidism, Cushing's, Acromegaly)



- 
- Bony Tenderness (Cushings[Osteoporosis], Hyperparathyroidism[Osteoclast], Acromegaly[Growth])
- Hypertrophic Cardiomyopathy (Acromegaly) → Murmurs, Displaced Apex Beat
- Pericardial Effusion + Pleural Effusion (Hypothyroidism)
- Molluscum Fibrosum in Axilla (Acromegaly)

- **Abdomen:**

- Central Adiposity (Type2Diabetes, Cushing's)
- Obesity (Type2Diabetes, Cushing's, Hypothyroidism, PCOS[Metabolic Syndrome])
- Striae (Cushing's)



- 
- Fat Atrophy @ Injection Sites (Diabetes)
- Hepatomegaly (Diabetes[fatty liver], PCOS[fatty liver], Acromegaly[GH stimulation])
- Hepatomegaly + Splenomegaly + Renal Enlargement (Acromegaly)
- Adrenal Masses (Cushing's, Addison's, Conn's, Pheochromocytoma)
- Ovarian Masses (PCOS)
- ↑Bowel Sounds (Hyperthyroidism); ↓Bowel Sounds (Hypothyroidism)

- **Legs:**

- Proximal Myopathy (Hyperthyroidism, Hypothyroidism, Cushing's, Acromegaly)
- Quadriceps Atrophy (Diabetes)
- Fat Atrophy @ Injection Sites (Diabetes)
- Pretibial Myxoedema (Hyperthyroidism); Non-Pitting Oedema (Hypothyroidism)



- Shiny, Hairless Skin + Pitting Oedema (Diabetes→PVD)
- Charcot's Joint (Diabetes)



- Arterial/Venous Ulcers (Diabetes)
- Hyperreflexia (Hyperthyroidism); Hung Reflexes (Hypothyroidism)
- Foot-Drop (Acromegaly[Common Peroneal Nerve Entrapment])

- **Feet:**

- Tendon Xanthomata (Hypothyroidism, Diabetes, Cushing's, Acromegaly, PCOS)
- Warm, Well Perfused, Erythema, ↓CRT, Sweaty (Hyperthyroidism, Acromegaly)
- Cool, Poorly Perfused, ↑CRT, Peripheral Cyanosis, Dry (Hypothyroidism)

- **Diabetic Leg & Foot Exam:**

○ **General Inspection:**

- Quadriceps Wasting (Diabetic Amyotrophy)
- Shiny Skin + Hair Loss + Pitting Oedema + Venous Stasis Ulcers
- Arterial Ulcerations
- Neuropathic Ulcerations
- *Necrobiosis Diabeticorum*
- Tendon Xanthomata
- Peripheral Perfusion + CRT + Peripheral Pulses
- Bunions (Hyperkeratosis on Pressure Points)
- Hallux Valgus, Pes Cavus, Loss of Transverse Arch, Hammer Toes
- Fungal Infections of Nails & Between Toes

○ **Neurological Testing (Patient's Eyes Closed)**

- Monofilament[light touch] on 10 Areas of Feet
- Monofilament[light touch] on All Dermatomes of Leg
- Vibration Sense – Start as distally as possible
- Proprioception Sense
- Pain Sense – Over All Dermatomes of Leg
- Muscle Power
- Reflexes

### **Diabetic Patient Focussed Examination:**

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - Obese Body habitus
  - Acanthosis Nigricans
  - Xanthelasma
  - Fat Atrophy @ Insulin Injection Sites
  - Endocrine Facies (Cushing's, Acromegaly, PCOS)
  - Pigmentation (Haemochromatosis)
- **Vital Signs:**
  - Tachycardia (if Dehydrated)
  - Hypotension (if Dehydrated)
  - Tachypnoeic (If DKA)
  - Febrile (If Infection)
- **Lower Extremities:**
  - **General Inspection:**
    - Quadriceps Wasting
    - Fat Atrophy @ Insulin Injection Sites
    - PVD: Hair Loss + Shiny Skin + Oedema + Venous Ulcers
    - Arterial Ulcers
    - Neuropathic Ulcers
    - Charcot's Joint
    - Pes Cavus, Loss of Transverse Arch, Hammer Toes, Hallux Valgus
    - Fungal Nail Infections
    - Infections/Cuts Between Toes
  - **Palpation:**
    - Muscle Wasting
    - Temperature
    - Perfusion (CRT & Pulses)
  - **Neurological Examination – Patient's Eyes Closed:**
    - **Soft Touch (Monofilament):**
      - All 10 Areas of the foot
      - All Dermatomes of the Leg
    - **Vibration:**
      - Tuning Fork on Toes
    - **Proprioception:**
      - Big Toe
    - **Pain:**
      - All Dermatomes of Leg (But NOT the sole of foot)
    - **Muscle Power:**
      - All movements
    - **Muscle Reflexes:**
      - Patellar Tendon
      - Achilles Tendon
      - Babinski (Positive if Upgoing)(Normal = Downgoing)
  - **Further Examinations & Tests:**
    - Cardiovascular Examination
    - Respiratory Examination
    - Eye Exam + Fundoscopy
    - BSL + Urine Dipstick + HBA1C
    - Candida Mouth Infections
    - Acetone Fetor
    - Carotid & Renal Bruits
    - Hepatomegaly (Fatty Liver/Haemochromatosis)

### Hyperthyroid Focussed Examination:

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - Weight Loss
  - Flushing
  - Sweating
  - Anxiety
  - Frightened Stare + Exophthalmos
  - Goitre
  - Hoarseness
- **Vital Signs:**
  - Tachycardia
  - Hyperthermia
  - Hypertension
- **Hands:**
  - Hands Warm, Well Perfused, ↓CRT, Sweaty, Erythematous
  - Hyperthyroid Acropathy (Clubbing, Swelling & Redness of Fingers)
  - Onycholysis/Plumber's Nails (Separation of Nails from Nailbed)
  - Hand Tremor
- **Arms:**
  - Proximal Myopathy (Weakness Abducting Arm)
- **Face:**
  - (Standard check for: Conjunctival Pallor, Central Cyanosis, Peripheral Cyanosis)
  - Frightened Stare + Exophthalmos + **LID LAG**
  - **Bitemporal Hemianopsia** (If Central Hyperthyroidism)
  - Pemberton's Sign
  - Fundoscopy (hypertensive changes)
- **Neck:**
  - **Thyroid Examination:**
    - **General Inspection:**
      - Thyroidectomy Scars
      - Goitre
      - Swallow Test
    - **Palpation (From Behind):**
      - Locate Thyroid (2cm below Laryngeal Prominence)
      - Palpate Size, Consistency, Nodules, Mobility, Thrills, Tenderness
      - Swallow test (Mobility)
    - **Percuss:**
      - Retrosternal goitre
    - **Auscultate:**
      - Thyroid Bruits
  - **Cervical Lymphadenopathy:**
    - Submental/Submandibular/Pre-Auricular/Post-Auricular/Occipital/Jugular Chain/Posterior Triangle)
- **Chest:**
  - Gynaecomastia
- **Abdomen:**
  - Hepatojugular Reflux (Retrosternal Goitre)
- **Legs:**
  - Pretibial Myxoedema
  - Proximal Myopathy
- **Feet:**
  - Warm, Well Perfused, ↓CRT, Erythema, Sweaty
- **Thank Patient "That concludes my examination, now I'd like to run some tests"**

## Hypothyroid Focussed Examination:

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - Hypothyroid Facies (Apathetic, Puffy Eyes, Loss of Lateral Eyebrows, Thinning of Hair)
  - Obesity
  - Oedema
  - Mental Slowness & ↓Affect
- **Vital Signs:**
  - Bradycardia
  - Hypothermia
  - Hypertension
- **Hands:**
  - Cool, Poorly Perfused, ↑CRT, Peripheral Cyanosis, Dry
  - Brittle nails
  - Xanthomata
  - Palmar Crease Pallor + Koilonychia (Menorrhagia)
  - Inverse Prayer Test for Carpal Tunnel
- **Arms:**
  - Proximal Myopathy (Abduction Weakness)
  - Hung Reflexes
- **Face:**
  - Conjunctival Pallor (Menorrhagia)
  - Central/Peripheral Cyanosis
  - Loss of Lateral 1/3 of Eyebrow
  - Periorbital Myxoedema
  - Xanthelasma
  - Fundoscopy (Hypertensive Changes)
- **Neck:**
  - **Thyroid Examination:**
    - **General Inspection:**
      - Thyroidectomy Scars
      - Goitre
      - Mobility (Swallow Test)
    - **Palpation (From Behind):**
      - Locate Thyroid Gland (2cm Below Laryngeal Prominence)
      - Goitre (Hyper/Hypo) – Size, Consistency, Tenderness, Mobility, Thrill
      - Swallow Test (Mobility)
    - **Cervical lymphadenopathy:**
      - Submental/mandibular, Pre/Post-Auricular, Occipital, Jugular Chain, Post-Triangle
  - Pemberton's Sign (Retrosternal Goitre)
  - ↑JVP
- **Chest:**
  - Percuss for Retrosternal Goitre
  - Pericardial Effusion (Soft Heart Sounds)
  - Pleural Effusion (Stony Dullness)
- **Abdomen:**
  - Abdominojugular Reflux (Retrosternal Goitre)
  - Obesity
- **Legs:**
  - Non-Pitting Oedema
  - Proximal Myopathy (Squat)
  - Hung Leg Reflexes
- **Feet:**
  - Cool, Poorly Perfused, ↑CRT, Pale, Dry
  - Xanthomata
- **Thank patient "that concludes my examination", I'll go organise further tests.**



### Cushing's Focussed Examination:

- **Introduction + Wash hands + Consent**
- **General Inspection:**
  - Central Adiposity
  - Obesity
  - Moon Facies
  - Buffalo hump
  - Striae
  - **Hirsutism**
- **Vital Signs:**
  - **Hypertension**
- **Hands:**
  - Warm & Well Perfused, Normal CRT
  - Standard: No Palmar Crease Pallor
  - **Xanthomata**
- **Arms:**
  - Easy Bruising
  - Poor Wound Healing
  - **Proximal Myopathy (Ab/Adduction)**
- **Face:**
  - Characteristic *Moon Facies*
  - Xanthelasma
  - Fundoscopy (Hypertensive & Hyperglycaemic Changes)
  - Bitemporal Hemianopsia (if Pituitary Adenoma)
  - Mouth Infections
- **Neck:**
  - Carotid Bruits (Hypercholesterolaemia)
  - **Acanthosis Nigricans**
- **Chest:**
  - **Gynaecomastia**
  - **Bony Tenderness (Osteoporosis)**
- **Abdomen:**
  - Central Adiposity
  - Striae
  - Adrenal Masses
- **Legs:**
  - Easy Bruising
  - Poor Wound Healing
  - **Proximal Myopathy (Squat)**
- **Feet:**
  - Tendon Xanthomata

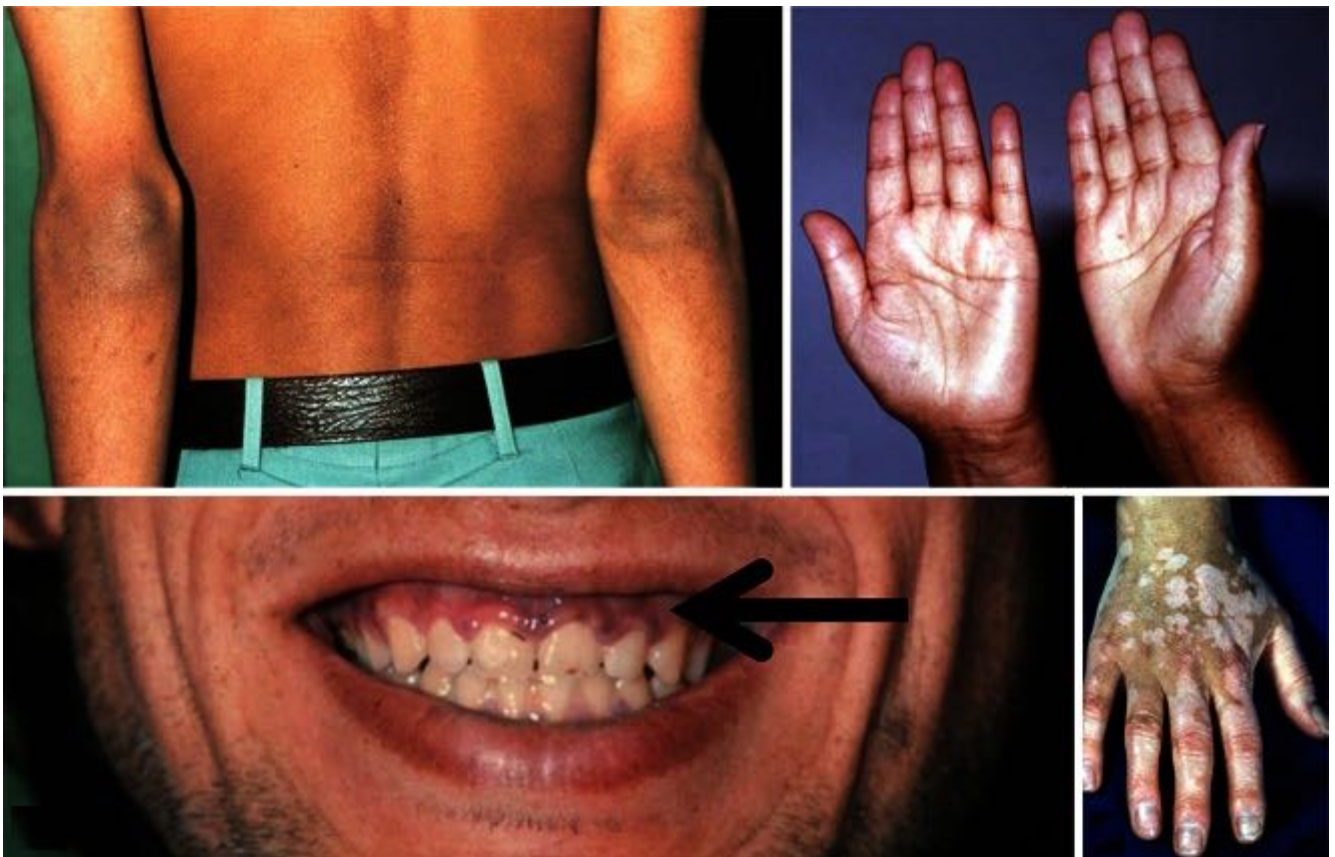
### Acromegaly Focussed Examination:

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - Acromegalic Facies – Frontal Bossing, Prominent Jaw/Brow Ridge/Nose/Lips/Tongue
- **Vital Signs:**
  - Hypertension
- **Hands:**
  - Large Spade-Hands
  - Warm & Well Perfused, ↓CRT, Palmar Erythema, Sweaty-Greasy Hands
  - Thickened Skin
  - Xanthomata
  - Osteoarthritis (Heberton's Nodules, Bouchard's Nodules)
- **Arms:**
  - Proximal Myopathy (Ab/Adduction)
- **Face:**
  - Acromegalic Facies – Frontal Bossing, Prominent Jaw/Brow Ridge/Nose/Lips/Tongue
  - Xanthelasma
  - Bitemporal Hemianopsia (Pituitary Adenoma)
  - Fundoscopy (Hypertensive Changes)
- **Neck:**
  - Acanthosis Nigricans
  - Carotid Bruits (due to ↑cholesterol)
- **Chest:**
  - Gynaecomastia
  - Bony Tenderness (Ribs & Spine) due to Osteoarthritis
  - Molluscum Fibrosum in Axillae (Skin Tags)
  - Hypertrophic Cardiomyopathy (Displaced Apex Beat, Murmurs & Signs of CCF)
- **Abdomen:**
  - Hepatomegaly
  - Splenomegaly
  - Renal Enlargement
- **Legs:**
  - Proximal Myopathy (Squat)
- **Feet:**
  - Enlarged Feet
  - Warm, Well-Perfused, ↓CRT, Erythema, Sweaty
  - Tendon Xanthomata
- **Thank patient “that concludes my examination”, I'll go organise further tests.**



### Addison's Disease Focussed Examination:

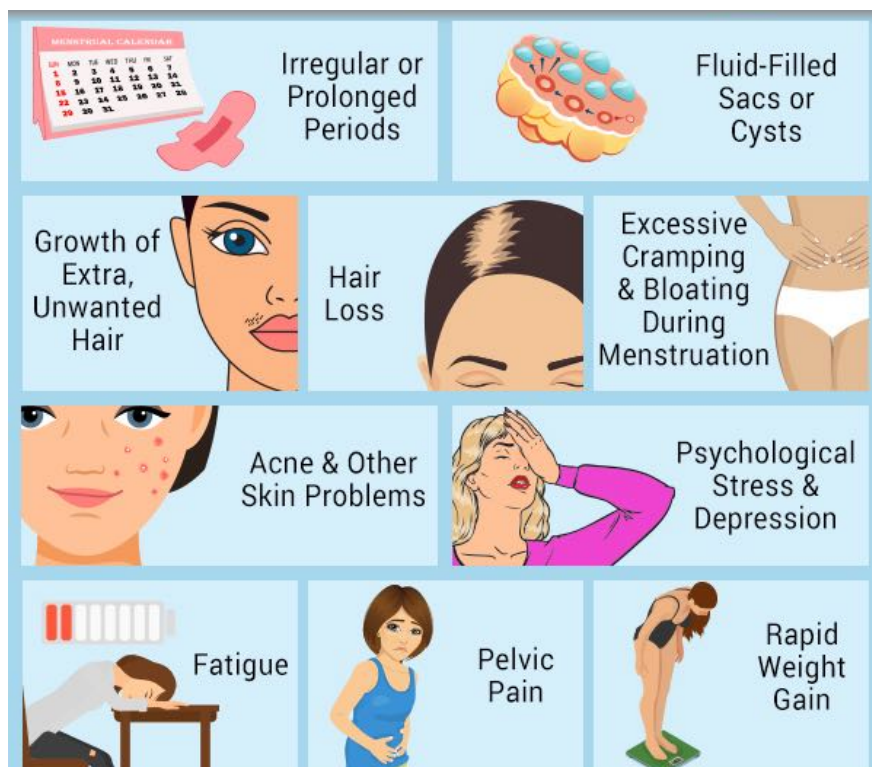
- **Introduction + Wash hands + Consent**
- **General Inspection:**
  - Wasting (Weight Loss)
  - Hyperpigmentation
  - Obvious Fatigue
  - **Mental Slowness**
- **Vital Signs:**
  - Hypotension (Hypovolaemia)
  - Tachycardia (Hypovolaemia)
- **Hands:**
  - Warm & Well Perfused, Normal CRT,
  - Palmar Crease Pigmentation
- **Arms:**
  - Generalised Muscle Weakness
  - **Muscle Wasting**
- **Face:**
  - Wasting (thin face)
  - Hyperpigmented + Pigmented Buccal Mucosae
  - Dry Mucosae (Dehydration)
- **Abdomen:**
  - Adrenal Masses
- **Legs:**
  - Generalised Muscle Weakness
  - **Muscle Wasting**
- **Feet:**
  - Warm & Well Perfused, Normal CRT,



Unattributable

### PCOS Focussed Examination:

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - o Hirsutism (Hair + Acne)
  - o Receding Hairline
  - o Obesity (Metabolic Syndrome)
  - o (Not Pregnant[infertile])
- **Vital Signs:**
  - o **Hypertension**
- **Hands:**
  - o Warm & Well Perfused
  - o No Palmar Crease Pallor
  - o Hirsutism (Hairy Dorsum of Hands)
  - o Xanthomata
- **Arms:**
  - o Hirsutism (↑Hair)
- **Face:**
  - o Xanthelasma
  - o Fundoscopy (Diabetic & Hypertensive Retinopathy)
  - o Hirsutism (Facial Hair & Acne)
  - o Deepening Voice
- **Neck:**
  - o Acanthosis Nigricans (Metabolic Syndrome)
  - o **Carotid Bruits (↑Cholesterol)**
- **Chest:**
  - o Hirsutism (Chest Hair)
- **Abdomen:**
  - o Ovarian masses/Tenderness
  - o PV Examination
  - o Central Adiposity (Metabolic Syndrome)
- **Legs:**
  - o Hirsutism (hair)
- **Feet:**
  - o Tendon Xanthomata



### Pituitary Disorders (Symptom Cluster):

- **Space-Occupying Lesions – (Eg: Adenomas):**
  - → Compression of Optic Chiasm → Visual Field Defect (Bitemporal Hemianopia)
  - → Compression of Extrinsic Ocular Muscles → Oculomotor Palsies (“Ophthalmoplegia”)
  - → Increased Intracranial Pressure → Headaches
- **Increased Hormone Secretion → Hyperpituitarism:**
  - **Anterior Pituitary:**
    - Eg: ↑PRL – Due to Prolactinoma → Galactorrhoea
    - Eg: ↑GH – Pit. Tumour → Gigantism (Kids)/Acromegaly(Adults)
    - Eg: ↑ACTH – Pit. Tumour → Cushing’s Disease
    - Eg: ↑FSH/LH – Pituitary/Hypothalamic Tumour → Precocious Puberty
    - Eg: ↑TSH – (Rare) Pit. Tumour → Primary Central Hyperthyroidism
  - **Posterior Pituitary:**
    - Eg: ↑ADH – Hypothalamic Osmoreceptor Dysfunction → SIADH
- **Decreased Hormone Secretion → Pan-Hypopituitarism:** (Eg: Sheehan’s Disease: Post-Partum Hypopituitarism) (Eg: Pituitary Apoplexy/ Infarction)
  - **Initially Asymptomatic**
  - **Later:**
    - 2° Hypothyroidism
    - 2° Adrenal Insufficiency (Similar to Addison’s)
    - 2° Dwarfism (GH Deficiency)
  - **(In Sheehan’s Syndrome):**
    - *Agalactorrhea* (No Lactation)
    - /Amenorrhoea after Delivery.

|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>SIADH:</u></b><br>(Hypothalamic Dysfunction → ↑ADH)                        | <ul style="list-style-type: none"> <li>- <b>1: Fluid Overload <i>Without</i> Oedema or Hypertension</b></li> <li>- <b>2: Hyponatraemia (Dilutional + ↑Excretion):</b> <ul style="list-style-type: none"> <li>○ If Severe → Cerebral Oedema →               <ul style="list-style-type: none"> <li>▪ Nausea/Vomiting</li> <li>▪ Headache/Confusion/Seizures/Coma</li> </ul> </li> </ul> </li> <li>- <b>3: High [Sodium] in Urine</b></li> <li>- <b>4: High Urine Osmolarity (relative to Plasma Osmolarity)</b></li> <li>- <b>5: Normal Renal &amp; Adrenal Function</b></li> </ul> |
| <b><u>Diabetes Insipidus:</u></b><br>(ADH Deficiency or Renal ADH Insensitivity) | <ul style="list-style-type: none"> <li>- Polydipsia (Extreme Thirst)</li> <li>- Polyuria (Excessive Urination)</li> <li>- Risk of Hypokalaemia</li> <li>- Risk of Dehydration (If water isn’t available)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                |



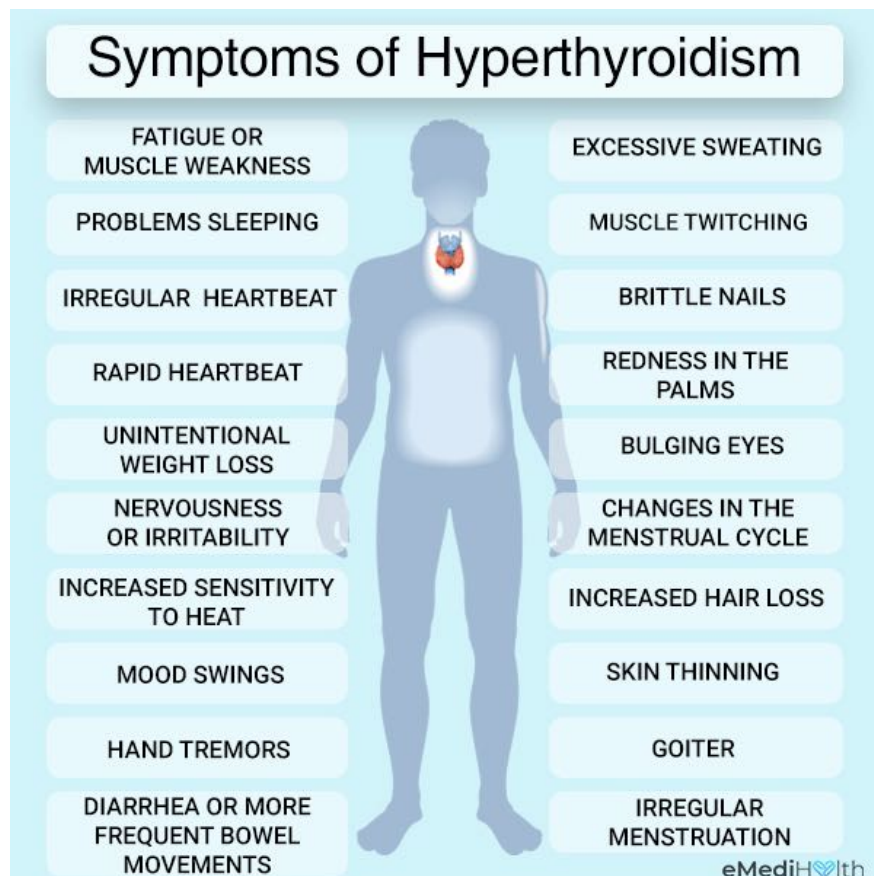
## **Hyperthyroidism & Thyrotoxicosis:** (Eg: Graves' Disease or Central ↑TSH)

### - **TYPICAL Symptoms/Presentation:**

- Fatigue,
- Weight Loss (Despite ↑Appetite)
- Diarrhoea
- Heat intolerance, ↑Sweating, Facial Flushing
- Irregular Menstruation
- Painless Goitre
- Anxiety, Tremor, Insomnia,
- Palpitations
- **(Note: May → CCF in Elderly)**

### - **TYPICAL Clinical Signs:**

- **Vitals:**
  - Tachycardia (Irregularly Irregular: AF), Hyperthermia, Hypertension,
- **Other:**
  - Weight Loss, Flushing, Anxiety, Sweating, Tremor, Alopecia, Frightened Facies
  - Hands Warm & Well Perfused, Palmar Erythema, Sweaty Hands, **Acropathy (Digital Clubbing & Swelling; Fingers & Toes) –GRAVES**, Onycholysis (Plummer's Nails = Separation of Nails from Nailbed), Hand Tremor,
  - Proximal Myopathy (Muscle Weakness) & Wasting, But HYPERREFLEXIA
  - Exophthalmos (Frightened Facies), Lid Lag,
  - Neck Scars (Thyroidectomy → ↑Exogenous Thyroid Hormone), Palpable Painless Goitre (Diffuse if Grave's Disease; Nodular if TMG), Mobile on Swallowing, Cervical Lymphadenopathy (Graves)
  - Thyroid Bruit
  - Check Pemberton's Sign (Retrosternal Goitre) & Percuss Across the Chest
  - Gynaecomastia
  - Pretibial oedema, HYPERREFLEXIA
- **Signs of Causes:**
  - Goitre
  - Pituitary Adenoma (Bitemporal Hemianopsia)



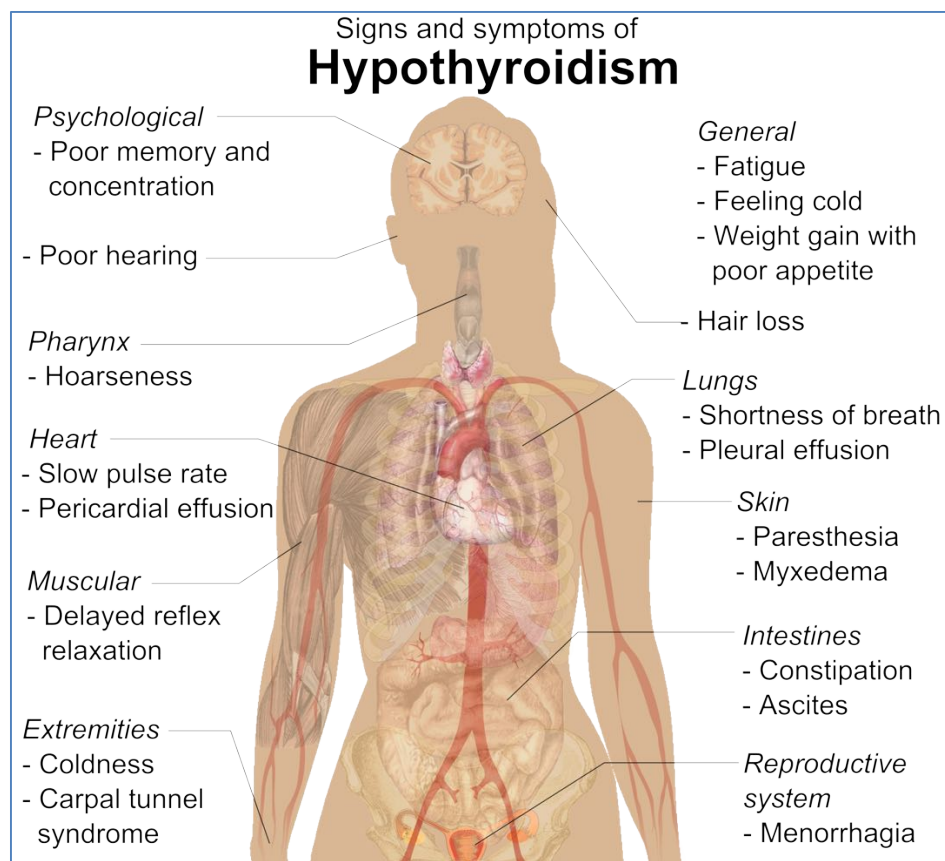
## **Hypothyroidism & Myxoedema:** (Hashimoto Thyroiditis, Iodine Deficiency or Central ↓TSH)

### - **TYPICAL Symptoms/Presentation:**

- **Early symptoms:**
- Fatigue, Depression
- Weight Gain (Despite ↓Appetite)
- Constipation
- Cold Intolerance, Dry Skin, Pale Skin
- Menorrhagia
- Painless Goitre
- Stupor, Memory Loss
- Muscle Pain
- Puffy Eyes, Hair Loss of Eyebrows

### - **TYPICAL Clinical Signs:**

- **Vitals:**
  - Bradycardia (Small Volume), Hypotension, Hypothermia
- **Other:**
  - Obesity, Oedema, Mental Sluggishness, Anhedonia, Slow Hypothyroid Speech, Hoarseness, Apathetic (Emotionless) Face,
  - Hands Cool, Dry & Poorly Perfused, ↑CRT, Peripheral Cyanosis, Brittle Nails, Palmar Crease Pallor (from Menorrhagia), Inverse Prayer Test for Carpal Tunnel, Xanthomata (↑Cholesterol)
  - Proximal Myopathy (Muscle Weakness) & Wasting, with “HUNG REFLEXES”
  - Facial Oedema, Periorbital Oedema, Alopecia, Loss of lateral 1/3 eyebrows, Xanthelasma (↑Cholesterol)
  - Painless Goitre, Mobile on Swallowing,
  - Pericardial Effusion, Pleural Effusions
  - Pedal Oedema, Hung Leg Reflexes, Peripheral Paraesthesia
- **Signs of Causes:**
  - Cretinism (Iodine Deficiency)



Häggström, Mikael (2014). "Medical gallery of Mikael Häggström 2014". WikiJournal of Medicine 1 (2). DOI:10.15347/wjm/2014.008. CC0, via Wikimedia Commons

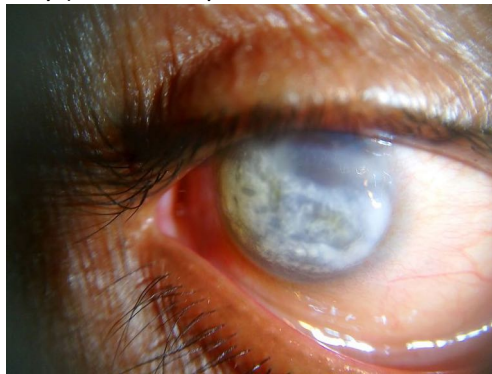
**Hyperparathyroidism:** ( $\uparrow$ PTH  $\rightarrow$  Hypercalcaemia & Hypophosphataemia)

- **TYPICAL Symptoms/Presentation:**

- (F >> M)
- **Asymptomatic** – Incidental Hypercalcaemia.
- **Symptomatic** – **“Bones, Moans, Stones & Abdominal Groans”**:
  - **Bone:** Pain/Osteoporosis/Fractures
  - **CNS:** Depression/Lethargy/Seizures
  - **Gallstones & Kidney Stones.**
  - **Abdo:** Constipation/Nausea/Ulcers
- **Other Organs:** Metastatic Calcification in Stomach/Lungs/Myocardium/Heart Valves/ & Vessels

- **TYPICAL Clinical Signs:**

- **Vitals:**
  - NA
- **Other:**
  - Mental State (Hypercalcaemia can  $\rightarrow$  Coma & Convulsions)
  - Signs of Dehydration (Hypercalcaemia can  $\rightarrow$  Polyuria)
  - Band Keratopathy (Calcium Deposition underneath the Corneal Epithelium)



Imrankabirhossain, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- Bony Tenderness (Shoulders, Sternum, Ribs, Spine, Hips)
- Pseudogout (Calcium Pyrophosphate deposition in the knee)
- Haematuria (Renal Stones)
- **Signs of Causes:**
  - $\uparrow$ PTH 2° to Chronic Renal Failure (CKD Signs + Osteoporosis)

**Hypoparathyroidism:** ( $\downarrow$ PTH  $\rightarrow$  Hypocalcaemia  $\rightarrow$  Tetany)

- **TYPICAL Symptoms/Presentation:**

- (Typically Post-Operative after Thyroidectomy)
- **\*Hypocalcaemia  $\rightarrow$  Neuromuscular \*\*Tetany\*\*:**
  - $\rightarrow$  Distal Paraesthesias
  - $\rightarrow$  Cramping & Spasms
  - $\rightarrow$  \*Laryngospasm (Emergency)
  - $\rightarrow$  Seizures
- (+/- CNS: Confusion/Depression/Psychosis)
- (+/- CVS: Characteristic Prolonged QT-Interval)

- **TYPICAL Clinical Signs:**

- **Vitals:**
  - NA
- **Other:**
  - **Trousseau's Sign** (Tetany of hand after 2mins of arm ischaemia with BP Cuff)
  - **Chvostek's sign** (Unilateral Facial Twitch when the Facial Nerve Outlet is Struck)
  - Hyperreflexia
  - Fragile Nails
  - Dry Skin
  - Tooth Deformities
- **Signs of Causes:**
  - Neck Scars (Thyroidectomy)

| <b>Disease:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>TYPICAL Symptoms/Presentation:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type 1 Diabetes:</b><br><i>Absolute Insulin Deficiency</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Juvenile Disease; Rapid onset:</b> <ol style="list-style-type: none"> <li><b>Polyuria, Polydipsia, Polyphagia</b></li> <li>Rapid Weight Loss Despite ↑ Appetite</li> <li>Nausea, Vomiting</li> <li>Fatigue</li> </ol><br><b>(+ LADA → Slowly Progressing “Type I Diabetes”):</b> <ol style="list-style-type: none"> <li>Typically Slim Adults (≈40yrs)</li> <li>Slow Progression to Insulin Dependency</li> <li>Same symptoms as D1M.</li> </ol>                                                                                                                                        |
| <b>Type 2 Diabetes:</b><br><i>Insulin Resistance And/Or Relative Insulin Deficiency</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Adult Disease; Slow, insidious onset:</b> <ol style="list-style-type: none"> <li><b>Polyuria, Polydipsia, Polyphagia</b></li> <li>Usually Overweight (Central Obesity)</li> <li>No Ketonuria</li> <li><b>Metabolic Syndrome</b> <ol style="list-style-type: none"> <li>1: Central Obesity</li> <li>2: Hypertension</li> <li>3: Impaired Glucose Tolerance</li> <li>4: Dyslipidaemia (↑TGLs, ↓HDL)</li> </ol> </li> </ol><br><b>(+MODY: Genetic – Autosomal Dominant):</b> <ol style="list-style-type: none"> <li>Young People, Non-Obese</li> <li>Sx of Hyperglycaemia – PPP</li> </ol> |
| <b>TYPICAL Clinical Signs Common To Diabetes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Vitals:</b> <ol style="list-style-type: none"> <li>Tachycardia (if Dehydration), Postural Hypotension (if Dehydration), Tachypnoea (if DKA)</li> </ol> <b>Other:</b> <ol style="list-style-type: none"> <li>Wasting (If Type 1), Obesity (If Type 2)</li> <li>Fat Atrophy @ Insulin Injection Sites</li> </ol> <b>Face:</b> <ol style="list-style-type: none"> <li>Visual Acuity, Fundoscopy (Retinopathy–Dots/Blots), Cataracts, Xanthelasma,</li> <li>Sweet Breath (DKA)</li> <li>Mouth Infections (Candida)</li> <li>Carotid Bruits (PVD)</li> <li>Acanthosis Nigricans</li> </ol> <b>Go from Face to Lower Limbs:</b> <ol style="list-style-type: none"> <li><b>Inspection:</b> Ulcers, Hyperkeratosis, Shiny Skin, Hair Loss, Fungal Nail Infections, Cellulitis, “Necrobiosis Lipoidica Diabeticorum” (Disgusting-looking Ulcer), “Charcot’s Joint” (Poor Proprioception), Hallux Valgus, Pes Cavus, Hammer Toes, Loss of Transverse Arch</li> <li><b>Palpation:</b> Cool Feet, ↑CRT, ↓Distal Pulses, Pedal Oedema,</li> <li><b>Neurological:</b> Monofilament (in 9 Areas of the Foot), Vibration, Proprioception, Pain (All Leg Dermatomes EXCEPT on Soles of Feet), Reflexes</li> </ol> <b>Signs of Causes:</b> <ul style="list-style-type: none"> <li>Cushings Facies, Acromegaly Facies, Pigmentation (Haemochromatosis)</li> </ul> <b>Further Examination:</b> <ul style="list-style-type: none"> <li>CVS (PVD), CKD (Renal), Full Neurological,</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Diabetic Ketoacidosis:** (Only in D1M or LADA)

- **TYPICAL Symptoms/Presentation:**
  - **Polyuria, Polydipsia, Polyphagia**
  - **Ketoacidosis →**
    - Hyperventilation
    - Headache/Altered Mental State
    - Vomiting → Dehydration, ↓ Na, ↓ Ca,
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (+/- Arrhythmia), Hypotension, Tachypnoea
  - **Other:**
    - Sweet Breath (Acetone Breath)
    - ALOC (↓ GCS)
    - Dehydration Signs (Loss of Skin Turgor, Enophthalmos, Dry Mucosae)
    - *Kussmaul's Breathing* (A form of Hyperventilation: Deep, Laboured Breathing)
  - **Signs of Causes:**
    - Diabetes (Insulin Injection Sites, Diabetes Bracelet, Peripheral Neuropathy, CKD)
  - **(Note: Hypokalaemia in DKA is a RED FLAG)**
  - **Treatment:**
    - IV access → Correct Dehydration
    - Insulin + Potassium Supp. → Correct Hyperglycaemia & Electrolytes
    - Treat underlying cause (Infection)

**HONC – Hyperosmolar Non-ketotic Crisis:** (Only in D2M)

- **TYPICAL Symptoms/Presentation:**
  - (Note: NO Ketogenesis – Even *Low Insulin* is enough to Inhibit Ketogenesis).
    - Hyperglycaemia
    - Polyuria (Osmotic Diuresis) & Dehydration
    - Hyperviscosity → ↑ Risk of Thrombosis
    - Mental Impairment (Stupor/Coma)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Dehydration), Hypotension (Dehydration)
  - **Other:**
    - Signs of Dehydration
    - Tremor
    - ALOC (Stupor/Coma)
    - Neurology – (Sensory/Motor Impairment, Focal Seizures, Hyporeflexia, Tremors)
  - **Treatment:**
    - IV Fluids
    - Insulin + Potassium (Since Insulin causes  $K^+$  Shift *Into Cells*)

**Hypoglycaemia:** (Eg: Insulinoma, Insulin Overdose, Missed Meal)

- **TYPICAL Symptoms/Presentation:**
  - Polyphagia
  - Tremor
  - ALOC (Confusion, Drowsiness, Seizures, etc)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia,
  - **Other:**
    - Sweating, Anxiety, Tremor, Palpitations
    - Confusion, Drowsiness, Visual Disturbances, Seizures, Coma
  - **Signs of Causes:**
    - Mental Illness (Suicide Attempt),
  - **Treatment:**
    - #1 – Give conservative dose of Oral/IV Glucose OR – IM/IV Glucagon



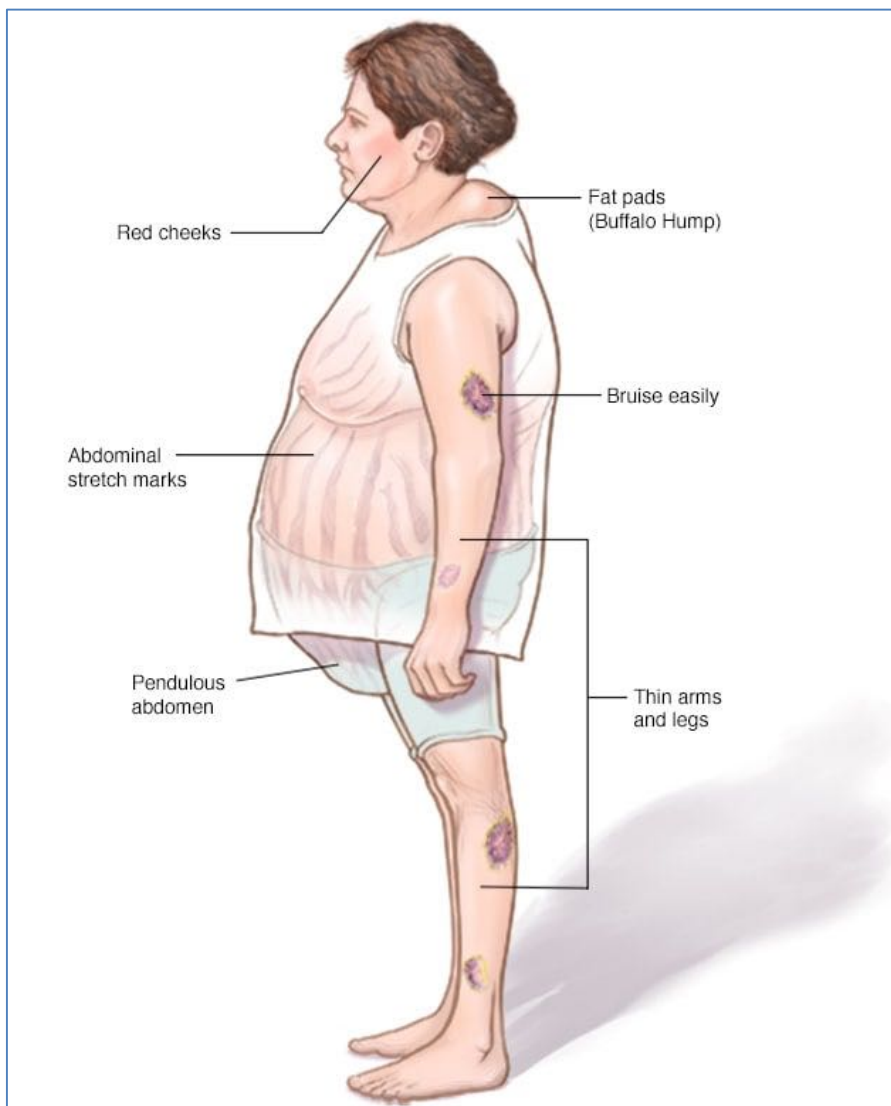
**Cushing's Disease:**(Hyper-Cortisolism)(*Central – Pituitary Adenoma; or Exogenous Steroids*)

- **TYPICAL Symptoms/Presentation:**

- Weight Gain
- Change in Appearance (Moon Facies)
- Mood Swings/Depression/Psychosis
- + Hirsutism & Menstrual Abnormalities

- **TYPICAL Clinical Signs:**

- **Vitals:**
  - Hypertension
- **Other:**
  - Moon Facies, Central Adiposity (Weight Gain), Striae, Buffalo Hump, Supraclavicular Fat Pads
  - Thin Skin, Easy Bruising, Poor Wound Healing
  - Hirsutism (in Females)
  - Visual Field Testing for Pituitary Tumour
  - Immunocompromise (Candida in Mouth, Fungal Nail Infections)
  - Ask Pt to Squat (Proximal Limb Muscle Atrophy & Weakness)
  - Bony Tenderness or Vertebral Bodies (Osteoporosis)
  - $\uparrow$ Cortisol  $\rightarrow$   $\uparrow$ Gluconeogenesis & Insulin Resistance  $\rightarrow$   $\rightarrow$  2<sup>o</sup>Diabetes:
    - Hyperglycaemia  $\rightarrow$  Polyuria, Polyphagia, Polydipsia
  - Osteoporosis, Backache
- **Signs of Causes:**
  - Bodybuilders? (Exogenous Steroid Use)
- **Diagnosis:** Negative Dexamethasone Suppression Test



**Conn's Disease:** (Hyper-Aldosteronism) (Pituitary Adenoma, Adrenal Hyperplasia, Carcinoma)

- **TYPICAL Symptoms/Presentation:**
  - (↑Aldosterone→Hypernatraemia(& Hypokalaemia))
  - **Renal Fluid Retention** →
    - **\*\*\*Hypertension (Often the ONLY Syx)**
    - **Hypernatraemia → Neuromuscular Syx** (Paraesthesia/Weakness/VisualΔ/Tetany)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - **\*\*\*Hypertension**
  - **Other:**
    - Paraesthesia/Weakness/VisualΔ/Tetany
    - Ballott Kidneys for Adrenal Tumours
  - **(Note: Hypertension can → LVH, ↑Risk of CVA & MI)**

**Phaeochromocytoma:** (Hyper-Adrenalinism) (Idiopathic or MEN2 Tumour Syndrome)

- **TYPICAL Symptoms/Presentation:**
  - **Symptoms:**
    - Palpitations/Tachycardia
    - Headache (Hypertension)
    - Sweating/Hot Flushes/Tremor
    - Anxiety
    - Nausea/Vomiting
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Paroxysmal Hypertension, Tachycardia,
  - **Other:**
    - Warm Sweaty Peripheries, ↓CRT, Facial Flushing, Pinpoint Pupils,
    - Cardiac Flow Murmurs (Hyperdynamic Circulation), ↑Bowel Sounds,
    - Ballott Kidneys for Adrenal Tumour
  - **Complications of Hypertension:**
    - Congestive Heart Failure & Pulmonary Oedema (Dyspnoea, PND, Orthopnoea, Inspiratory Creps, RV Heave, ↑JVP)
    - Myocardial Infarction
    - Ventricular Fibrillations
    - CVAs

**Addison's Disease:** (CHRONIC Adrenal Insufficiency) (Autoimmune Adrenalitis)

- **TYPICAL Symptoms/Presentation:**
  - (↓↓Aldosterone & ↓↓Cortisol)
  - **Insidious Onset:**
    - Weakness, Fatigue, Lethargy, Depression
    - Anorexia, Weight Loss,
    - Vomiting, Diarrhoea
    - Skin Hyperpigmentation
    - Polyuria → Dehydration (↓Aldosterone)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Postural Hypotension (Hypovolaemia)
  - **Other:**
    - Cachexia, Pigmentation (Palmar Creases, Elbows, Gums, Buccal Mucosa) [due to melanocyte-stimulating activity of ACTH],
    - Generalised Weakness
    - Dehydration Signs (Loss of Skin Turgor, Enophthalmos, Dry Mucosae)
  - **Diagnosis:**
    - Clinical Diagnosis
    - **Synacthen Test** (Measure Cortisol & Aldosterone 30mins after ACTH Injection)
    - Hypoglycaemia (Due to Glucocorticoid (Counter-Regulatory) Deficiency)

### **Waterhouse-Frederichson Syndrome:**

(ACUTE Adrenal Insufficiency due to Meningococcal Sepsis → Adrenal Infarction)

#### **TYPICAL Symptoms/Presentation:**

- **Abrupt & Severe Clinical Course**
  - ↓Aldosterone → Na & H<sub>2</sub>O Loss → Hypovolaemic Shock
- (Death in Hours-Days unless Treated)
- **+ Symptoms of Meningococcal:**
  - Fever
  - Headache
  - Photophobia
  - Neck Stiffness

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Fever (Due to Meningococcal)
- **Other:**
  - Kernig's Sign (Pain on Hip Flexion & Knee Extension) & Brudzinski's Sign (Neck Flexion causes Involuntary Hip & Knee Flexion), Photophobia, Headache,
  - Petechiae (Skin, Mucosae, Conjunctiva)
- **Signs of Causes:**
  - Typically Meningococcal Septicaemia
  - ∴ Neck stiffness
  - ∴ DIC

### **Polycystic Ovaries PCOS:** (Hyper-Androgenism) (Genetic – Sex-Limited Autosomal Dominant)

#### **TYPICAL Symptoms/Presentation:**

- (↓Conversion of Androgens → Oestrogen → Hyperandrogenism)
  - **1: Infertility:** Due to Anovulation
  - **2: Menstrual Changes:** Amenorrhoea → Infertility
  - **3: Masculinisation:** Acne, Hirsutism (↑Hair), Deepening Voice
  - **4: Metabolic Syndrome ("Synd. X"):** Insulin Resistance (+/- Obesity, D2M, ↑Cholesterol)

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Hypertension
- **Other:**
  - Hirsutism (Facial hair, deepening voice, acne)
  - Metabolic Syndrome (Obesity, Xanthelasma, Xanthomata, Acanthosis Nigricans, Polyuria, Polydipsia, Polyphagia)
  - Palpate Abdomen for Ovarian Masses

## **THE GASTROINTESTINAL EXAMINATION**

## THE GASTROINTESTINAL EXAMINATION

### Full GI/Abdo Exam:

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - **Alertness/Orientation**
    - ↓ in Hepatic Encephalopathy (Ammonia)
    - ↓ in Uraemic Encephalopathy
  - **Pain/Distress** (Acute Abdomen, Appendicitis, Pancreatitis, Cholecystitis, Diverticulitis, Bowel Obstruction, Perforation, etc)
  - **Body Habitus:**
    - Obesity (Fatty liver, Diabetes, Ascites, GORD)
    - Cachexia (Malignancy, Malabsorption, Cohn's/UC)
  - **Colour:**
    - Jaundice (Hepatitis, Cholelithiasis, Liver Failure, Cirrhosis, Haemolysis)

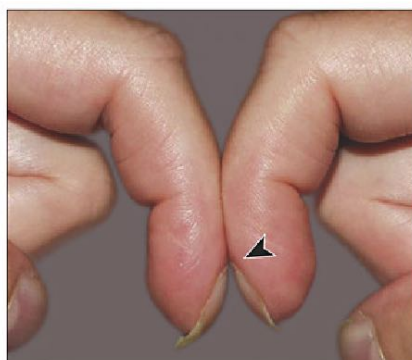


- Pallor (Anaemia, Malignancy, Malabsorption, GI Bleeding)
  - Pigmentation (Haemochromatosis)
  - Bruising, Bleeding, Petechiae (Liver Failure, Haematological Malignancy)
- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (Anaemia, Blood Loss, Pain, Infection)
    - AF (Dilated Cardiomyopathy in Alcoholism)
  - **Blood Pressure:**
    - Hypertension (Renal Failure, Pain)
    - Hypotension (Blood Loss, Shock)
    - Postural Hypotension (Anaemia)
  - **Respiratory Rate:**
    - Tachypnoea (Pain, Anaemia)
  - **Temperature:**
    - Fever (Infection)
- **Hands:**
  - **Perfusion + CRT**
    - Warm & Sweaty (Carcinoid) + (Hyperthyroid, Pheochromocytoma, Acromegaly)
    - Cool & Dry (Shock, Hypovolaemia)
  - **Clubbing (Crohn's Disease & Ulcerative Colitis) + (Cardiac/Resp Disease & ↑Thyroid Acropathy)**

Normal



Clubbed





- **Leukonychia & Muercke's Lines** (Hypoalbuminaemia – Liver Failure, Nephrotic Syndrome)



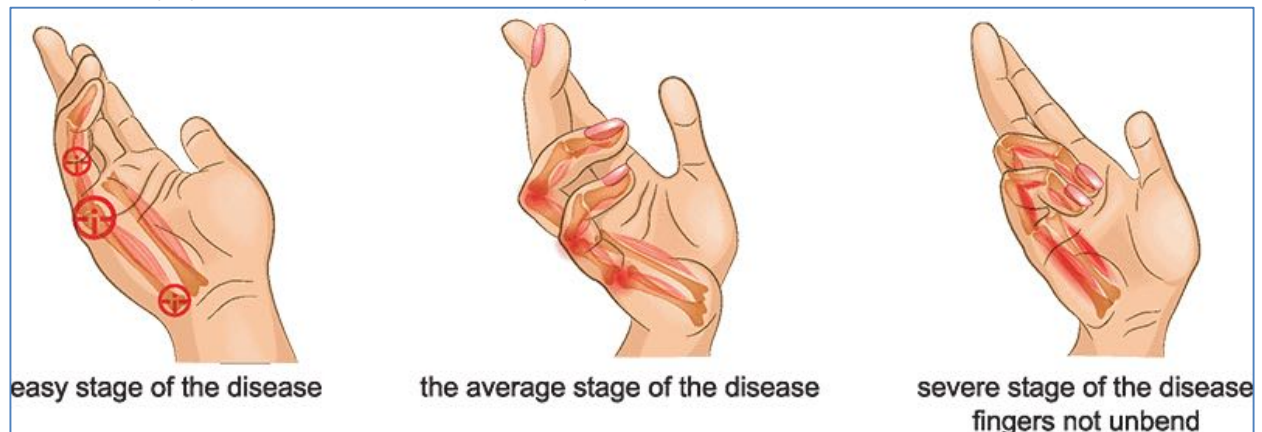
- **Koilonychia** (Iron Deficiency – Blood Loss, GI Bleed, Malabsorption)



- **Blue Lunulae** (Wilson's Copper Disease)



- **Palmar Crease Pallor + Pale Nails** (Anaemia)
- **Palmar Erythema** (↑Oestrogen in Liver Disease)
- **Dupuytren's Contracture** (Alcoholic Hepatitis/Cirrhosis)



<https://www.drbesb.com/dupuytren-contraction/>

- **Xanthomata** (↑Cholesterol – Fatty Liver, Nephrotic Syndrome, Diabetes)
- **Hepatic Flap** (Asterixis – Hepatic Encephalopathy)

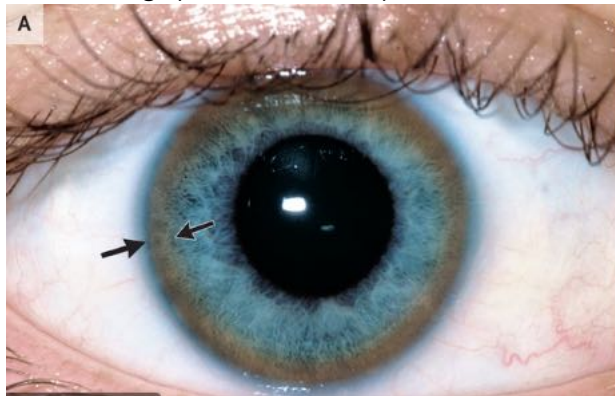
- **Arms:**

- Bruising, Petechiae
- Scratch Marks (Uraemic Pruritis)
- Uraemic Frost
- Acanthosis Nigricans in Axilla (GI Malignancy)

- **Face:**

○ **Eyes:**

- Conjunctival Pallor (Anaemia)
- Scleral Icterus (Jaundice)
- Keyser Fleischer Rings (Wilson's Disease)



<https://www.nejm.org/doi/full/10.1056/nejmicm1101534>

- Iritis (Crohn's/Ulcerative Colitis)
- Xanthelasma (↑Cholesterol – Biliary Obstruction, PBC, Cirrhosis, Diabetes)

○ **Mouth:**

- Hydration
- Parotid Gland Enlargement (Alcoholism)
- Central/Peripheral Cyanosis
- Mucosal Ulcers (Crohn's/UC)
- Glossitis/Angular Stomatitis (Anaemia, Alcoholism B12, Malabsorption B12)



- 
- Peutz Jegher's Pigmentation
- Fetor Hepaticus
- Mucosal Petechiae
- Leucoplakia (Spirits, Smoking, Sepsis, Syphilis, Shit teeth)



- **Neck:**

- Supraclavicular Lymph Nodes (Virchow's Node = GI/Lung Malignancy)

- **Chest:**

- Gynaecomastia (↑Oestrogen – Liver Failure)
- >3 Spider Naevi (↑Oestrogen – Liver Failure)



▪

- **Abdomen:**

○ **Inspection:**

- Abdominal Distension (Ascites, Obstruction)
- Scars
- Visible Masses (Cancer, Hernias)
- Visible Peristalsis (Obstruction)
- Bruising, Petechiae (Liver Failure)
- **Cullen's & Grey Turner's Sign (Pancreatitis, Haemoperitoneum)**
- Caput Medusa (Portal HTN, Cirrhosis)
- **Striae**
- Vesicles (Shingles)

○ **Palpation:**

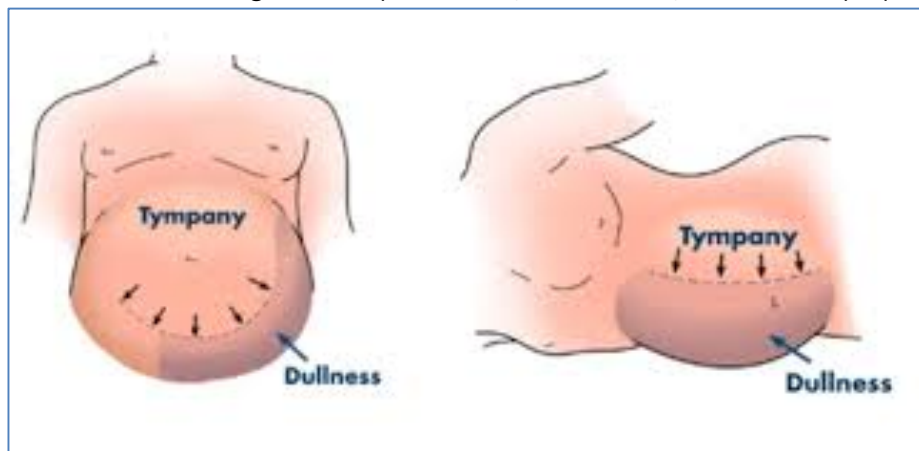
- Light Palpation – Tenderness, **Guarding, Rigidity, Rebound? (Peritonitis)**
- Deep Palpation – Masses?
- Hepatomegaly (Fatty Liver, Portal HTN, Hepatitis, Hepatocellular Carcinoma, Polycystic Liver)
  - Small Liver (Cirrhosis)
  - Pulsatile Liver (Tricuspid Regurgitation)
- Splenomegaly (Infection, Haematological Malignancy)
- **Ballott Kidneys**
- Aortic Aneurysm
- Para-Aortic Lymph Nodes (Malignancy, Infection)

○ **Special Tests:**

- **Cholecystitis:** Murphey's Sign
- **Appendicitis:** Rovsing's Sign, Pain @ Mcburney's Point, Psoas Sign, Obturator Sign.
- **Pyelonephritis/Renal Stones:** Murphey's Kidney Punch

○ **Percussion:**

- Ascites & Shifting Dullness (Portal HTN, Liver Failure, Renal Failure) + (Heart Failure)



<https://depts.washington.edu/physdx/liver/tech.html>

- Percuss for Splenomegaly & Hepatomegaly

○ **Auscultation:**

- Bowel Sounds (Absent in Ileus)
- Renal Bruits

○ **+ Deferred PR Exam for Cancer, Blood, Malena.**

- **Legs:**

- Pitting Oedema (Liver Failure, Renal Failure)
- Bruising, Petechiae (Liver Failure)
- Varicosities (Portal Hypertension)

- **Feet:**

- Perfusion & CRT
- Xanthomata (↑Cholesterol – Biliary Obstruction, PBC, Cholelithiasis)
- **Leukonychia & Muercke's Lines** (Hypoalbuminaemia – Liver Failure, Nephrotic Syndrome)
- **Clubbing (Crohn's Disease & Ulcerative Colitis)**

**Mallory Weiss Syndrome Tear:** = Oesophageal laceration – (Longitudinal tear)

- **TYPICAL Symptoms/Presentation:**
  - Dysphagia & Odynophagia
  - Chest Pain
  - Haematemesis
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal (Tachy if Pain/Dehydration/Blood Loss), Bradypnoea (if Alkalotic from Vomiting)
  - **Signs of Causes:**
    - Obesity (Gluttony), Subconjunctival Haemorrhages (Severe Coughing), Low BMI (Bulimia), Destroyed dentition (Bulimia)
    - Note: Nothing to do with Alcohol.

**Oesophageal Varices (Due to Portal Hypertension):** (Often 2<sup>o</sup> to alcoholic cirrhosis)

- **TYPICAL Symptoms/Presentation:**
  - Hematemesis
  - Haematochezia
  - May Rupture → Massive Bleeding
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal unless Haemorrhage (Tachycardia, Tachypnoea, Hypotension)
  - **Other:**
    - Portal HTN (Caput Medusa, Ascites, Hepatomegaly, Splenomegaly, Pedal Oedema)
    - If Cirrhosis (Palmar Erythema, Mee's/Meurkhe's/Leukonychia, Jaundice, Xanthomata, Xanthelasma, Scleral Icterus, Hepatic Fetor, Hepatic Flap, Ascites, Pedal Oedema)
  - **Signs of Causes:**
    - Chronic Alcoholism (Dupuytren's Contracture, Cerebellar Dysfunction)

**Gastritis:** = Inflammation of the Stomach Lining

- **TYPICAL Symptoms/Presentation:**
  - Epigastric Pain
  - Nausea/Vomiting
  - Indigestion (Dyspepsia)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (if infection), all else normal
  - **Other:**
    - Epigastric Tenderness
  - **Signs of Causes:**
    - Infection
    - Pernicious Anaemia (B12 Deficiency → Peripheral Neuropathy & Macrocytic Anaemia)
    - Alcohol Abuse (Dupuytren's Contracture, Macrocytic Anaemia, Cerebellar Dysfunction)

**Peptic Ulcer Disease:** (NSAIDs, H.Pylori, or Gastrinoma/ZE-Syndrome)

- **TYPICAL Symptoms/Presentation:**
  - Burning Epigastric Pain (Relieved by Food)
  - Haematemesis/Melena
  - Nausea & Vomiting
  - (If Perforated → Acute Peritonitis & Shock)
  - (If Pyloric Stenosis → Intractable Vomiting)
  - **Note: May → Gastric Ca. → Weight Loss**
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal unless Perforated (Tachycardia, Tachypnoea, Hypotension)
    - If Anaemic from GI Bleeding (Tachycardia, Tachypnoea)
  - **Other:**
    - But may have weight-loss
    - If Anaemic from NSAIDs → Pale Nails, Koilonychia (↓Fe), Palmar Pallor, SC Pallor, Atrophic Glossitis.
    - If Ruptured → Peritonism (Guarding, Rigidity, Rebound, Shoulder-tip Pain, Shallow Breathing, Cullen's Sign, Grey Turner's Sign, Peripheral Shutdown, ALOC)

**Gastric & Duodenal Cancers:** ("Fungating" – H.Pylori) ("Leather Bottle" – Familial)

- **TYPICAL Symptoms/Presentation:**
  - **Asymptomatic until Advanced Disease**
  - **Early Symptoms:** Epigastric Pain, Nausea/Vomiting, Anorexia/Weight Loss, Adenopathy, Anaemia.
  - **Late Symptoms:** Malignant Ascites/Jaundice, Symptoms of Brain/Bone/Lung Mets
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia (If Anaemia or Hypovolaemic), Tachypnoea (If Anaemia)
  - **Other:**
    - Virchow's Node (L-Supraclavicular LN), Acanthosis Nigricans in Axillae,
    - Anaemia (Pale Nails, Palmar Pallor, SC Pallor, Atrophic Glossitis)
    - Metastases (Oesophagus, LN, Liver, Lungs)
  - **Signs of Causes:**
    - Peutz Jegher's Syndrome (Mouth Pigmentation, Clubbing)
    - Alcoholism, Smoking

**Coeliac Disease:** (SI Gluten Hypersensitivity)

- **TYPICAL Symptoms/Presentation:**
  - Can Present at ANY Age (Peaks = Infancy, in 50's)
  - Fatigue, Malaise
  - Diarrhoea/Steatorrhea
  - Abdo Pain/Discomfort/Bloating
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Weight Loss, Mouth Ulcers, Angular Stomatitis, Atrophic Glossitis,



**Radiation Enteritis:** (Fibrosis from Radiotherapy)

- **TYPICAL Symptoms/Presentation:**
  - **Acute Symptoms** – Nausea, Vomiting, Diarrhoea, Abdo Pain. (Improves within 6wks of Radiation)
  - **Chronic Symptoms** – (Symptoms for >3mths) Pain due to Obstruction, Malabsorption, Diarrhoea, Tenesmus
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Hypovolaemia/Anaemia), Tachypnoea (Anaemia), Hypotension (Hypovolaemia)
  - **Other:**
    - Radiation Tattoos, Radiation Fibrosis of Skin (Abdomen/Back/Perineum), Anal Fissures (Diarrhoea), Surgical Scars, Abdominal Distension

**Whipples Disease:** (Chronic Bacterial Infection: *Tropheryma Whipplei*)

- **TYPICAL Symptoms/Presentation:**
  - **Initially** – Arthritis & Arthralgia
  - **YEARS Later** → Fever, Abdo Pain, Diarrhoea, Weight Loss → MALABSORPTION
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia
  - **Other:**
    - Lymphadenopathy, Poly Arthritis, Steatorrhea, Oedema, Anaemia (Fe/B12), Weight Loss
    - Can → Brain Damage (Mental Changes/Memory Loss)
    - Can → Endocarditis (Heart Murmur)

**Acute Bacterial Diarrhoeal Diseases:**

(Enterotoxigenic E-Coli = Traveller's)

(S-aureus & Salmonella = Food Poisoning) (Shigella = Dysentery)

- **TYPICAL Symptoms/Presentation:**
  - Hyperacute (<24hrs) – Probably ETEC Toxin
  - Sub-Acute (<3-5days) – Probably Infective Gastroenteritis (Food Poisoning)
  - (If Chronic, more likely to be Parasitic than Bacterial)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea, Hypotension (Hypovolaemic)
  - **Other:**
    - Peripheral Shutdown, ↑CRT, Low-Volume Tachycardia, Dry Mucosae, Enophthalmos, Loss of Skin Turgor
  - **Signs of Causes:**
    - Dysentery (Blood & Pus in Stools) = Shigella

**Acute Viral Diarrhoea:**

- **TYPICAL Symptoms/Presentation:**
  - Vomiting
  - + Watery Diarrhoea
  - + Fever
  - (Typically In a Child/Infant)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea, Hypotension (Hypovolaemic)
  - **Other:**
    - Typically In a Child/Infant → Irritability, Poor Feeding
    - Peripheral Shutdown, ↑CRT, Low-Volume Tachycardia, Dry Mucosae, Enophthalmos, Loss of Skin Turgor
  - **Signs of Causes:**
    - Hippie Mother Not Vaccinating!



## **Chronic Diarrhoea** (Usually Parasitic/ Irritable Bowel/ or Malignancy)

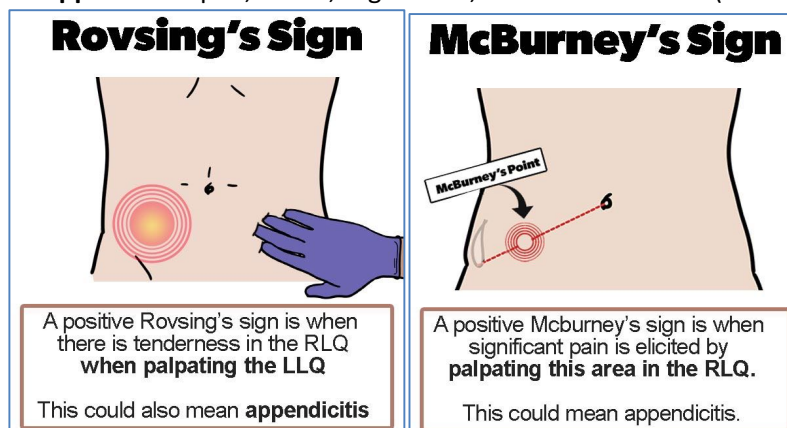
- **TYPICAL Symptoms/Presentation:**
  - o Long-Term Diarrhoea
  - o Weight Loss
  - o Fatigue
- **TYPICAL Clinical Signs:**
  - o **Vitals:**
    - Normal
  - o **Other:**
    - Anaemia (Microbleeding → Pale Nails, Koilonychia, Palmar Pallor, SC Pallor, Atrophic Glossitis), Anal Fissures.
    - If CD/UC – Clubbing, Mucosal Ulcers, Red Eyes
  - o **Signs of Causes:**
    - Immunocompromise
    - (Also Think Coeliac Disease & Colon Ca)

## **Intestinal Tuberculosis**

- **TYPICAL Symptoms/Presentation:**
  - o \*\*Fever + Night Sweats
  - o \*\*Weight Loss
  - o \*Ileocecal Area is most commonly affected → RIF Abdominal Pain
- **TYPICAL Clinical Signs:**
  - o **Vitals:**
    - Fever, Tachycardia, Tachypnoea (if Pulmonary TB)
  - o **Other:**
    - Palpable Masses, Generalised Peritonitis, Bowel Obstruction, Anaemia
  - o **Signs of Causes:**
    - Immunocompromised (HIV/Drugs)
    - Pulmonary TB

## **Appendicitis:**

- **TYPICAL Symptoms/Presentation:**
  - o Initially Umbilical Pain → Severe **RIGHT** Iliac Fossa Pain
  - o Fever
  - o Nausea/Vomiting/Anorexia/Diarrhoea(occasionally)
  - o (Ie: Similar to Diverticulitis, but on the Right)
- **TYPICAL Clinical Signs:**
  - o **Vitals:**
    - Fever, Tachycardia,
  - o **Other:**
    - R-Iliac Fossa Pain/Tenderness/Guarding
    - **Rovsing's Sign:** Referred Rebound Pain in the RIF when the LIF is Pressed.
    - **Psoas Sign:** RIF Pain on Flexion of the Hip
    - **Obturator Sign:** RIF Pain on Internal Rotation of the Hip
    - **McBurney's Sign:** Deep tenderness at McBurney's point
  - o **If Ruptured Appendix:** Sepsis, Shock, High Fever, General Peritonitis (Guarding/Rigidity/Rebound)



### Pseudomembranous Colitis:

Clostridium Difficile Overgrowth due to Antibiotic → ↓ Gut Flora (C-Difficile is Directly Cytotoxic)

#### - TYPICAL Symptoms/Presentation:

- Onset within 2days of Antibiotics; Persists for 2wks After.
- Fever,
- Abdo Cramps
- Profuse Water Diarrhoea (<10/day)
- Faecal Urgency
- **Note: Can Perforate**
- **Note: Can → Toxic Megacolon**

#### - TYPICAL Clinical Signs:

- **Vitals:**
  - Fever, Tachycardia (Infection & Dehydration), Tachypnoea (infection), Hypotension (Dehydration)
- **Other:**
  - Abdo Pain, Haematochezia
  - (If Perforation – Peritonitis, Shoulder-tip Pain, Cullen's/Grey-Turner's, Shock)
  - (If Toxic Megacolon – Abdominal Distension, Abdo Tenderness, Septic Shock, Loss of Bowel Sounds)
- **Signs of Causes:**
  - History of Antibiotic Usage

### Diverticulosis/Diverticulitis:

#### - TYPICAL Symptoms/Presentation:

- **Note: Diverticulosis is Asymptomatic**
- **Note: Diverticul-ITIS is Symptomatic:**
  - Severe **LEFT** Iliac Fossa Pain
  - Fever
  - Constipation
- (Ie: Similar to Appendicitis, but on the Left)

#### - TYPICAL Clinical Signs:

- **Vitals:**
  - Fever, Tachycardia,
- **Other:**
  - LIF Tenderness/Guarding/Rigidity
  - GI Bleeding → Anaemia (Pale Nails, Koilonychia, Palmar/SC Pallor, Glossitis)
  - (Note: If Perforation → Peritonitis, Shoulder-Tip Pain, Sepsis, Shock, Death)
- **Signs of Causes:**
  - Opioid Addicts (Chronic Constipation), Paraplegic, Multiparity.

### Crohn's Disease: (Mouth → Anus)(Patchy)

#### - TYPICAL Symptoms/Presentation:

- Typically Starts @ Ileocecal Valve (RIF)
- **Common Symptoms (Both CD & UC):**
  - \*\*Abdominal Pain/Severe Internal Cramps
  - \*\*Vomiting/Diarrhoea \*(Porridge-like, Fatty)
  - \*\*Rectal Bleeding
- (+ Fever, Weight Loss)

#### - TYPICAL Clinical Signs:

- **Vitals:**
  - **Fever,**
- **Other:**
  - Clubbing
  - **Mouth Ulcers & Anus Involvement,**
  - GI Bleeding → Anaemia (Pale Nails, Koilonychia, Palmar/SC Pallor, Glossitis)
- **Signs of Causes:**
  - Autoimmune (Arthritis, Iritis, Pyoderma Gangrenosum, Primary Biliary Cirrhosis)

### Ulcerative Colitis: (Typically affects Colon)(Continuous)

- **TYPICAL Symptoms/Presentation:**
  - Typically Starts @ Rectum (LIF)
  - **Common Symptoms (Both CD & UC):**
    - \*\*Abdominal Pain/Severe Internal Cramps
    - \*\*Vomiting/Diarrhoea \*(Bloody & Mucus – but NO Pus. {Not Dysentery})
    - \*\*Rectal Bleeding
  - (+ Tenesmus)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Afebrile,
  - **Other:**
    - Clubbing
    - No Mouth or Anus Involvement,
    - GI Bleeding → Anaemia (Pale Nails, Koilonychia, Palmar/SC Pallor, Glossitis)
  - **Signs of Causes:**
    - Autoimmune (Arthritis, Iritis, Pyoderma Gangrenosum, Primary Biliary Cirrhosis)

### Irritable Bowel Syndrome/‘Spastic Colon’:

(Umbrella Term)

- **TYPICAL Symptoms/Presentation:**
  - Chronic Abdo Pain/Discomfort
  - Alternating Bowel Habits (+ Tenesmus)
  - (All Investigations Normal)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Normal
  - **Signs of Causes:**
    - \*Stress/Anxiety/Depression, Chronic Pain, Gut Hypersensitivity.

### Hirschsprung’s Disease (“Congenital Aganglionic Megacolon”):

(Immotile Section of Colon)

- **TYPICAL Symptoms/Presentation:**
  - **Presentation within 2-3days after Birth:**
    - Delayed Meconium (First Defecation)
    - Abdo Distension
    - Vomiting
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Abdominal Distension & Tenderness, Poor Feeding, Irritability, Anorexia
  - **Signs of Causes:**
    - Baby

**Meckel's Diverticulum:** (Congenital SI True Diverticulum)

- **TYPICAL Symptoms/Presentation:**
  - (Note: Majority are Asymptomatic)
  - **Presentation @ 2yrs Old:**
    - Malena (Bleeding)
    - Severe Central Abdo Pain (Obstruction/Volvulus/Intussusception)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Pain/if infected), Tachypnoea (Pain), Hypertension (Pain)
  - **Other:**
    - Abdominal Distension, Abdominal Tenderness, Peritonitis & Sepsis (perforation), Loss of Bowel Sounds, Visible Peristalsis, Cullen's/Grey-Turners (Haemoperitoneum)
  - **Signs of Causes:** Baby

**Colonic Polyps:** (Note: Common in Autosomal Dominant Peutz-Jeger's Syndrome)

- **TYPICAL Symptoms/Presentation:**
  - Asymptomatic in Early Stages
  - (+/- Change in Bowel Habits)
  - (+/- Sx of Anaemia)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - None
    - If Anaemia (Koilonychia, Pale Nails, PC/SC Pallor, Atrophic Glossitis)
  - **Signs of Causes:**
    - Peutz-Jeger's Syndrome (Clubbing + Melanin Pigmentation on Lips & Hands)

**Carcinoid Tumour of the Intestines:**

(Serotonin Neuroendocrine Tumour)

- **TYPICAL Symptoms/Presentation:**
  - Hot Flushes
  - Watery Diarrhoea
  - Abdominal Pain
  - Palpitations
  - (3 Common Sites = Appendix, Terminal Ileum, Rectum; Also the R-Sided Heart Valves)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Hypotension (Systemic Vasodilation; & Hypovolaemia), Tachycardia (Hypovolemia)
  - **Other:**
    - Cardiac Abnormalities – Pulmonary Stenosis or Tricuspid Regurgitation → Tender Pulsatile Hepatomegaly.
    - "Pellagra" (Sign of Niacin/B3 Deficiency) → Delusions, Confusion, Scaly Skin Sores.

### **Bowel Cancer (Sporadic & Inherited):**

#### **TYPICAL Symptoms/Presentation:**

- **Bowel Ca. Common Symptoms:**
  - (Asymptomatic in Early Stages)
  - Change in Bowel Habit & Stool Shape
  - Blood Mixed Within Stool (+/- Anaemia)
  - Abdominal Cramping & Bloating
  - Fevers/Night Sweats
  - Weight Loss
  - Fatigue
  - (Late → Bowel Obstruction +/- Metastasis)
- **HNPCC; Amsterdam Criteria:**
  - 1: Must have 3 Affected Relatives
  - 2: >One Relatives must be a 1<sup>st</sup>-Degree
  - 3: FamHx must span >2 Generations
  - 4: 1<sup>+</sup> cases diagnosed @ <50yrs.
- **FAP:**
  - APC Gene Mutation

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Tachycardia (if Anaemia/Hypovolaemic/Perforation/Septic)
  - Tachypnoea (if Anaemia/Perforation/Septic)
  - Hypotension (If Perforation/Septic/Shock)
  - Febrile (If Septic)
- **Other:**
  - Acanthosis Nigricans in Axillae (Sign of GI Malignancy)
  - **If Anaemia** (Koilonychia, Pale Nails, PC/SC Pallor, Atrophic Glossitis)
  - **If Shock** (↑CRT, Cool Peripheries, Peripheral Cyanosis, Low-Vol Pulse, Dry Mucosae)
  - **If Perf/Peritonism** (Cullen's/Grey-Turner's Signs, Tenderness/Guarding/Rebound, Rigidity, Shoulder-Tip Pain)
  - **If Metastasis to Liver** (Obstructive Jaundice, Clubbing, Leukonychia, Muercke's Lines, Mee's Lines, Xanthomata, Palmar Erythema, Scleral Icterus, Hepatic Flap, Hepatic Encephalopathy, Xanthelasma, Hepatic Fetor, Hepatomegaly, Gynaecomastia, Spider Naevi, Abdominal Distension, Caput Medusa, Ascites, Pedal oedema, Bruising, Scratch marks)
- **Signs of Causes:**
  - Note: Both HNPCC & FAP → ↑Risk of other Cancers (Endometrial/Gastric/Ovarian)

### **Fulminant Hepatic Failure (Acute Liver Failure):**

(Alcohol/Drugs/Chronic Hepatitis/Biliary Obstruction/Etc)

#### **TYPICAL Symptoms/Presentation:**

- Jaundice, Pruritis
- Bleeding, Bruising
- RUQ Pain
- Fetor Hepaticus
- Cerebral Oedema, Vomiting
- Hepatic Encephalopathy (within 2 wks)
- Death without transplant.

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - May be Febrile
- **Other:**
  - Jaundice, Scratch marks,
  - Bruising/Petechiae/Purpura,
  - Hepatic Flap, Hepatic Encephalopathy, Hepatic Fetor
  - RUQ Tenderness
  - Oedema

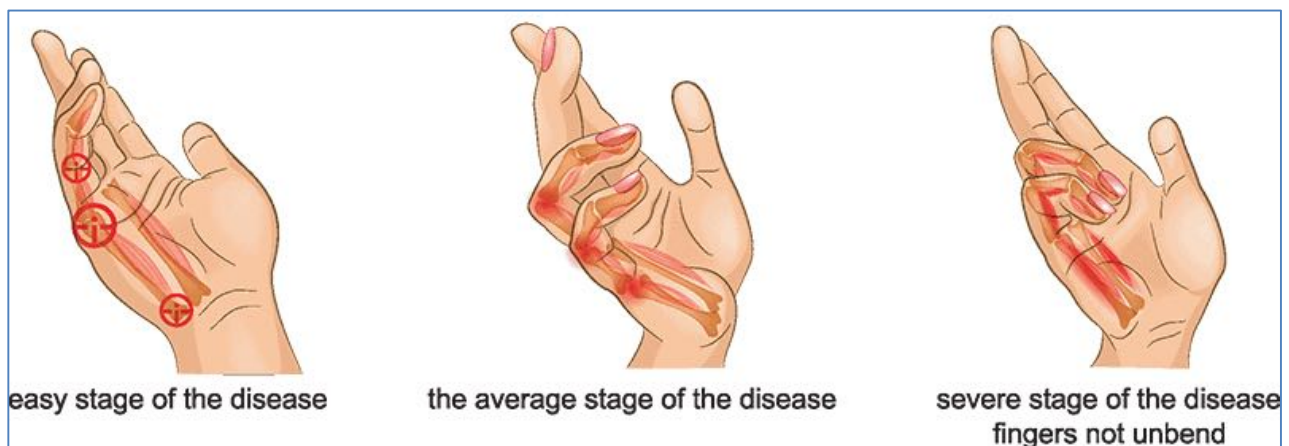
## **Alcoholic Liver Disease (Alcoholic Hepatitis):**

### **TYPICAL Symptoms/Presentation:**

- Jaundice
- RUQ Tenderness
- Palpitations
- Gynaecomastia, Spider Naevi, Testicular Atrophy
- Abdo Distension
- (Ataxia)

### **TYPICAL Clinical Signs:**

- **Vitals:**
  - AF (If Dilated Cardiomyopathy)
- **Other:**
  - Fatty Liver Changes → Hepatomegaly, RUQ Tenderness
  - If Portal Hypertension → Hepatomegaly, Splenomegaly, Ascites, Pedal Oedema
  - CLD (Clubbing, Leukonychia, Muercke's Lines, Mee's Lines, Xanthomata, Palmar Erythema, Bruising, Scratch marks, Scleral Icterus, Hepatic Flap, Hepatic Encephalopathy, Xanthelasma, Hepatic Fetor, Hepatomegaly, Gynaecomastia, Spider Naevi, Abdominal Distension, Caput Medusa, Testicular Atrophy, Ascites, Pedal oedema)
- **Signs of Causes:**
  - Dupuytren's Contracture
  - \*\* ↓Thiamine → Wernicke/Korsakoff Syndrome, Cerebellar Degeneration, Peripheral Neuropathy
  - Dilated Cardiomyopathy
  - Pancreatitis
  - Anaemia (Macrocytic – B12 Deficiency) (Microcytic – if GI Blood Loss)



<https://www.drbesb.com/dupuytren-contraction/>



## **Hepatic Cirrhosis (Chronic Liver Failure):**

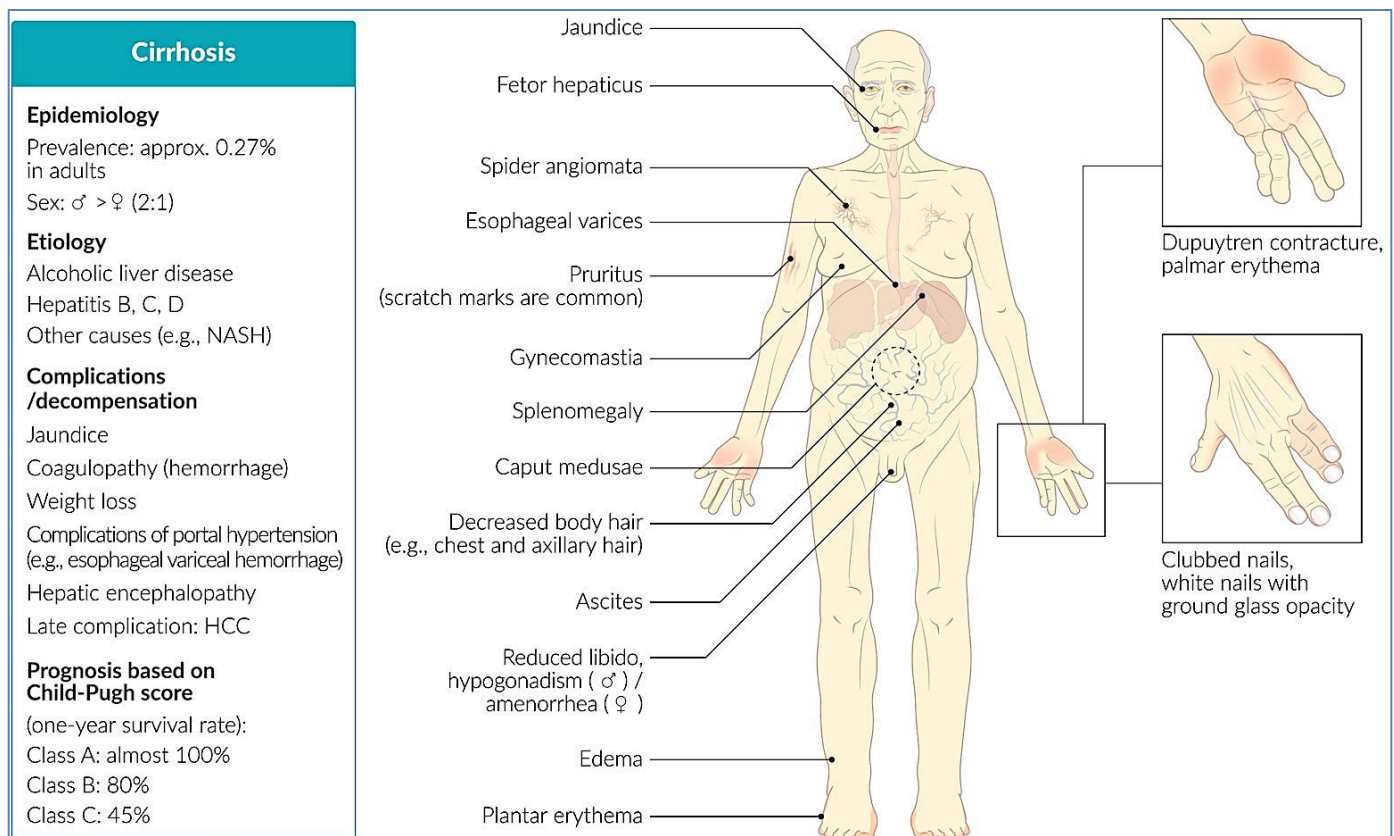
(Alcohol/Drugs/Chronic Hepatitis/Biliary Obstruction/Haemochromatosis/ Wilson's Disease/Etc)

### **- TYPICAL Symptoms/Presentation:**

- May be Asymptomatic
- RUQ Pain
- Pruritis (Jaundice), Bruising (Liver Failure)
- Abdominal Distension, Ankle Swelling, Caput Medusa (Portal Hypertension)
- Gynaecomastia, ↓ Libido, Amenorrhoea, Palmar Erythema, Spider Naevi (Oestrogen Excess)
- Confusion/Forgetfulness/Drowsiness/Flap/ Coma/Seizures (Hepatic Encephalopathy)

### **- TYPICAL Clinical Signs:**

- **Vitals:**
  - Normal
- **Other:**
  - Jaundice, Ascites, Pedal Oedema
  - Clubbing, Leukonychia, Muercke's Lines, Mee's Lines, Xanthomata, Palmar Erythema, Bruising, Scratch marks, Scleral Icterus, Hepatic Flap, Hepatic Encephalopathy, Xanthelasma, Hepatic Fetor, Shrunken Nodular Liver (Micro or Macro-Nodular depending on Aetiology), Splenomegaly, Gynaecomastia, Spider Naevi, Abdominal Distension, Caput Medusa, Testicular Atrophy, Ascites, Pedal oedema)
- **Signs of Causes:**
  - IVDU, Alcoholism, Tattoos, Skin Pigmentation (Haemochromatosis), Cachexia (Ca), Xanthelasma/mata (Chronic Biliary Obstruction)



<https://www.amboss.com/us/knowledge/Cirrhosis>

### Hepatitis-Viruses → Acute Hepatitis:

(Hep A & E)(Faecal-Oral)(Acute ONLY)

- TYPICAL Symptoms/Presentation:

- Epidemics Common
- Acute Hepatitis ONLY:
  - Viraemia → Flu-like Symptoms (Fever, Malaise, Anorexia, Nausea, Arthralgia)
  - Jaundice after 10days
- (\*\*Note: *20% Mortality of Hep-E in Pregnancy*)

- TYPICAL Clinical Signs:

- Vitals:
  - Fever, Tachycardia
- Other:
  - Jaundice (Intrahepatic Cholestasis ∴ Pale Stools & Dark Urine)
  - +/- Hepatomegaly
  - +/- Splenomegaly
  - +/- Tender Lymphadenopathy

### Hepatitis-Viruses → Chronic Hepatitis:

(Hep B, B+D, & C)(Blood-Transmission)(Acute→Chronic)

- TYPICAL Symptoms/Presentation:

- Acute → Non-Specific Viral Symptoms:
  - Viraemia → Flu-like Symptoms (Fever, Malaise, Anorexia, Nausea, Arthralgia)
  - (90% of Hep B → Full Recovery)
  - (10% of Hep C → Full Recovery)
- Chronic → Chronic Hepatitis Symptoms:
  - May have Non-Specific Viral Syx if Reactivation (Eg: "Chronic Active Hep B")
  - OR...May be Completely Asymptomatic until Cirrhosis → Liver Failure
- Cirrhosis → Liver Failure:
  - Abdominal Distension +Ankle Swelling
  - Pruritis (Jaundice), Bruising
  - Gynaecomastia, ↓Libido, Amenorrhoea (Oestrogen Excess)
  - Confusion/Drowsiness/Coma/Flap (Hepatic Encephalopathy)

- TYPICAL Clinical Signs:

**Acute:**

- Vitals:
  - Fever, Tachycardia
- Other:
  - Jaundice (Intrahepatic Cholestasis ∴ Pale Stools & Dark Urine)
  - +/- Hepatomegaly, +/- Splenomegaly, +/- Tender Lymphadenopathy

**Chronic:**

- Vitals:
  - (If Active → Fever, Tachycardia)
  - If Subclinical → Normal Vitals.
- Other:
  - Jaundice (Both Types),
  - Small, Nodular Liver
  - Signs of Portal HTN – (Telangiectasias, Caput Medusa, Ascites, Pedal Oedema, Hepatomegaly, Splenomegaly, Gynaecomastia)
  - Hep. Encephalopathy

**(10% of Hep B & 90% of Hep C) → Cirrhosis → Liver Failure:**

- Signs:
  - Clubbing, Leukonychia, Muercke's Lines, Mee's Lines, Xanthomata, Palmar Erythema, Bruising, Scratch marks, Scleral Icterus, Hepatic Flap, Hepatic Encephalopathy, Xanthelasma, Hepatic Fetor, Gynaecomastia, Spider Naevi, Abdominal Distension, Caput Medusa, Testicular Atrophy, Ascites, Pedal oedema.

### **Primary Biliary Cirrhosis (AKA: "Chronic, Non-Suppurative Destructive Cholangitis"):**

(Genetic Autoimmune)

- **TYPICAL Symptoms/Presentation:**
  - Mid-Aged Females
  - Insidious Onset
  - Pruritis, *then* (Cholestatic) Jaundice
  - Fatigue
  - Hepatomegaly
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Jaundice, Ascites, Pedal Oedema
    - Clubbing, Leukonychia, Muercke's Lines, Mee's Lines, Xanthomata, Palmar Erythema, Bruising, Scratch marks, Scleral Icterus, Hepatic Flap, Hepatic Encephalopathy, Xanthelasma, Hepatic Fotor, Shrunken Nodular Liver (Micro or Macro-Nodular depending on Aetiology), Splenomegaly, Gynaecomastia, Spider Naevi, Abdominal Distension, Caput Medusa, Testicular Atrophy, Ascites, Pedal oedema)
  - **Signs of Causes:**
    - Commonly Mid-Age Female.

### **Gilbert's Syndrome:**(Genetic; Benign)

- **TYPICAL Symptoms/Presentation:**
  - Asymptomatic
  - Occasional Mild Jaundice (*Associated with Fasting/Infection /Stress/Exertion*).
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Normal
  - **Other:**
    - Occasional Jaundice Precipitated by Stress/Infection/Exertion/etc.
  - **Signs of Causes:**
    - Jaundice + Young + Family Hx of Jaundice

### **Haemochromatosis – (Primary – Genetic; or Secondary)**

- **TYPICAL Symptoms/Presentation:**
  - **Initially Asymptomatic**
  - **Early –** (Fatigue, Arthralgia, Loss of Libido)
  - **Later –** (Skin Bronzing, **Abdo Pain, Hepatomegaly, Liver Cirrhosis**)
- **TYPICAL Clinical Signs:**
  - Skin "Bronzing"
  - **Cirrhosis –** (Jaundice, Bruising, Ascites, Oedema, etc.)
  - **Other Complications:**
    - **Heart** - Cardiomyopathy
    - **Endocrine Glands** – Failure of gland:
    - **Joints** - Arthritis (Iron Deposition in the Joints)
  - **Signs of Causes:**
  - (Acquired – Transfusions/Supplements/Haemolysis)

**Liver Abscesses:** (Infection – Typically E-Coli – 2° to Sepsis)

- **TYPICAL Symptoms/Presentation:**
  - Fever, Rigors, Malaise,
  - Anorexia, Weight Loss
  - Vomiting, Abdo Pain
  - (Similar to TB)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea, Hypotension (if Septic Shock)
  - **Other:**
    - Sepsis, Jaundiced,
    - Tender, Enlarged Liver
  - **Signs of Causes:**
    - Intra-Abdominal Sepsis

**Hepatocellular Carcinoma:** (Primary Liver Tumour)

- **TYPICAL Symptoms/Presentation:**
  - (Note: EXTREME RISK in Chronic Carriers of HBV, HCV)
  - (Note: EXTREME RISK in Cirrhotics)
    - RUQ Pain
    - Fever, Anorexia, Weight Loss
    - Ascites
  - Note: Suspect HCC if you see these signs in a Cirrhotic.
  - (Note: ↑ Serum a-Fetoprotein)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever
  - **Other:**
    - RUQ Tenderness/Mass, Hepatomegaly,
    - Anorexia/Weight Loss
    - Ascites/Peripheral Oedema
    - (+/- Obstructive Jaundice)
  - **Signs of Causes:**
    - Cirrhosis (Shrunken, Nodular Liver + *Signs of Chronic Liver Failure*)
    - HBV/HCV Infected (IVDU/Homosexual/Tattoos/Prostitutes/etc)

**Liver Metastases:** (Secondary Liver Ca's)

- **TYPICAL Symptoms/Presentation:**
  - RUQ Pain
  - Jaundice (if Obstructive)
  - Fever, Anorexia, Weight Loss
  - + Previous Hx of Cancer
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever
  - **Other:**
    - RUQ Pain
    - Fevers/Sweats
    - Confusion
    - Jaundice
    - (\*\*Will eventually progress to show signs of *Chronic Liver Failure*)
  - **Signs of Causes:**
    - Typically from Colorectal Ca, Breast Ca, Lung Ca, Melanoma.

### **Biliary Colic:** (AKA: "**Cholelithiasis**")

- **TYPICAL Symptoms/Presentation:**
  - **Note:** Cholelithiasis is typically Asymptomatic.
  - **HOWEVER** if Gallstones get Stuck → Biliary Colic:
    - **(= Biliary Pain for <3hrs)**
    - Severe, Colicky RUQ Pain
    - Radiation to the R-Shoulder
    - Fat Intolerance → Clay Stools
  - **(Note: Can → Perforation or Obstructive Pancreatitis)**
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Pain → Tachycardia, Tachypnoea, Hypertension
  - **Other:**
    - Positive Murphey's Sign (RUQ pain in deep inspiration)
    - Obstructive Jaundice (Pale Stools, Dark Urine)
  - **Signs of Causes:**
    - Xanthelasma/Xanthomata (Hypercholesterolaemia), Pregnancy, Obesity, Rapid Weight LOSS, Cystic Fibrosis

### **Acute Cholecystitis:**

- **TYPICAL Symptoms/Presentation:**
  - **(= Biliary Pain for >3hrs)**
  - Severe, Colicky RUQ Pain
  - Radiation to the R-Shoulder
  - **\*Pain Associated/Exacerbated with FOOD**
  - **\*Nausea, Vomiting**
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea, Hypertension
  - **Other:**
    - Positive Murphey's Sign (RUQ pain in deep inspiration)
    - Obstructive Jaundice (Pale Stools, Dark Urine)
    - Peritoneal Involvement – Rigidity/Guarding/Rebound
    - RUQ Mass (Swollen Gallbladder)
  - **Signs of Causes:**
    - Xanthelasma/Xanthomata (Hypercholesterolaemia), Pregnancy, Obesity, Rapid Weight LOSS, Cystic Fibrosis

### **Acute Pancreatitis:**

(50% - Gallstones; 40% - Alcohol; 10% other)

- **TYPICAL Symptoms/Presentation:**
  - Epigastric/Abdo Pain – Following Large Meal OR Alcohol
  - Vomiting
  - Shock
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (Shock), Hypotension (Shock), Tachypnoea (Shock)
  - **Other:**
    - Epigastric Tenderness,
    - **\*\*Peritonitis** → Guarding, Rigidity, Rebound + Referred Shoulder Tip Pain
    - If Haemorrhage → (Cool Dry Peripheries, ↑CRT, Peripheral Cyanosis, Low-Vol Tachy, Dry Mucosae, Cullen's & Grey-Turner's Signs)
  - **Signs of Causes:**
    - Alcoholism (Ataxia, Anaemia, Dupuytren's, Wernicke-Korsakoff Syndrome)
    - Cholelithiasis (Xanthelasma, Xanthomata)

### **Chronic Pancreatitis: \*\*Alcohol Abuse**

- **TYPICAL Symptoms/Presentation:**
  - Intermittent Epigastric Pain
  - Weight Loss (Malabsorption)
  - Steatorrhea
  - →Secondary Diabetes
  - (Can → → Pancreatic Cancer)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - NA
  - **Other:**
    - Jaundice
    - Weight Loss
    - Epigastric Tenderness
  - **Signs of Causes:**
    - Alcoholism (Ataxia, Anaemia, Dupuytren's, Wernicke-Korsakoff Syndrome)

### **Pancreatic Adenocarcinoma:**(Ductal Tumour):

- **TYPICAL Symptoms/Presentation:**
  - **Note: Asymptomatic until Advanced Disease**
    - Pain – Mid Epigastrium → Back
    - Migratory Venous Thrombosis (Trousseau Sign)
    - **\*\*Anorexia & Weight Loss**
    - **\*\*Extreme Fatigue**
    - **\*\*Depression**
    - + Steatorrhea, Malabsorption
    - +/- Jaundice
  - (Note: 25% 1yr Survival; 5% 5yr Survival)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever
  - **Other:**
    - Obstructive Jaundice, Epigastric Pain (+/- Radiation to Back), Courvoisier's Palpable Gallbladder,
    - Low BMI, Migratory Venous Thrombosis (Trousseau Sign of Malignancy – Due to Hypercoagulability),
    - 2° Diabetes
  - **Signs of Causes:**
    - Smoking, Alcohol, Chronic Pancreatitis (Eg: CF/Alcohol), Diabetes
    - Familial Syndromes – Eg: BrCA, Peutz-Jeger's Syndrome, Hereditary Pancreatitis



## Summary of Important Points for Gastrointestinal OSCEs:

### Causes of Dysphagia:

- **Types of Dysphagia:**
  - o **With Solids Only** = **Mechanical Obstruction** – (Hiatus Hernia/Strictures/Plummer Vinson Web from Iron Deficiency/Tumours)
  - o **With Solids & Liquids** = **↓ Motility** – (Achalasia/Neural[Vagus Nerve]/**Scleroderma**)
  - o **With Liquids Only** = **Pharyngeal Disorders** – (Globus Pharyngeus)
- **Causes of Dysphagia:**
  - o **Oesophagitis:** (Infection in Immunocompromised Or Eosinophilic/Allergic)
  - o **Achalasia:** (Oesophageal *Aperistalsis*)
  - o **Mallory Weiss Syndrome Tear:** = Oesophageal laceration – (Gluttony, Coughing, Bulimia)
  - o **Oesophageal Varices:** (Portal HTN, Cirrhosis)
  - o **Hiatus Hernia:**
  - o **GORD, Barrett's Oesophagus:** (Obesity, Pregnancy, Alcoholism)
  - o **Oesophageal Cancer:** (Smoking, Alcoholism, Chronic GORD)

### Causes of Epigastric Pain:

- **Gastritis:** (Alcoholism, Infection, Pernicious B12 Anaemia)
- **Peptic Ulcer Disease:** (NSAIDs, H.Pylori, or Gastrinoma/ZE-Syndrome)
- **Gastric & Duodenal Cancers:** (H.Pylori or Familial)
- **Acute Pancreatitis:** (50% - Gallstones, 40% - Alcohol)
- **Chronic Pancreatitis:** (\*\*Alcohol Abuse)

### Causes of Acute Abdomen:

- **Appendicitis:**
- **Diverticulosis/Diverticulitis:**
- **Acute Cholecystitis:**(AKA: "**Cholelithiasis**")
- **Acute Pancreatitis:** (50% - Gallstones, 40% - Alcohol)
- **Chronic Pancreatitis:** (\*\*Alcohol Abuse)
- **Pseudomembranous Colitis:** (Clostridium Difficile Overgrowth due to Antibiotic → ↓Gut Flora (C-Difficile is Directly Cytotoxic))
- **Bowel Obstruction**

## Causes of Jaundice:

| <u>Background Info on Jaundice:</u>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Disease:</u>                                      | <u>TYPICAL Symptoms/Presentation:</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                         |
| <b>Unconjugated (Prehepatic/Haemolytic) Jaundice</b> | Jaundice + Neonate<br>Jaundice + Dyspnoea + Fatigue<br>Jaundice + Young + Family Hx of Jaundice<br>Jaundice + Young + Malaise<br>Jaundice Epidemic<br>Jaundice + Recent Shellfish Consumption<br>Jaundice + Hx of IVDU/Injections/Tattoos<br>Jaundice + Sodomy/Prostitution<br>Jaundice + Elderly + Weight Loss                                                                                                                                                                                    | → Physiological Neonatal Jaundice<br>→ Haemolytic Anaemia<br>→ Gilbert's Disease.<br>→ Hepatitis<br>→ Hep A Virus<br>→ Hep A Virus<br>→ Hep B/C Viruses<br>→ Hep B Virus<br>→ Carcinoma |
| <b>Conjugated (Posthepatic) Jaundice</b>             | Jaundice + Elderly + Weight Loss<br>Jaundice + Abdo Pain<br>Jaundice + 50yr old Woman<br>Jaundice + Dyspepsia + Steatorrhea<br>Jaundice + Fevers/Rigors                                                                                                                                                                                                                                                                                                                                            | → Carcinoma<br>→ Biliary Obstruction (Gallstones)<br>→ Primary Biliary Cirrhosis<br>→ Head of Pancreas Tumour<br>→ Cholangitis or Liver Abscess.                                        |
| <b>General Signs of Liver Disease:</b>               | Jaundice, Bruising, Scratch marks, Ascites, Oedema, Hepatic Encephalopathy, Hepatic Flap<br>Leukonychia, Muercke's Lines, Xanthomata, Palmar Erythema, Xanthelasma, Feter Gynaecomastia, Spider Naevi, Testicular Atrophy, Nodular Liver, Hepato/Splenomegaly, Abdominal Distension, RUQ Tenderness, Caput Medusa, Ascites, Pedal oedema<br><b>Signs of Causes:</b> IVDU, Alcoholism, Tattoos, Skin Pigmentation (Haemochromatosis), Cachexia (Ca), Xanthelasma/mata (Chronic Biliary Obstruction) |                                                                                                                                                                                         |

## Inflammatory (Autoimmune) Bowel Diseases:

| <u>Inflammatory (Autoimmune) Bowel Diseases:</u>                        |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Disease:</u>                                                         | <u>TYPICAL Symptoms/Presentation:</u>                                                                                                                                                                                                                                                                                                                 | <u>TYPICAL Clinical Signs:</u>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Crohn's Disease:</b><br>(Mouth → Anus)<br>(Patchy)                   | <ul style="list-style-type: none"><li>- Typically Starts @ Ileocecal Valve (RIF)</li></ul> <b>Common Symptoms (Both CD &amp; UC):</b> <ul style="list-style-type: none"><li>- **Abdominal Pain/Severe Internal Cramps</li><li>- **Vomiting/Diarrhoea<br/>*(Porridge-like, Fatty)</li><li>- **Rectal Bleeding</li></ul><br>(+ Fever, Weight Loss)      | <b>Vitals:</b> <ul style="list-style-type: none"><li>- Fever,</li></ul> <b>Other:</b> <ul style="list-style-type: none"><li>- Clubbing</li><li>- Mouth Ulcers &amp; Anus Involvement,</li><li>- GI Bleeding → Anaemia (Pale Nails, Koilonychia, Palmar/SC Pallor, Glossitis)</li></ul> <b>Signs of Causes:</b> <ul style="list-style-type: none"><li>- Autoimmune (Arthritis, Iritis, Pyoderma Gangrenosum, Primary Biliary Cirrhosis)</li></ul> |
| <b>Ulcerative Colitis:</b><br>(Typically affects Colon)<br>(Continuous) | <ul style="list-style-type: none"><li>- Typically Starts @ Rectum (LIF)</li></ul> <b>Common Symptoms (Both CD &amp; UC):</b> <ul style="list-style-type: none"><li>- **Abdominal Pain/Severe Internal Cramps</li><li>- **Vomiting/Diarrhoea *(Bloody &amp; Mucus – but NO Pus. {Not Dysentery})</li><li>- **Rectal Bleeding</li></ul><br>(+ Tenesmus) | <b>Vitals:</b> <ul style="list-style-type: none"><li>- Afebrile,</li></ul> <b>Other:</b> <ul style="list-style-type: none"><li>- Clubbing</li><li>- No Mouth or Anus Involvement,</li><li>- GI Bleeding → Anaemia (Pale Nails, Koilonychia, Palmar/SC Pallor, Glossitis)</li></ul> <b>Signs of Causes:</b> <ul style="list-style-type: none"><li>- Autoimmune (Arthritis, Iritis, Pyoderma Gangrenosum, Primary Biliary Cirrhosis)</li></ul>     |

### **Clinical Evaluation of a Pt with an “Acute Abdomen”:**

- **Conditions Requiring Laparotomy:**
  - **Organ Rupture (Spleen/Aorta/Ectopic)**
    - → Shock
  - **Peritonitis (Perforated PUD/DUD/Diverticulum/Appendix/Bowel/Gallbladder)**
    - → Prostration, Shock, Lying Still, Tenderness (Guarding/Rebound/Percussion), Abdo Rigidity, No Bowel Sounds.
- **Conditions NOT Requiring Laparotomy:**
  - **Local Peritonitis (Diverticulitis, Cholecystitis, Salpingitis, Appendicitis)**
    - → Lying Still, Tenderness (Guarding/Rebound/Percussion), Abdo Rigidity
  - **Colic**
    - → Restlessness, Regularly Waxing/Waning Pain.
- **Tests to Perform:**
  - U&E, FBC, Amylase, LFT, CRP, ABG, Urinalysis
  - Erect CXR (look for Air under Diaphragm)
- **Immediate Priorities:**
  - Resuscitation Before Surgery!! – Note: Anaesthesia compounds shock!!

### **Pathogenesis & Signs of Hepatic Encephalopathy (Hepatic Coma)**

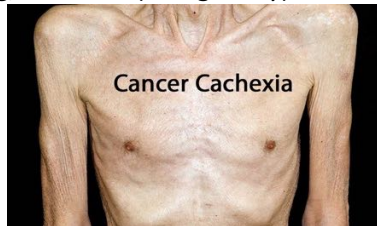
- Hepatic Encephalopathy (↑↑ Ammonia in Bloodstream due to ↓↓ Liver Breakdown)
  - → Forgetfulness/Confusion/Irritability
  - → Tremor/Asterixis (Due to ↓ Proprioception)
  - → Seizures/Coma

## **THE HAEMATOLOGICAL (HEMATOLOGICAL) EXAMINATION**

## THE HAEMATOLOGICAL (HEMATOLOGICAL) EXAMINATION

### FULL HAEMATOLOGY EXAM:

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - o Wasting, Cachexia (Malignancy)



- - o Mediterranean Origin & Frontal Bossing (Thalassaemia)
  - o African Origin (Sickle Cell Anaemias)
  - o Gross Pallor (Anaemia & All Haem Malignancies)
  - o Jaundice & Scratch Marks (Haemolytic Anaemia)
  - o Cyanosis
  - o Facial Plethora (Polycythaemia, SVC obstruction from Lymphoma)
  - o Bruising, Bleeding, Petechiae, Purpura (Thrombocytopenia, Haemophilia & All Haem Malignancies)
  - o Neck Swellings (Lymphoma & All Haem Malignancies)
  - o Abdominal Distension (Organomegaly due to Infection & All Haem Malignancies)
- **Vital Signs:**
  - o **Pulse:**
    - Tachycardia (Infection, Anaemia)
  - o **Blood Pressure:**
    - Postural Hypotension (Anaemia)
  - o **Respiratory Rate:**
    - Tachypnoea (Anaemia)
  - o **Temperature:**
    - Fever (Malignancy, Infection)
- **Hands:**
  - o Warm/Well Perfused + CRT
  - o Palmar Crease Pallor (Anaemia)
  - o Peripheral Cyanosis
  - o Koilonychia (Spoon Nails) and Pale Nails (Iron Deficiency Anaemia)



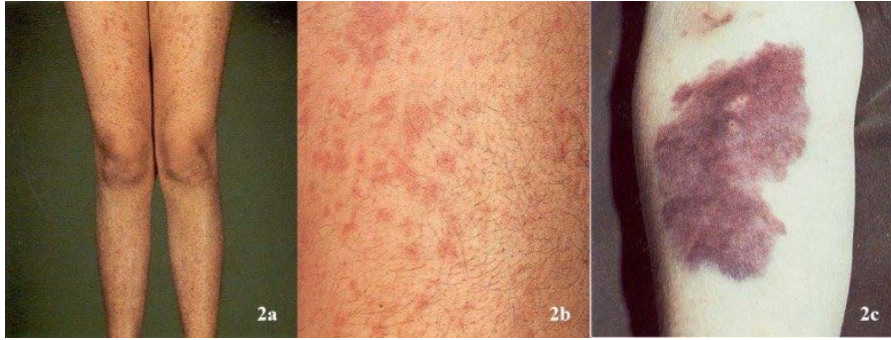
- - o Palmar Erythema (Polycythaemia, Chronic Leukaemias)
  - o Blue Lunulae (Wilson's Disease)



- - o Digital Infarcts & Raynaud's (Rheumatoid, Multiple Myeloma)
  - o Arthropathy (Rheumatoid, Connective Tissue Disease, Haemophilia)
  - o Gouty Tophi (↑ Cell Turnover in Malignancy, Myeloproliferative Diseases & Hb Disorders)
  - o Dupuytren's Contracture (B12 Deficiency, Alcoholism & Megaloblastic Anaemia)

- **Arms:**

- Bruising, Petechiae, Purpura (Haemophilia, Thrombocytopenia & All Haem Malignancies)



- Scratch Marks (Jaundice – Haemolytic Anaemia)
- Epitrochlear & Axillary Lymph Nodes (All Haem Malignancies)
  - Supraclavicular, Infraclavicular, Parasternal, Subpectoral, Subscapular, Central, Lateral

- **Face:**

- Conjunctival Pallor, Angular Stomatitis, Atrophic Glossitis (Anaemia)



- Scleral Icterus
- Central/Peripheral Cyanosis
- Facial Plethora (Polycythaemia)
- Pemberton's Sign (SVC Obstruction due to Lymphoma)



- Buccal Petechiae (Thrombocytopenia, Haemophilia, Bleeding Disorders & All Haem Malignancies)
- Candidal Mouth Infections (All Haem Malignancies)
- Gum Hypertrophy + Bleeding (AML - Acute Myeloid Leukaemias)



- Gum Hypertrophy (Methotrexate Chemotherapy)
- Enlarged Tonsils (MALT Lymphoma)
- Wasting (All Haem Malignancies)



- **Neck:**
  - Cervical Lymph Nodes (Submental/Submandibular/Pre/Post-Auricular/Occipital/Jugular/Post-Triangular) (In All Haem Malignancies)
    - Firm, Non-Tender, Immobile = Malignancy
    - Tender, Mobile = Infection
    - Site
    - Size (>1cm)
    - Consistency (Hard – Carcinoma; Rubbery – Lymphoma)
    - Tenderness (Infection)
    - Mobility (Fixed if Malignant)
  - Neck Swellings/Masses
  - **\*\*Check for Dysphagia/Odynophagia (Plummer Vinson Oesophageal Web from Iron def. Anaemia)**
  - **Thyroid Examination (As Hypothyroidism can → menorrhagia → anaemia)**
  - BCC/SCCs from Immunosuppression (Chemotherapy)
  - Radiotherapy Tattoos
- **Chest:**
  - Bruising, Petechiae, Purpura (Bleeding Disorders, Haemophilia, Thrombocytopenia & All Haem Malignancies)
  - Chest Infections (Immunosuppression)
  - **Systolic Flow Murmur (Severe Anaemia)**
  - **\*\*\*Bony Tenderness (ALL Haematological Malignancies) - Ribs, Clavicles, Spine, Hips.**
- **Abdomen:**
  - Bruising, Petechiae, Purpura (Bleeding Disorders, Haemophilia, Thrombocytopenia & All Haem Malignancies)
  - Hepatomegaly, Splenomegaly (All Haem Malignancies)
  - Para-Aortic Lymph Nodes
  - **Ascites**
  - **Per Rectal Examination – Blood on Stool**
  - **Palpate Kidneys (Signs of Renal Failure → Anaemia of Chronic Disease)**
- **Inguinal:**
  - **Inguinal Lymph Nodes (All Haem Malignancies)**
  - **Testicular Masses in Children (ALL – Acute Lymphoblastic Leukaemia)**
  - **DRE (Lymphoma)**
- **Legs:**
  - Bruising, Petechiae, Purpura (Bleeding Disorders, Haemophilia, Thrombocytopenia & All Haem Malignancies)
  - Scratch Marks (Haemolytic Jaundice)
- **Feet:**
  - Peripheral Perfusion & CRT
  - Koilonychia & Pale Nails (Iron Deficiency Anaemia)
  - Sole Erythema (Polycythaemia)
  - Gouty Tophi (↑ Cell Turnover in Myeloproliferative Diseases & Hb Disorders)



- **Thank patient “that concludes my examination”, I’ll go organise further tests.**

### **Focussed Anaemia Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - o Alert & Oriented
  - o Racial Origin (African = Sickle Cell Anaemia; Mediterranean = Thalassemia)
  - o Gross Pallor (Anaemia), Jaundice (Haemolytic Anaemia), or Cyanosis.
  - o Scratch Marks (Jaundice from Haemolytic Anaemia)
  - o Bruising, Petechiae, Purpura (Thrombocytopenia any Haem Malignancies)
- **Vital Signs:**
  - o **Pulse:**
    - Tachycardia
  - o **Blood Pressure:**
    - Postural Hypotension
  - o **Respiratory Rate:**
    - Tachypnoea
  - o **Temperature:**
    - Normal
- **Hands:**
  - o Warm & Well Perfused + CRT
  - o Koilonychia (Iron Deficiency Anaemia)
  - o Pale Nails (General Anaemia)
  - o Leukonychia (Chronic Kidney Disease → Hypoalbuminaemia)
  - o Palmar Crease Pallor
  - o Dupuytren's Contracture (Megaloblastic Anaemia from Alcoholism)
- **Arms:**
  - o Bruising/Petechiae/Purpura (Thrombocytopenia due to Myelosuppression which also → Anaemia)
  - o Scratch Marks (Haemolytic Jaundice → Anaemia)
- **Face:**
  - o Conjunctival Pallor
  - o Scleral Icterus (Haemolytic)
  - o Central/Peripheral Cyanosis
  - o Angular Stomatitis (General Anaemia)
  - o Atrophic Glossitis (General Anaemia)
  - o Gum Bleeding (AML → As a cause of Iron Deficiency Anaemia)
  - o Buccal Petechiae/Purpura (Thrombocytopenia due to Myelosuppression which also → Anaemia)
- **Neck:**
  - o Cervical Lymphadenopathy (Because All Haem Malignancies cause Anaemia)
  - o **\*\*Check for Dysphagia/Odynophagia (Plummer Vinson Oesophageal Web from Iron def. Anaemia)**
  - o Thyroid Examination (As Hypothyroidism can → menorrhagia)
- **Chest:**
  - o Bony Tenderness (All Haem Malignancies)
  - o Systolic Flow Murmurs (hyperdynamic circulation)
- **Abdomen:**
  - o Cullen's/Grey Turner's Signs of Haemoperitoneum (Haemorrhagic Anaemia)
  - o Tenderness (Peptic Ulcer Disease, Cohn's, Ulcerative Colitis)
  - o Hepatomegaly & Splenomegaly (All Haem Malignancies cause Anaemia)
  - o **Per Rectal Examination – Blood on Stool**
  - o Palpate Kidneys (Signs of Renal Failure → Anaemia of Chronic Disease)
- **Legs:**
  - o Peripheral Perfusion
- **Feet:**
  - o Peripheral Perfusion + CRT
  - o Koilonychia (Iron deficiency Anaemia)
  - o Pale Nails (General Anaemia)
- **Thank patient "that concludes my examination", I'll go organise further tests.**

### **Focussed Leukaemia Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - Alert & Orientated?
  - Wasting, Weight Loss
  - Gross Pallor (Marrow Overcrowding - All Haem Malignancies)
  - Lethargy
  - Obvious masses (Neck/Abdomen)
  - Bruising, Bleeding, Petechiae, Purpura (Thrombocytopenia - All Haem Malignancies)
- **Vital Signs:**
  - **Pulse:**
    - Normal
    - Tachycardia (If Anaemia)
  - **Blood Pressure:**
    - Normal
    - Postural Hypotension (if Anaemia)
  - **Respiratory Rate:**
    - Normal
    - Tachypnoea (If Anaemia)
  - **Temperature:**
    - Febrile (All Haem Malignancies, Infection)
- **Hands:**
  - Peripheral Perfusion + CRT
  - Pale Nails & Palmar Crease Pallor (Marrow Overcrowding Anaemia)
  - **Palmar Erythema (Chronic Leukaemias)**
  - **Gouty Tophi (High Turnover in Myeloproliferative)**
- **Arms:**
  - Bruising, Petechiae, Purpura (Marrow Overcrowding Thrombocytopenia)
  - Epitrochlear & Axillary Lymph Nodes (All Haem Malignancies)
    - Supraclavicular, Infraclavicular, Parasternal, Subpectoral, Subscapular, Central, Lateral
- **Face:**
  - Gross/Conjunctival Pallor (Marrow Overcrowding Anaemia)
  - Central/Peripheral Cyanosis
  - Gum Hypertrophy & Bleeding (Acute Myeloid Leukaemia)
  - Gum Hypertrophy (Methotrexate Side Effect)
  - Candidal Mouth Infections (Neutropenia - All Haem Malignancies)
- **Neck:**
  - Cervical Lymphadenopathy (All Haem Malignancies)
  - Meningism (Due to CNS Infiltration – esp. ALL)
- **Chest:**
  - Bruising, Petechiae, Purpura (Thrombocytopenia)
  - Bony Tenderness (Bony Infiltration - All Haem Malignancies)
  - Chest Infections (Immunocompromise – From Chemo or Neutropenia)
- **Abdomen:**
  - Bruising, Petechiae, Purpura (Thrombocytopenia - All Haem Malignancies)
  - Distension
  - Hepatomegaly, Splenomegaly (All Haem Malignancies)
  - Para-Aortic Lymph Nodes
- **Legs:**
  - Inguinal lymph Nodes
  - Testicular Masses (Acute Lymphoblastic Leukaemia)
  - Bruising, Petechiae, Purpura (Thrombocytopenia - All Haem Malignancies)
- **Feet:**
  - Peripheral Perfusion + CRT
  - Pale Nails (Anaemia)
- **Thank patient “that concludes my examination”, I’ll go organise further tests.**

### **Focussed Lymphoma Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - Alert & Orientated
  - Wasting, Weight Loss
  - Neck Swellings
  - Facial Plethora (SVC Obstruction)
  - Gross Pallor (Marrow Infiltration)
  - Bruising, Petechiae, Purpura (Marrow Infiltration)
- **Vital Signs:**
  - **Pulse:**
    - Normal
    - Tachycardia (if Anaemia due to BM Infiltration)
  - **Blood Pressure:**
    - Normal
    - Postural Hypotension (if Anaemia due to BM Infiltration)
  - **Respiratory Rate:**
    - Normal
    - Tachypnoea (If Anaemia due to BM Infiltration)
  - **Temperature:**
    - Fever
- **Hands:**
  - Peripheral Perfusion + CRT
  - Peripheral Cyanosis
  - Palmar Crease Pallor (Anaemia of BM Infiltration)
  - Pale Nails (Anaemia of BM Infiltration)
  - Scratch Marks (Pruritis of Lymphoma)
- **Arms:**
  - **Epitrochlear Lymph Nodes** (All Haem Malignancies)
  - **Axillary Lymph Nodes** (Supra/Infraclavicular/Parasternal/Pectoral/Subscapular/Central/Lateral)
    - \*Painful
- **Face:**
  - Conjunctival Pallor (Anaemia of BM Infiltration)
  - Scleral Icterus
  - Wasting
  - Central/Peripheral Cyanosis
  - **Enlarged Tonsils (MALT Lymphoma)**
  - **Candidal Mouth Infections (Immunocompromise - All Haem Malignancies)**
- **Neck:**
  - **Painful Cervical Lymphadenopathy (All Areas)**
- **Chest:**
  - Chest Infections (Immunocompromise - All Haem Malignancies)
  - Systolic Flow Murmurs (if Anaemia of BM Infiltration)
  - Bony Tenderness (Bony Infiltration - All Haem Malignancies)
- **Abdomen:**
  - Hepatomegaly (All Haem Malignancies)
  - Splenomegaly (All Haem Malignancies)
  - Para-Aortic Lymphadenopathy
  - Abdominal Masses (MALT Lymphoma)
  - Bruising, Petechiae, Purpura (Thrombocytopenia of BM Infiltration)
- **Legs:**
  - Bruising, Petechiae, Purpura (Thrombocytopenia of BM Infiltration)
- **Feet:**
  - Peripheral Perfusion + CRT
  - Fungal Nail Infections (Pancytopenia of BM Infiltration)
- **Thank patient “that concludes my examination”, I’ll go organise further tests.**

### **Focussed Multiple Myeloma Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - Wasting, Weight Loss
  - Pallor (Anaemia of BM Infiltration)
  - Bruising, Petechiae, Purpura (Thrombocytopenia of BM Infiltration)
  - Abdominal Masses (Hepatomegaly +/- Splenomegaly)
- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (Anaemia of BM Infiltration)
  - **Blood Pressure:**
    - Postural Hypotension (Anaemia of BM Infiltration)
  - **Respiratory Rate:**
    - Tachypnoea (Anaemia of BM Infiltration)
  - **Temperature:**
    - Fever (Due to Infection/Malignancy)
- **Hands:**
  - Peripheral Perfusion + CRT
  - Raynaud's Phenomenon & Digital Infarcts (Blood Hyperviscosity)
  - Palmar Crease Pallor & Pale Nails (Anaemia of BM Infiltration)
- **Arms:**
  - Bruising, Petechia, Purpura (BM Infiltration → Thrombocytopenia)
  - Epitrochlear Lymphadenopathy (All Haem Malignancies)
  - Axillary Lymphadenopathy (All Haem Malignancies)
- **Face:**
  - Conjunctival Pallor
  - No Central or Peripheral Cyanosis
  - Mucosal Candida Infections (Immunocompromise & Neutropenia)
- **Neck:**
  - Cervical Lymphadenopathy (All Haem Malignancies)
- **Chest:**
  - **BONY TENDERNESS (Especially in Ribs & Lumbar Spine)**
  - Bruising, Petechiae, Purpura (BM infiltration → Thrombocytopenia)
- **Abdomen:**
  - Bruising, Petechiae, Purpura (BM infiltration → Thrombocytopenia)
  - Massive Hepatomegaly +/- Splenomegaly
  - Signs of Renal Failure (Due to Ig-Deposition in Kidneys) → Ascites, Uraemic Fetor, etc
- **Legs:**
  - Peripheral Neuropathy (Due to Spinal Cord Compression)
  - Bruising, Petechiae, Purpura (BM infiltration → Thrombocytopenia)
  - Oedema (If Renal Failure)
- **Feet:**
  - Peripheral Perfusion + CRT
  - Peripheral Neuropathy (Due to Spinal Cord Compression)
  - Fungal Nail Infections (Immunocompromise)
- **Thank patient "that concludes my examination", I'll go organise further tests.**

## **THE MUSCULOSKELETAL EXAMINATION**



## THE MUSCULOSKELETAL EXAMINATION

### Musculoskeletal History:

- **PC – Injury/Pain/Stiffness/Movement Limitation.**
- **HxPC:**
  - **Acute or Chronic problem?**
  - **Mechanism of injury – Utmost Importance!!**
  - **Problems experienced:**
    - **Pain Vs Tenderness.** (+ Be mindful of referred pain)
      - NILDOCARF
    - **Movement limitation**
      - **Pain:**
        - Tendinopathy
        - Impingement
        - Sprain
        - Labral
      - **Mechanical Block:**
        - Stiffness/Creakiness + Time of day & Duration
        - Labral
        - Frozen shoulder
    - **Instability** (clicking/clunking)
    - **Joint swelling**
    - **Night pain?** (Lying on affected joint)
      - Eg: Rotator cuff
      - Eg: AC joint injury
      - Eg: Collateral Knee Ligaments
      - Eg: Bone Cancers
  - **Associated Rheumatologic Symptoms:**
    - Fever, Weakness, Fatigue, Weight Loss, Conjunctivitis/Iritis.
  - **Progression with time (Trending Better/worse/same)**
- **PmHx, Meds, FamHx, SocHx, Systems Review**
- **Red Flags? (Ie: Things that if you miss → mortality/morbidity)**
  - Open fractures
  - Neurovascular compromise
  - Cauda equina syndrome
  - Infections (Joints/Bones)
  - Acute Compartment Syndrome
  - Cancer
  - Temporal arteritis (high risk in pts with PMR – polymyalgia rheumatica)
  - Serious/life-threatening conditions that present with symptoms mimicking muscular pain (Eg: MI)
- **Yellow Flags – Conditions ‘masquerading’ as musculoskeletal conditions:**
  - Eg: Psychological – depression & back pain.

### **GALS Screen (Gait, Arms, Legs, Spine):**

- **Introduction, Wash Hands, Consent,**
- **Adequate Exposure!!! (Remove Shirt, Pants & Shoes)**
- **Gait – (Observe for Fluidity, Symmetry, Limp, Compensation, Foot Drop[L4,L5]):**
  - Walk to the other side of the room, Turn around, and walk back to me.
- **Arms:**
  - **Shoulder & Elbow:**
    - Raise hands up to the ceiling, then down behind your head. Push your shoulders backwards.
    - Touch your fingers on your shoulders, and raise your elbows as high as you can.
    - Lock your elbows into your side, and turn your arms outward as far as you can. (External Rotation)
    - Run your thumb up the middle of your back. (Internal Rotation)
    - Push your hands as far back behind you as you can. (Extension)
    - Touch your left thigh with your right hand (and Vice Versa) (Adduction)
    - Supinate & Pronate your Hands
  - **Hands & Fingers:**
    - Splay your fingers wide + **Resist me squeezing them.**
    - Fingers together + **Resist me pulling them apart.**
    - Grip my fingers as hard as you can + **Assess Grip Strength**
    - Make a fist
    - Touch each of your fingers with your thumb.
    - **+ Metacarpophalangeal Squeeze Test (For Tenderness)**
- **Legs:**
  - **3x Half Squats** (Patellofemoral Joint)
  - **1x Full Squat** (Knee Joint Proper)
  - Ankle Eversion & Inversion
  - Dorsiflexion & Plantarflexion
  - Splay Toes & Scrunch Toes.
- **Spine:**
  - (Feet shoulder-width apart)
  - Observe for Lordosis, Kyphosis, Scoliosis.
  - Bend from Lumbar Spine and reach for your toes
  - Bend from Lumbar Spine and look at the ceiling
  - Bend Sideways & Slide your hand down your thigh
  - Neck Flexion (Chin on chest)
  - Neck Extension (look at ceiling)
  - Lateral Neck Flexion (Ear on Shoulder)
  - Neck Rotation (Look to the side)
- **Reporting on a GALS Screening.**
  - G – Normal/Abnormal + Comments on Findings
  - A – Normal/Abnormal + Comments on Findings
  - L– Normal/Abnormal + Comments on Findings
  - S– Normal/Abnormal + Comments on Findings
  - (Eg: Abnormal Arm (limited abduction of the shoulder to ?degrees))
- **“Thankyou that concludes my examination”**

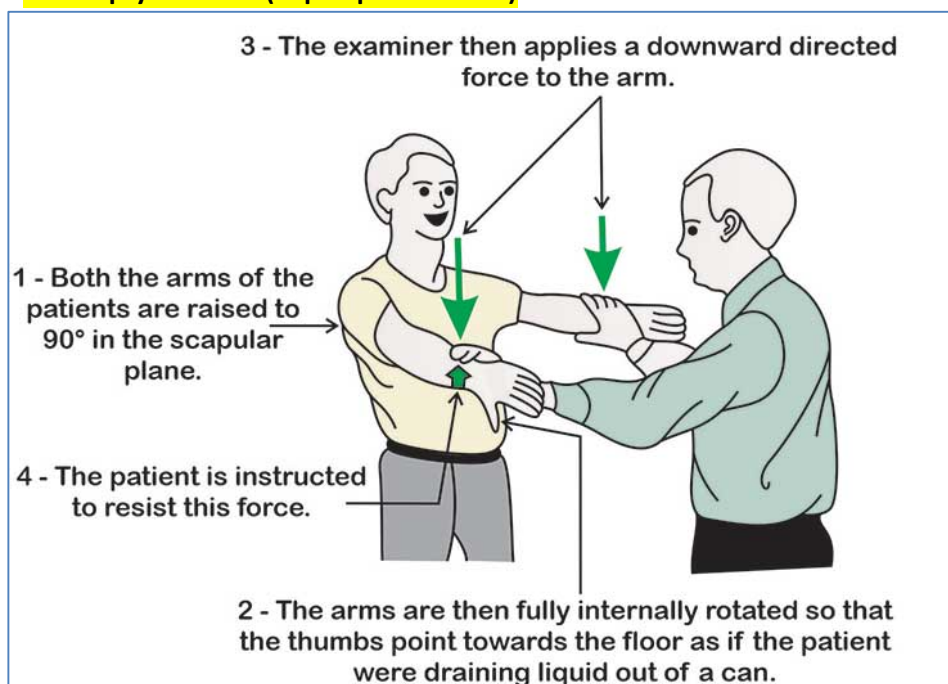
## Focussed Joint Exam Algorithm (Shoulder or Knee):

- **Look (Visual Inspection)**
  - Symmetry? (Size, Shape, Position, Height)
  - Scars
  - Redness
  - Bruising
  - Lumps
  - Atrophy
  - Swelling
- **Feel (Palpate Relevant Anatomy for Tenderness, Heat, Swelling, Crepitus):**
  - **Objective Findings:**
    - Tenderness
  - **Subjective Findings:**
    - Warm (Inflammation, Trauma, Infection, Tumour)
    - Swelling (Effusion, Tumour)
    - Crepitus (Osteoarthritis, Tendinopathy, Fracture)
- **Move (Active +/- Passive if Required):**
  - Symmetry?
  - **Active Movement in All Planes:**
    - Range of Movement
    - Fluidity
    - Pain with Movement
  - If a pt can't do something, ask them why? (Blockage/weakness/Neuro)
  - (Passive – Only for Movements that were Limited)
- **Measure (Range of Movement):**
  - Done at the same time as "Move"
  - Compare with opposite side (ballpark – doesn't have to be objective)

## **+/- Special Tests:**

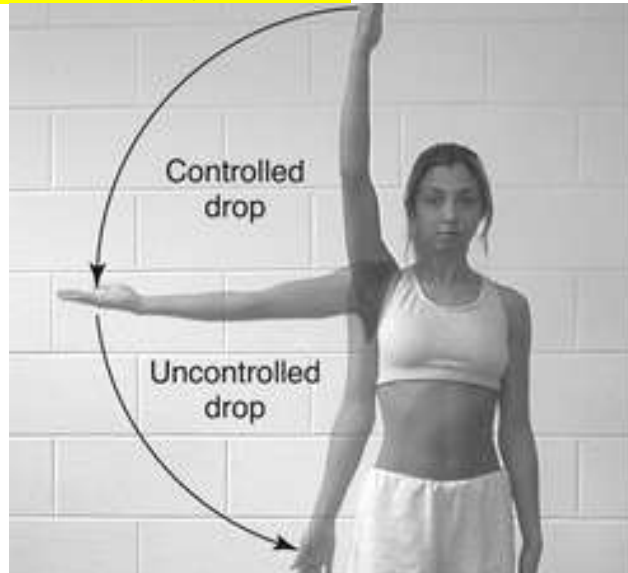
### **Shoulder:**

#### **Empty Can Test (Supraspinatus Tear)**



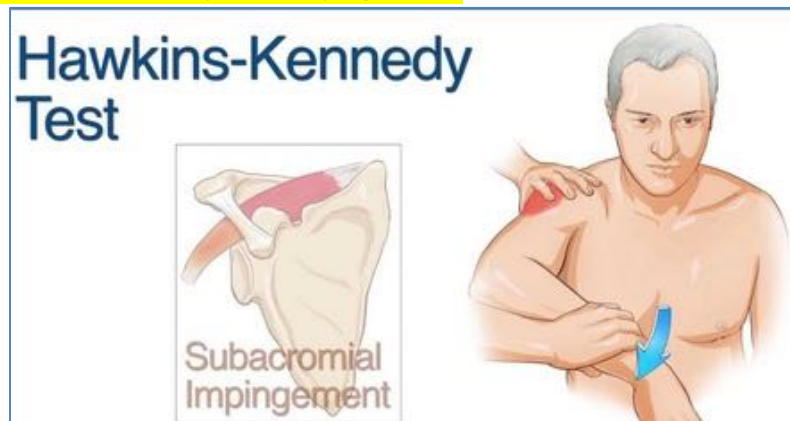
<https://physio-study.com/empty-can-test/>

▪ **Drop Arm Test (Supraspinatus Tear)**



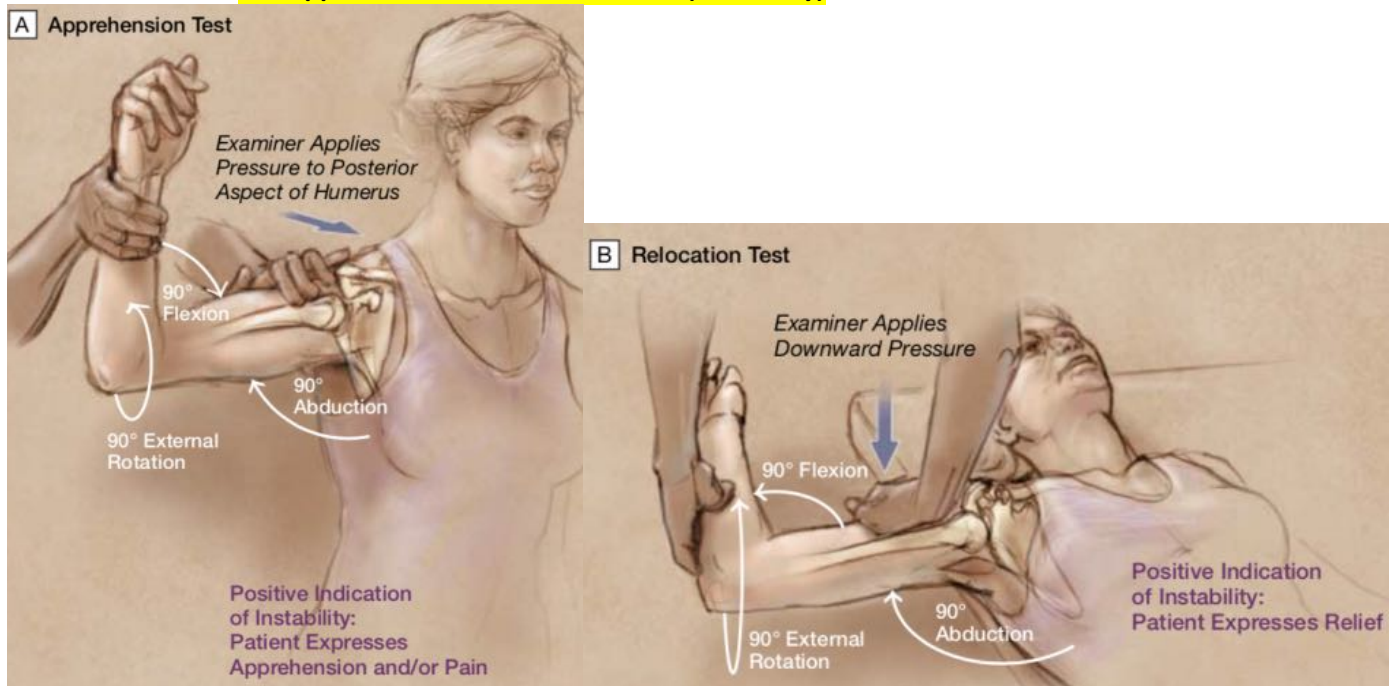
<https://medical-dictionary.thefreedictionary.com/drop+arm+test>

▪ **Hawkin's Kennedy Test (Impingement)**



<https://www.youtube.com/watch?v=hgzQcLuaFLw>

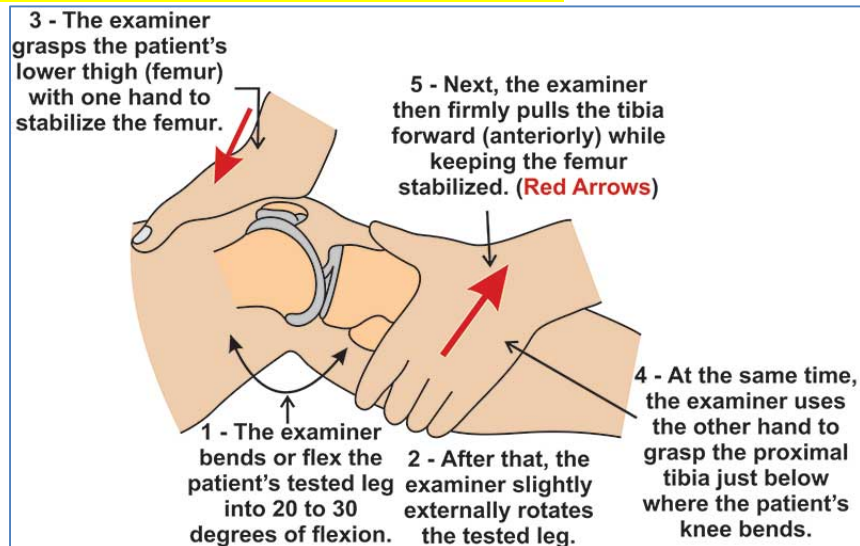
▪ **Apprehension & Relocation Test (Instability)**



<https://stanfordmedicine25.stanford.edu/the25/shoulder.html>

○ **Knee:**

▪ **Lachman's Test (Anterior Cruciate Ligament)**



<https://physio-study.com/lachman-test/>

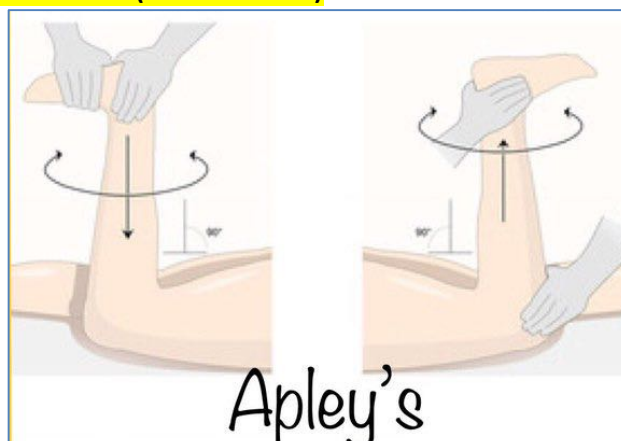
▪ **Thessaly Test (Meniscal Tear)**



Figure 2. The Thessaly test.

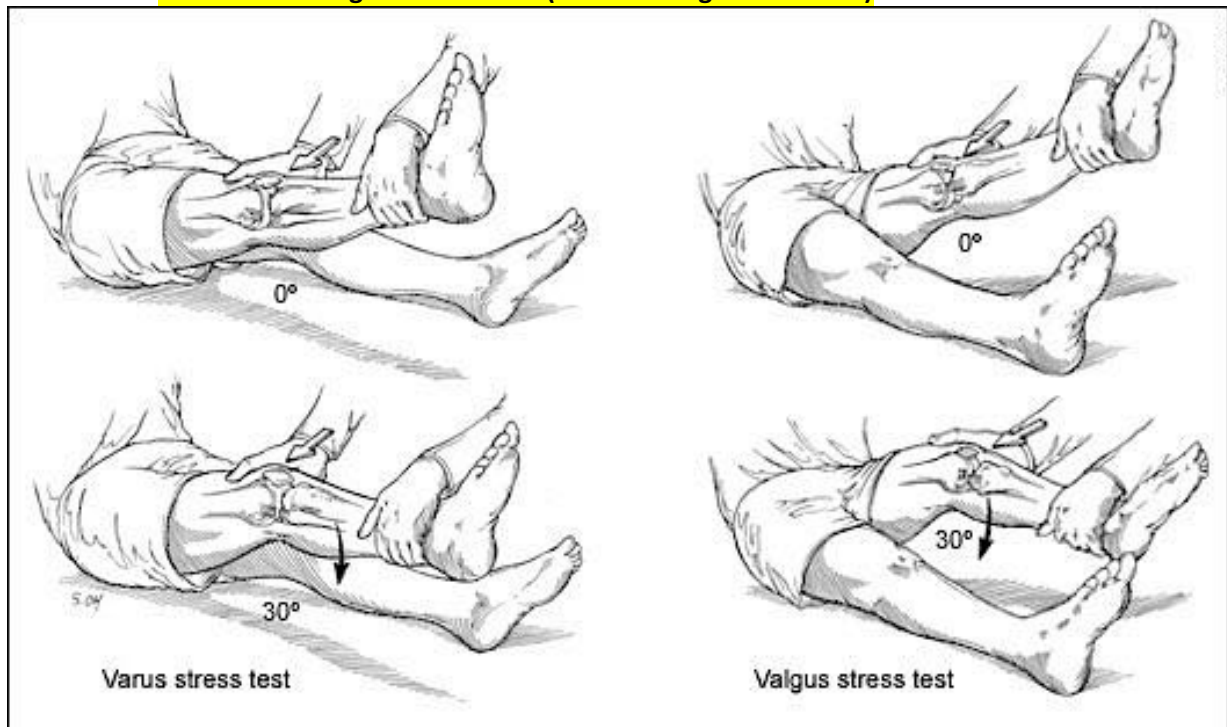
<https://www.aafp.org/afp/2018/1101/p576.html>

▪ **Apley's Grind Test (Meniscal Tear)**



[https://twitter.com/peds\\_sports\\_med/status/1041163674436730881?lang=ca](https://twitter.com/peds_sports_med/status/1041163674436730881?lang=ca)

▪ **Varus & Valgus Stress Tests (Collateral Ligament Tears)**



*Am Fam Physician.* 2003 Sep 1;68(5):907-912. <https://www.aafp.org/afp/2003/0901/p907.html>

▪ **Half Squats (For Patellofemoral Maltracking Syndrome)**



## **Focussed Shoulder Examination:**

- **1: EXPOSE THE PATIENT FULLY!!**
- **Look (Visual Inspection)**
  - Symmetry? (Size, Shape, Position, Height)
  - Scars
  - Redness, Bruising
  - Lumps, Swelling
  - Atrophy
- **Feel (Palpate Relevant Anatomy for Tenderness, Heat, Swelling, Crepitus):**
  - Palpate BOTH joints (Say you'd like to examine the other side)
  - Palpate Definite Anatomical Structures for Tenderness:
    - Sterno-clavicular Joint
    - Clavicle
    - Acromio-clavicular Joint
    - Gleno-humoral Joint Line
    - Spine of Scapula
    - Edges of the Scapula
  - Feel for Heat, Swelling, & Crepitus
  - Eg: "Non tender to firm palpation"
- **Move (Active +/- Passive if Required):**
  - Symmetry?
  - Active Movement in All Planes:
    - Range of Movement
    - Fluidity
    - Pain with Movement
  - If a pt can't do something, ask them why? (Blockage/weakness/Neuro)
  - (Passive – Only for Movements that were Limited)
  - Eg: "Symmetrical and full ROM in all planes"
- **Measure (Range of Movement):**
  - Done at the same time as "Move"
  - Compare with opposite side (ballpark – doesn't have to be objective)
  - "Limited active extension of l-elbow to 150deg"
- **(Clinical Presentations Requiring Special Tests):**
  - Tears – Eg: Supraspinatus Tear:
    - Special Tests:
      - "Drop Arm Test" – Passive abduction of patient's arm → let go and get them to lower arm as slow as possible. Arm Drops at around Parallel = Positive.
      - "Empty Can Test" – Passive abduction of Patient's arm with full internal rotation (Ie: Emptying can). Push down and get them to resist your movement. Pain &/or Weakness = Positive.
  - Impingement – Eg: Of Subacromial Contents:
    - Subacromial contents get squashed between the humeral head and the acromion. (Eg: Supraspinatus, Subacromial Bursa)
    - Special test:
      - "Hawkins test" – Pain = Positive, No Pain = Negative
        - Passive Internal Rotation of Shoulder with elbow in 90deg and humerus at parallel to the ground.
  - Instability – Eg: Anterior Instability of Shoulder (Ie: Post dislocation)
    - Eg: Dislocation/Post Dislocation/Stretch of Anterior Capsule of the Shoulder
    - Special Test:
      - "Apprehension & Relocation Test" – Apprehension (not pain) in 1; Less apprehension in 2 = Positive
        - 1: Patient lies on bed while you Abduct & Externally Rotate their arm to precipitate apprehension (Ie: High five position)
        - 2: Same thing, but with pressure on the anterior capsule. (This should be much more bearable for the patient.)

## **Focussed Knee Examination:**

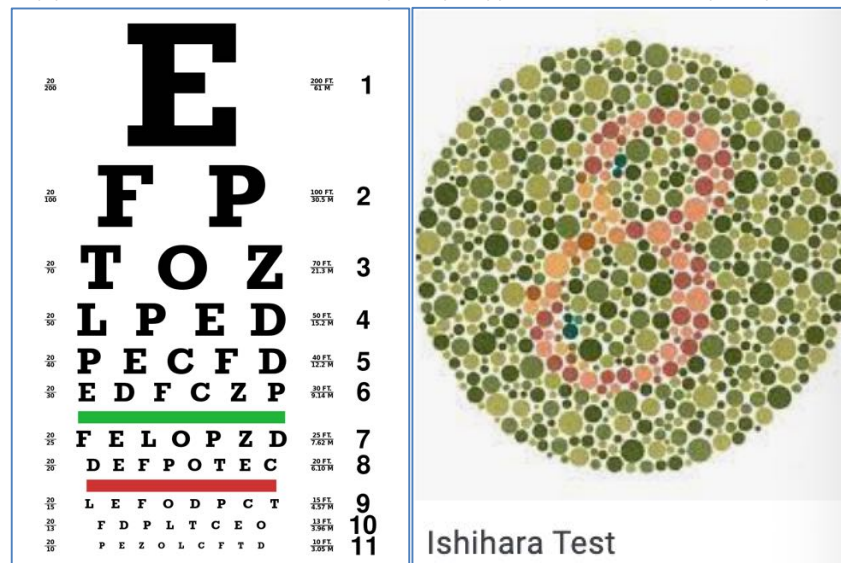
- **1: EXPOSE THE PATIENT FULLY!!**
- **Look (Visual Inspection)**
  - Symmetry? (Size, Shape, Position)
  - Scars
  - Redness, Bruising
  - Lumps, Swelling
  - Atrophy/Wasting
- **Feel (Palpate Relevant Anatomy for Tenderness, Heat, Swelling, Crepitus):**
  - Palpate BOTH Knees (Say you'd like to examine the other side)
  - Palpate Definite Anatomical Structures for Tenderness:
    - Quadriceps Tendon
    - Patella (+ Patella "Tap" test for Effusion)
    - Patella Tendon
    - Tibial Tuberosity
    - Joint Line
    - Collateral Ligament Attachments.
  - Feel for Heat, Swelling, & Crepitus
  - Eg: "Non tender to firm palpation"
- **Move (Active +/- Passive if Required):**
  - Symmetry?
  - 3x Half Squats (Patellofemoral Joint)
  - 1x Full Squat (Knee Joint Proper)
    - Range of Movement
    - Fluidity
    - Pain with Movement
  - If a pt can't do something, ask them why? (Blockage/weakness/Neuro)
  - (Passive – Only for Movements that were Limited)
  - Eg: "Symmetrical and full ROM in all planes"
- **Measure (Range of Movement):**
  - Done at the same time as "Move"
  - Compare with opposite side (ballpark – doesn't have to be objective)
  - "Limited active extension of l-elbow to 150deg"
- **(Clinical Presentations Requiring Special Tests):**
  - Acute Knee pain:
    - Cruciate Ligament Tear ("went to change direction, felt something pop, and was swollen within hours" = ACL)
      - Lachman's Test – Pt supine, examiner's knee under patient's knee, stabilise femur with one hand, move tibia ant-&-post firmly. Positive if displacement > other side.
      - Anterior Drawer – (Note: Only really useful in an anaesthetised, paralysed pt. NB. A conscious pt will guard). Pt supine, knee bent to 90degrees. Sit on patient's foot. With 2 hands apply anterior force to the tibial head. Positive if Anterior Displacement is > than other side.
    - Meniscal Tear
      - \*\*Thessaly Test (For Meniscal Tear) – Hold hands standing. Stand on 1 foot. Bend the weight-bearing leg 20°. Do the Twist. Pain = Positive. (Note: Pt must point to source of pain – same place both times)
    - Collateral Ligament Sprains/Tears
      - Valgus & Varus Stress Tests –Pt supine, support knee from below, 30deg flexion & apply valgus & varus stresses. (Positive = Laxity/gaping.)
    - Patellofemoral Mal-Tracking Syndrome. Tracking of the Patella through its groove isn't right; or groove isn't straight; or vastus medialis isn't working properly.
      - → Pain with stairs/squatting
      - Half squats - Good indication of patella-femoral function

## THE NEUROLOGICAL EXAMINATIONS

## THE NEUROLOGICAL EXAMINATIONS

### Full Cranial Nerve Exam:

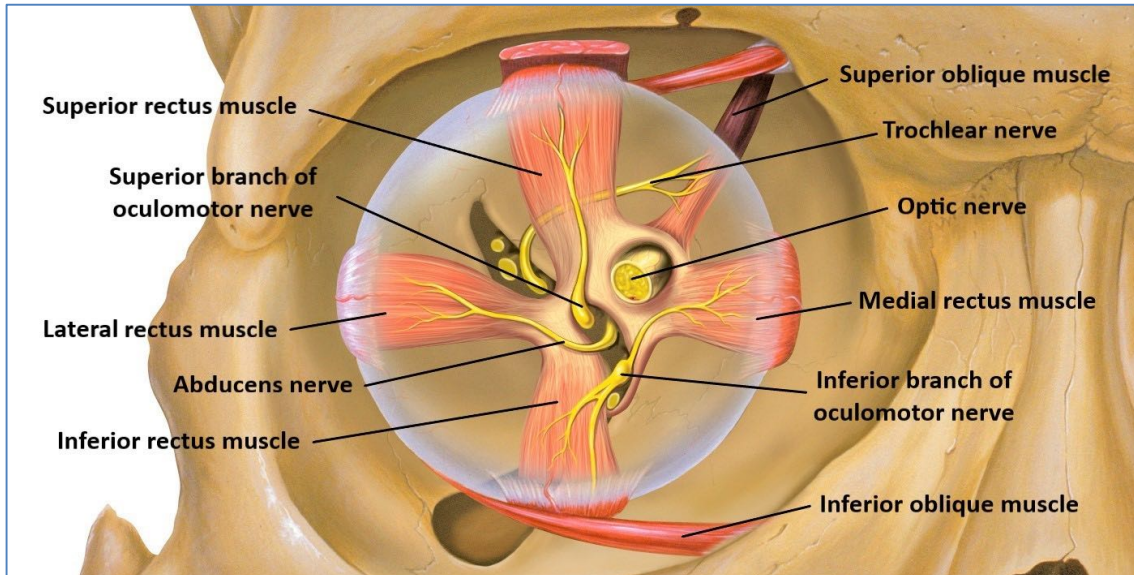
- **General Inspection:**
  - Altered Level of Consciousness?
  - Patient Alert? Oriented to Person Place & Time?
  - Facial Asymmetry?
  - Evidence of Trauma?
  - Fasciculations/Tremors?
  - Muscle Wasting?
  - Speech Impediments (Dysphasia?/Dysarthria?/Dysphonia)
  - **Ptosis?**
  - **Inability to Close the Eye?**
  - **Facial Muscle Wasting?**
  - **Facial Sweating?**
- **I - Olfactory:**
  - "Have you noticed any change in smell or taste lately?"
- **II - Optic:**
  - "Have you noticed any changes in your vision lately?"
  - Obtain Corrected Visual Acuity in Each Eye Separately, then both eyes. (Snellen's Chart)
  - Assess for Colour Blindness (Ishihara Charts)
  - 6 Point Visual Field Testing (confrontation position. Use hat pin and cover ipsilateral eyes)
  - Pupil Response to Light (Direct & Consensual + Swinging Torch Test)
  - Fundoscopy (Cataracts/Diabetic Retinopathy/Hypertensive Retinopathy)



<https://rakhospital.com/specialties/ophthalmology/>

- **III - Oculomotor, IV - Trochlear & VI - Abducens:**

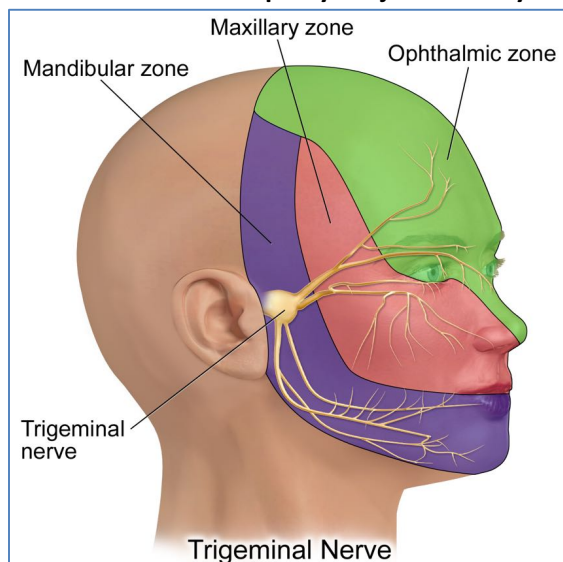
- “Keeping your head still, follow this hat pin with your eyes”
- “Let me know if at any time you begin to see double”
- Do the 6-speed gearbox positions
- Asymmetrical Movement of the Eyes
- Nystagmus (Note: This is a sign of Cerebellar Pathology – and always points to the side of the lesion)
- **Oculomotor:** Superior Rectus, Medial Rectus, Inferior Rectus, Inferior Oblique
- **Trochlear:** Superior Oblique
- **Abducens:** Lateral Rectus



File:Eye orbit anterior.jpg: Patrick J. Lynch, medical illustratorderivative work: Shyhedgehuginlabcoat, CC BY 2.5  
<<https://creativecommons.org/licenses/by/2.5>>, via Wikimedia Commons

- **V - Trigeminal:**

- **“I’m going to be testing your facial sensation now using a sensory pin”**
- **“First I’m going to test your “Sharp” (Pain) sensation, and this is what it feels like (on sternum)”**
  - Pain: Ophthalmic Division, Maxillary Division, Mandibular Division
- **“next i’m going to test your “light touch” sensation, and this is what it feels like (on sternum)”**
  - Light Touch (Cotton Wool): Ophthalmic Division, Maxillary Division, Mandibular Division.
- **“Ok now look straight ahead with your eyes wide open thank you”**
  - Corneal Reflex (Cotton wool): touch rolled cotton wool onto the corneas from the sides
- **“Next I’m going to test the motor function of the trigeminal nerve”**
  - “Clench your jaw please” → Feel the Masseter Muscle
  - “Open your jaw and resist me closing it” → Pterygoids
  - “Just open your jaw halfway and relax, and I’m going to tap it with this” → Jaw jerk reflex



BruceBlaus, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons



- **VII – Facial Nerve:**

- “Wrinkle your forehead and don’t let me flatten it”
- “Close your eyes and don’t let me open them”
- “Smile”
- “Puff out your cheeks and don’t let me squash them”
- **(Note: UMN Lesions – you lose the Lower Quadrant of the face on the Contralateral Side)**
- **(Note: LMN Lesions – you lose the Whole Half of the face on the Ipsilateral Side)**



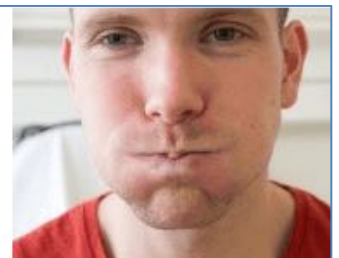
Crease up the forehead



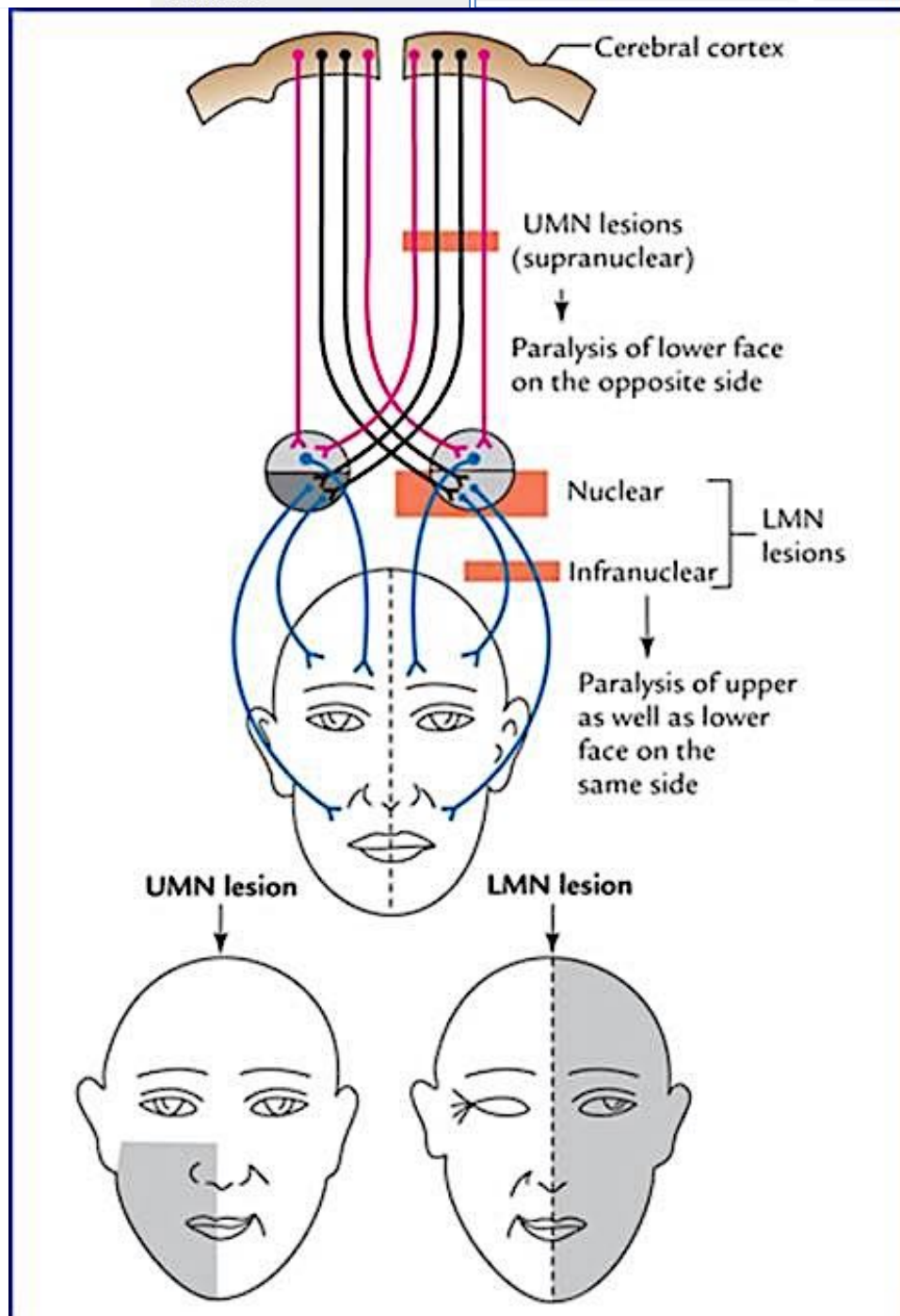
Keep eyes closed against resistance



Reveal the teeth



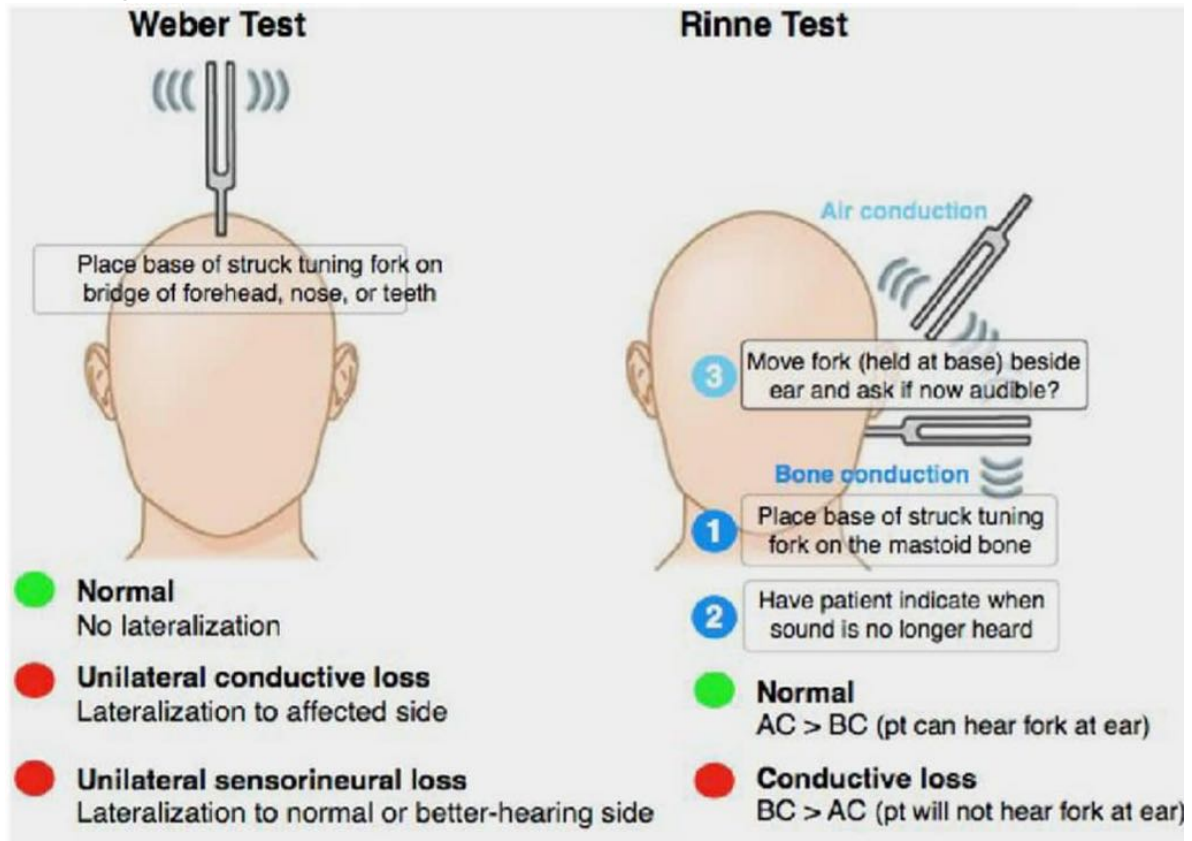
Puff out the cheeks





- **VIII – Vestibulocochlear Nerve:**

- “Next I’m going to test your hearing and balance”
- **Romberg’s Test** (“Stand feet together and close your eyes”)
- **Weber’s Test** (“Tell me if you hear the sound more in one ear, or is it equal in both?”) – Comment on lateralisation. If it lateralises to one ear, then there is **CONDUCTIVE DEAFNESS** in that ear.
- **Rinne’s Test** (“Tell me when you cannot hear the sound any more”) – then move the blades close to the ear canal – “Can you hear it now?”
- **Whisper Test** (“No repeat what I whisper” – 69, 100) Whilst distracting the other ear with rubbing fingers.



Unattributable

- **IX – Glossopharyngeal & X – Vagus:**

- “Open your mouth and say AH”
  - Look for asymmetrical elevation of the Uvula.
- Mention that you’d also do the gag reflex.
- “Can you say your name please” – Assess for Hoarseness
- “Can you cough please” – Assess for bovine cough.

- **XI – Accessory Nerve:**

- “Turn your head to the side and resist me moving it” (Contralateral Sternocleidomastoids)
- “Shrug your shoulders and resist me pushing them down” (Trapezius)



- **XII – Hypoglossal:**

- “Poke your tongue out as far as you can”
- Assess for Asymmetry (The tongue will point to the side of the lesion)



**“Thankyou that concludes my examination”**

### Focussed Cerebellar Examination:

- **Introduction + Wash Hands + Consent**
- **General Inspection:**
  - Patient Alert & Orientated?
  - Tremor? Fasciculations
  - Wasting?
  - Evidence of Head Injury?
- **Speech:**
  - Say "British Constitution" (Listen for Dysphasia, Dysarthria, Dysphonia)
- **Horizontal Nystagmus:**
  - Oscillations of the eyeball when looking from one side to the other.
  - **Note: Nystagmus always points to the side of the cerebellar lesion.**
- **Standing Coordination:**
  - **Romberg's Test – "Stand with feet together and close your eyes for me" – SUPPORT THE PATIENT:**
    - Loss of balance with eyes closed = Proprioceptive Dysfunction (Dorsal Column)
    - Loss of balance with eyes open = Cerebellar Dysfunction



#### ROMBERG'S TEST

Normal: upright posture maintained with eyes closed (only very minimal swaying may be observed)

- **GAIT:**
  - Walk Normally (Note any wide-based gait = Cerebellar Dysfunction)
- **Heel to toe Walking:**
  - Inability to do this = Cerebellar Dysfunction
- **Pronator Drift - "Can you open your palms, raise your arms out to the front and close your eyes":**
  - **Upward Drift** = Cerebellar Dysfunction
  - **Downward Drift** = Pyramidal
- **Rebound – "Can you raise your arms out to the front as quick as you can and then stop them":**
  - Rebound = Cerebellar Dysfunction
- **Dysdiadochokinesis:**
  - "Clap your hands like this as fast as you can"
- **Finger nose Test (Past Pointing):**
  - Touch your nose
  - Touch my finger



- **Supine Coordination:**

○ **Heel-Shin Test:**

- “Place your heel on your knee and slide it down your shin...then lift off...then back to knee & repeat.”



○ **Finger Toe Test (Past Pointing):**

- Touch My finger
- Touch the bed
- Etc.

○ **Dysdiadochokinesis:**

- “Tap my hands with your feet alternating as fast as you can.”

○ **Clonus:**

- Flex the knee, externally rotate the hip, and Rapidly Dorsiflex (Sustained Rhythmic Contraction = Clonus = Cerebellar Dysfunction)

- **Truncal Ataxia:**

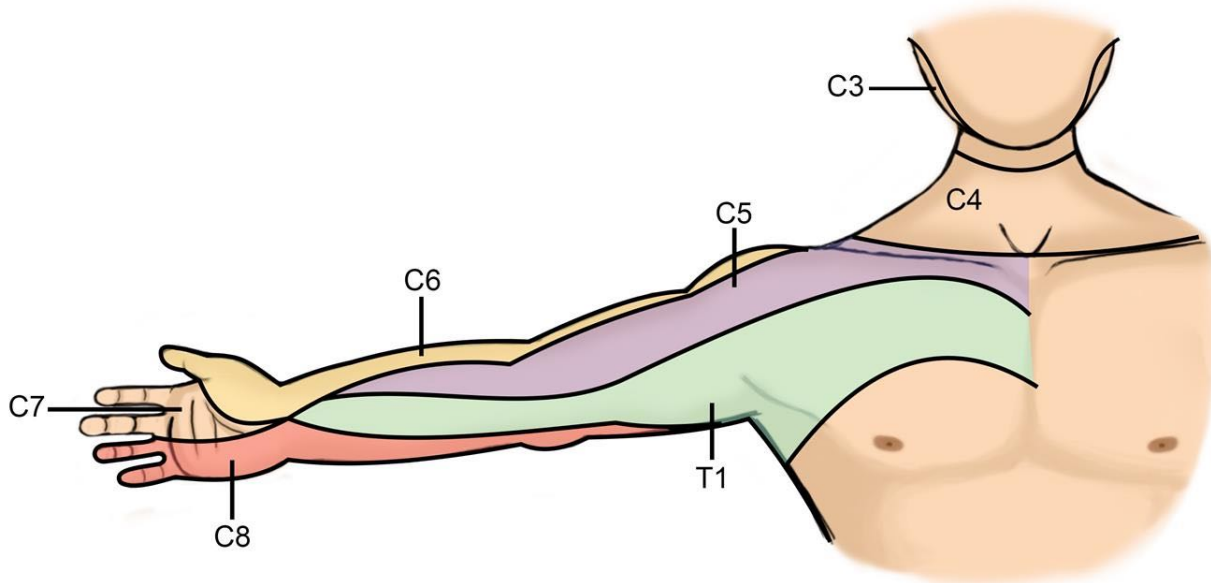
- “Have a seat...Cross your arms...now stand up without using your hands”

**“Thankyou that concludes my examination”**

## **Focussed Upper Limb Neurological Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - "Can you please take your shirt off"
  - "Patient appears alert and oriented, and in no apparent pain or distress"
  - Wasting (LMN Lesion)?
  - Tremors (Parkinson's/Benign)? Fasciculations (LMN Lesion)?
  - Scars? Deformities?
  - Bruising? Injury?
  - Asymmetry?
  - "Are you left or right handed?"
- **Vital Signs**
- **Motor Function:**
  - **Muscle Bulk/Wasting of:**
    - Intrinsic Hand Muscles
    - Forearm
    - Biceps/Triceps
    - Shoulders
  - **Tone – Hypertonia (UMN Lesion), Hypotonia (LMN Lesion), Cogwheeling/Lead pipe Rigidity (Parkinson's):**
    - Shoulder Abduction/Adduction
    - Shoulder Flexion/Extension
    - Elbow Flexion/Extension
    - Supination/Pronation
    - Wrist Flexion/Extension
    - Finger Flexion/Extension
  - **Power – Graded 0→5 (0=None; 1=Flicker; 2=Gravity Limited; 3=Gravity Unlimited; 4=Fatigue; 5=Full)**
    - Shoulder Abduction (C5, C6) "Pick up sticks"
    - Shoulder Adduction (C7, C8) "Lay them straight"
    - Elbow Flexion (C5, C6) "Pick up sticks"
    - Elbow Extension (C7, C8) "lay them straight"
    - Wrist Flexion (C6, C7) "Point to heaven"
    - Wrist Extension (C7, C8) "lay them straight"
    - Grip Strength (C7, C8)
    - Finger Adduction & Abduction (C8, T1)
    - Thumb Opposition (C8, T1)
  - **Reflexes – Hyperreflexia (UMN Lesion); Hyporeflexia (LMN Lesion):**
    - Triceps Reflex (C6, C7)
    - Biceps Reflex (C5, C6)
    - Brachioradialis (C5, C6)
  - **Coordination/Cerebellar Function:**
    - **Pronator Drift** – "Can you open your palms, raise your arms out to the front and close your eyes" (**Downward = Pyramidal; Upward = Cerebellar**)
    - **Rebound** – "Keep your hands there & your eyes closed, and don't let me move your arms" OR "Quickly raise your arms to the front and stop".
    - **Dysidiadochokinesis** – "Clap your hands like this as fast as you can"
    - **Past-Pointing (Cerebellar Dysfunction)** – "Touch your nose, Touch my finger"
- **Sensory Function – (Standardise on Sternum first):**
  - **OVER ALL DERMATOMES:**
    - C3 (Corporals Patch)
    - C4 (Corporals Patch)
    - C5 (Biceps)
    - C6 (Thumb)
    - C7 (Middle Finger)
    - C8 (Pinky)
    - T1 (Medial Forearm)
    - T2 (Medial Bicep)

## Upper Extremity Dermatome Anterior View



- **1: Pain (Spinothalamic)**
  - EYES CLOSED
  - (Temperature – Not done)
- **2: Vibration (Dorsal Column)**
  - Start Distally → Move Proximally
- **3: Proprioception (Dorsal Column)**
  - EYES CLOSED
- **4: Light Touch (Both Pathways)**
  - EYES CLOSED

**“Thankyou that concludes my examination”**



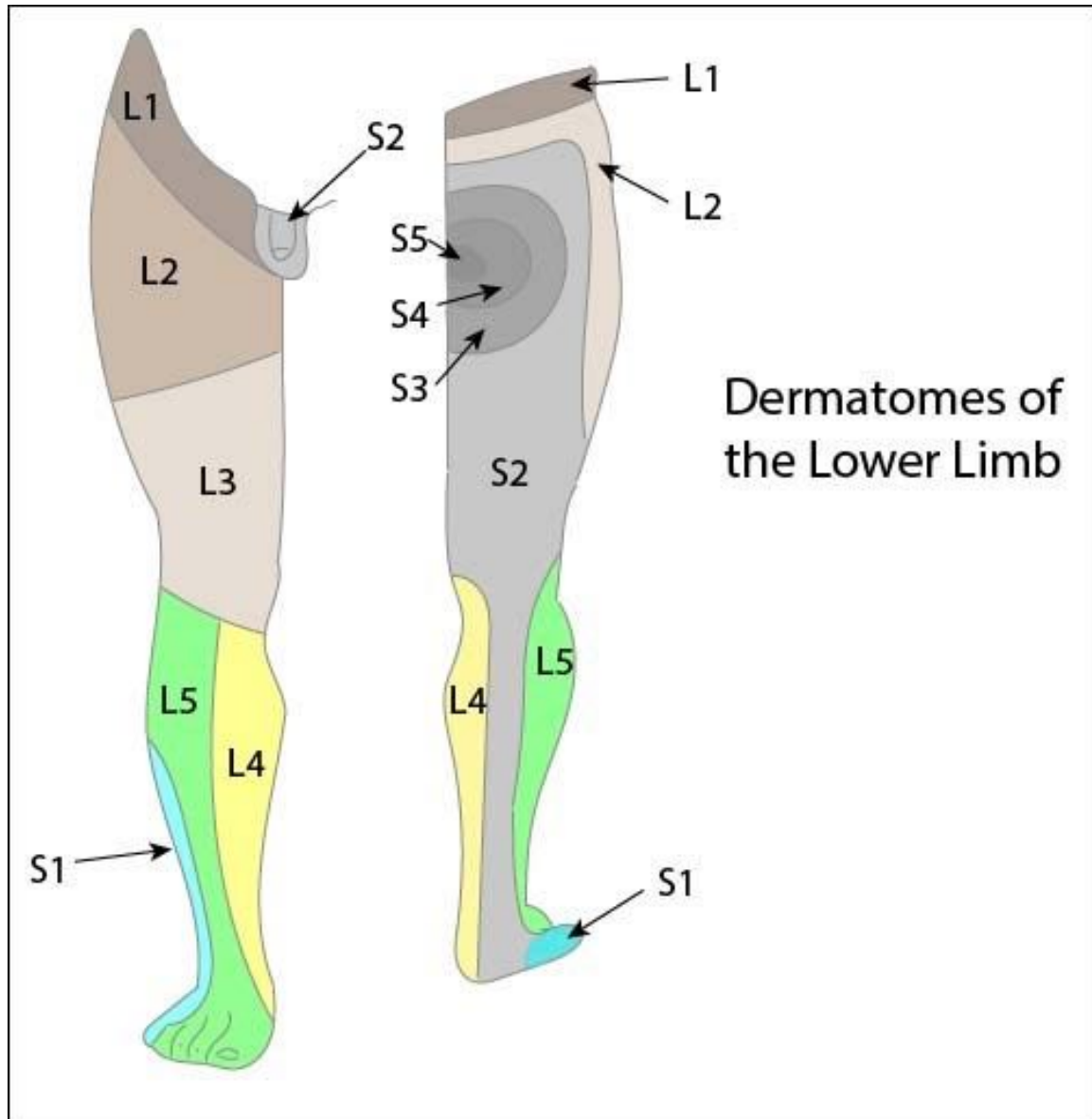
### **Focussed Lower Limb Neurological Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - **Standing:**
    - Muscle Wasting? (LMN)
    - Fasciculations? (LMN)
    - Scars? Deformities?
    - Injury?
    - Involuntary Movements?
  - **GAIT – “Can you walk to the other end of the room, turn round, and walk back”. Look for:**
    - **Foot Drop L4 + L5 Lesion** (Common Peroneal/Fibular Nerve Palsy)
    - **Cannot Walk on Toes S1 Lesion.**
    - **Shuffling Gait** (Parkinson’s)
    - **Wide Based Gait** (Cerebellar)
    - **Poor Heel-Toe Walking** (Cerebellar)
  - **Romberg’s Test – “Stand with feet together and close your eyes for me” – SUPPORT THE PATIENT:**
    - Loss of balance with eyes closed = Proprioceptive Dysfunction (Dorsal Column)
    - Loss of balance with eyes open = Cerebellar Dysfunction
- **Motor Function – On Examination Couch:**
  - **Muscle Bulk:**
    - Feel for any wasting (LMN Lesion)
  - **Tone – (Hypertonia = UMN Lesion; Hypotonia = LMN Lesion; Cog/Lead pipe Rigidity = Parkinson’s):**
    - Hip Flexion
    - Knee Flexion & Extension
    - Ankle Flexion & Extension
    - Toe Flexion & Extension
  - **Power:**
    - Hip Flexion (L2, 3, 4) “Enforce the Law” (With Roundhouse Kick)
    - Hip Adduction (L2, 3, 4) “Enforce the Law” (With Roundhouse Kick)
    - Hip Abduction (L2, 3, 4) “Enforce the Law” (With Roundhouse Kick)
    - Hip Extension (L5, S1) + (S2) “Tense your bum”
    - Knee Extension (L3, L4) “Kick the Ball”
    - Knee Flexion (L5, S1) “Kick your bum”
    - Dorsiflexion (L4, L5) “Walk on Fire”
    - Plantar Flexion (S1) “Find your son”
  - **Reflexes:**
    - Patellar Tendon (L3, L4) “Kick the Ball”
    - Achilles Tendon (S1) “Find your son”
    - Plantar Reflex/(Babinski’s Positive if UMN Lesion)
  - **Coordination/Cerebellar Function:**
    - **Dysdiadochokinesis:**
      - “Tap your feet against my hands as quickly as possible”
    - **Finger-Toe Test (Past Pointing Test):**
      - “touch my finger, touch the bed, touch my finger”
    - **Heel-Shin Test:**
      - Place your heel on your knee and slide it down your shin, lift off, and place back on knee...Repeat.
    - **+ Test for CLONUS (Cerebellar Dysfunction)**

- **Sensation:**

○ **OVER ALL DERMATOMES – With EYES CLOSED:**

- L1 Garter Band 1
- L2 Garter Band 2
- L3 Anterior Knee
- L4 Medial Calf
- L5 Lateral Calf
- S1 Lateral Foot (little toe)
- S2 Posterior Thigh



<https://www.earthslab.com/anatomy/facial-nerve/>

- Pain
- Vibration
- Proprioception
- Light Touch

- **Special tests for Meningitis:**

- **Kernig's Sign** – Neck pain on hip flexion & knee extension
- **Brudzinski's Sign** – Involuntary Hip Flexion & knee Extension on Neck Flexion
- **Neck Stiffness** – Pain on Neck Flexion

**"Thankyou that concludes my examination"**

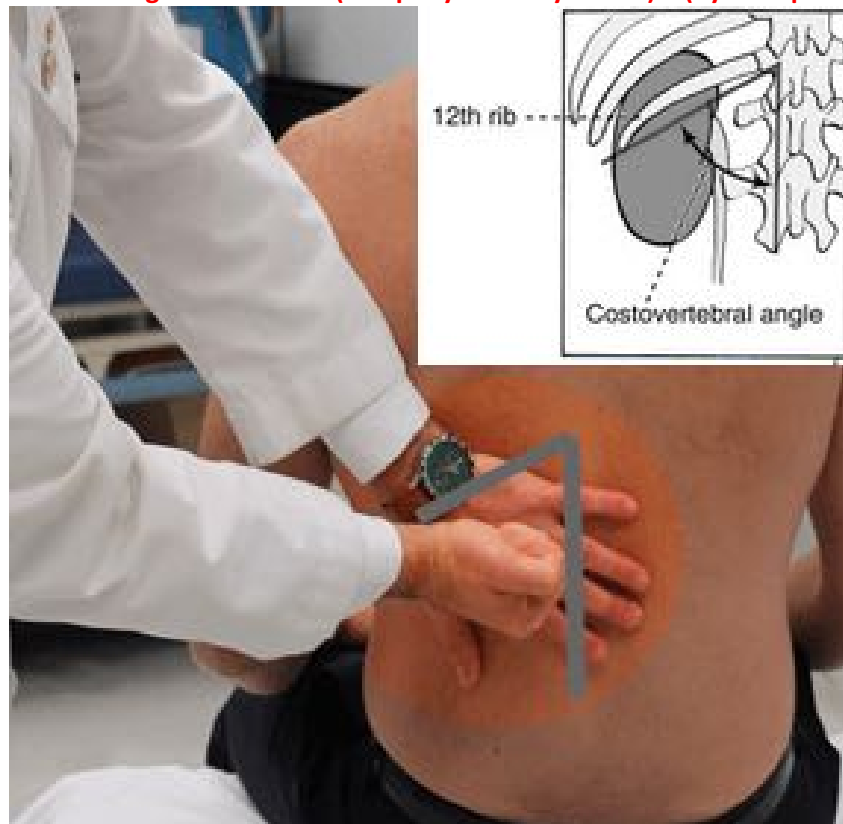
## **THE RENAL SYSTEM EXAMINATION**

## THE RENAL SYSTEM EXAMINATION

### Full Renal Exam:

- **Introduction, Wash Hands, Consent,**
- **Expose Patient & General Inspection:**
  - Alert & Orientated? (Uraemic Encephalopathy, **UTI in Elderly**)
  - Acute pain or distress (Renal Colic, Pyelonephritis)
  - Signs of Fluid Overload (Ascites, Peripheral Oedema)
  - Signs of Dehydration (Causing Pre-Renal Failure)
  - Facial Oedema (Nephrotic Syndrome)
  - Obesity
  - Scars
  - In-Dwelling catheter
  - **Uraemic Fetor/Tinge (Ammonia Smell from Hyperuricaemia)**
  - **Easy bruising on Arms, Legs & Trunk**
- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (If Infection, Hypovolaemia, Anaemia)
  - **Blood Pressure:**
    - Hypertension (**Nephritic**, Fluid Overload, Polycythaemia, Polycystic Kidney)
    - Postural Hypotension (Anaemia, Hyponatraemia, Addison's, Diabetic Neuropathy)
  - **Respiratory Rate:**
    - Tachypnoea (if Renal Acidosis)
    - Bradypnoea (if Renal Alkalosis)
  - **Temperature:**
    - Fever (Infection, Malignancy)
- **Hands:**
  - Peripheral Perfusion + CRT
  - **Mee's Lines [single horizontal bands] (Arsenic/Heavy metal Poisoning)**
  - **Muercke's Lines [paired horizontal bands] & Leukonychia (Hypoalbuminaemia due to Nephrotic Syndrome)**
  - Palmar Crease Pallor & Pale Nails (Anaemia, Nephritic Syndrome)
  - Palmar Crease Pigmentation (Addison's Disease → Hyponatraemia & Hypovolaemia)
  - Xanthomata (Diabetes, Obesity, Metabolic Syndrome)
  - **Gouty Tophi (Hyperuricaemia)**
  - **Vasculitic Changes (Eg: Digital Infarcts – Sign of Autoimmune Glomerulonephritis)**
- **Arms:**
  - Uraemic Tinge/Uraemic Frost (Hyperuricaemia)
  - Scratch Marks (Uraemia)
  - AV-Fistulae (ESRD - Dialysis)
  - Scars
  - Asterixis (Uraemic Encephalopathy)
- **Face:**
  - **Butterfly Malar Rash of SLE (Cause of Renal Failure)**
  - Periorbital Oedema (Nephrotic Syndrome)
  - Conjunctival Pallor (Anaemia, Nephritic Syndrome)
  - Central/Peripheral Cyanosis
  - Mouth Infections:
    - (Strep Pharyngitis can → PSGN Nephritic Syndrome)
    - (Also Immunocompromise from Nephrotic Syndrome)
    - Thrush (Immunocompromise from diabetes)
  - Xanthelasma (Diabetes, Nephrotic Syndrome)
  - **Band Keratopathy (calcification of cornea due to 2° Hyperparathyroidism due to Vit D Deficiency)**
  - Dehydration (Nephrotic Syndrome)
  - **+Fundoscopy (Diabetic Retinopathy)**
  - **Gum Hyperplasia (Methotrexate – Transplant Medication)**

- **Neck:**
  - Carotid Bruits (Hypercholesterolaemia, PVD, Diabetes)
  - Vas Cath (Haemodialysis)
  - **Parathyroidectomy Scars (due to 2° Hyperparathyroidism due to Vit-D Deficiency)**
  - **Acanthosis Nigricans (Diabetes, Metabolic Syndrome, Obesity)**
- **Chest:**
  - **CCF & Pulmonary Oedema** (Due to Fluid Overload – Ie: Nephrotic Syndrome, CKD, ESRD)
- **Abdomen:**
  - **Inspection:**
    - Distension (Ascites in Nephrotic, Nephritic, Peritoneal Dialysis)
    - Surgical Scars
    - Peritoneal Dialysis Port
    - Visible Masses
  - **Palpation:**
    - Renal masses (Polycystic Kidneys, Carcinoma)
    - Insulin Injection Sites
    - **Polycystic Liver → Hepatomegaly (Polycystic Kidneys)**
    - **Abdominal Aorta (Aneurysm → Pre-Renal Failure)**
    - Enlarged Bladder (Obstructive Uropathy)
  - **Auscultation:**
    - Renal Bruits (Renal Artery Stenosis, Pre-Renal Failure)
    - Bowel Sounds Present
  - **Percussion:**
    - Shifting Dullness (Ascites – Nephrotic, Nephritic, CKD)
- **Back:**
  - Surgical Scars
  - **Costovertebral Angle Tenderness (Murphey's Kidney Punch) – (Pyelonephritis/Stones)**



<https://epomedicine.com/clinical-medicine/costovertebral-or-renal-angle-tenderness/>

- **Bony Tenderness**
- **Sacral Oedema**
- **Legs:**
  - Pitting Oedema (Nephrotic, Nephritic, CKD, Diabetes, PVD)
  - Scratches/Uraemic Frost (Hyperuricaemia)

- **Feet:**
  - Xanthomata (Hypercholesterolaemia)
  - Gouty Tophi (Hyperuricaemia)
  - Mee's Lines (Arsenic/Heavy metal Poisoning)
  - Meurkhe's Lines & Leukonychia (Hypoalbuminaemia – Nephrotic Syndrome)
  - Peripheral Perfusion + CRT
  - Peripheral Pulses
  - **(+ Diabetic Neuropathy & Foot Examination)**
- **+ Urine Dipstick**
  - Blood (Nephritic, Pyelonephritis, Renal Stones)
  - Protein (Nephrotic, Nephritic)
- **Signs of Intra-Renal Failure (Nephrotics & Nephritics):**
  - **Nephrotic (Less Serious) (Autoimmune [adults] or Post-URTI [child]):**
    - Polyuria
    - Massive Proteinuria → Hypoalbuminaemia, Oedema & Periorbital Oedema
    - Compensative Hyperlipidaemia (Xanthomata/Xanthelasma)
    - Immunosuppression (Loss of IgG in Urine)
    - Hypercoagulability (Loss of AT3 in Urine)
    - No Haematuria
  - **Nephritic (More Serious) (Post URTI [Adults] or Post Strep-Pharyngitis [Children]):**
    - Anuria/Oliguria (↓GFR)
    - Modest Proteinuria (*Note: NORMAL ALBUMIN*) → Oedema
    - Anaemia (PainLESS Haematuria)
    - Hypertension



## **THE RESPIRATORY EXAMINATION**

## THE RESPIRATORY EXAMINATION

### Full Respiratory Exam:

- **Introduction, Wash Hands, Consent**
- **General Inspection:**
  - Alert/Orientated
  - Pain/Distress
  - **Body Habitus:**
    - Obesity (Blue Bloaters)
    - Cachexia (Pink Puffers, Malignancy)
    - Chest Deformities (Pectus Excavatum, Pectus Carinatum, Barrel Chest, Kyphosis, Lordosis)
    - Chest Scars
  - **Respiratory Distress?**
    - Tripoding
    - Pursed-Lip Breathing (Obstructive Lung Disease)
    - Dyspnoea
    - Audible Wheeze/Cough/Stridor (Obstructive Lung Disease)
    - Intercostal Recession & Tracheal Tug (Restrictive Lung Disease)
  - **Colour:**
    - Cyanosis (Blue Bloaters)
    - Plethora (Pink Puffers)
    - Pallor (Anaemia)

### **Blue Bloater**

#### **Chronic Bronchitis**



#### **Symptoms**

- Chronic , productive cough
- Purulent sputum
- Hemoptysis
- Mild dyspnea initially
- Cyanosis (due to hypoxemia)
- Peripheral edema (due to cor pulmonale)
- Crackles, wheezes
- Prolonged expiration
- Obese

#### **Complications**

- Secondary polycythemia vera due to hypoxemia
- Pulmonary hypertension due to reactive vasoconstriction from hypoxemia
- Cor pulmonale from chronic pulmonary hypertension

### **Pink Puffer**

#### **Emphysema**



#### **Symptoms**

- Dyspnea
- Minimal cough
- Increased minute ventilation
- Pink skin, Pursed-lip breathing
- Accessory muscle use
- Cachexia
- Hyperinflation, barrel chest
- Decreased breath sounds
- Tachypnea

#### **Complications**

- Pneumothorax due to bullae
- Weight loss due to work of breathing

- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (Anaemia, Cyanosis, Infection)
  - **Blood Pressure:**
    - Typically Normal
    - Postural Variation (Severe Anaemia)
    - Pulsus Paradoxus (Severe COPD, Asthma, Tamponade, Pneumothorax)
  - **Respiratory Rate:**
    - Tachypnoeic (Anaemia & All Lung Diseases)
  - **SPO2:**
    - May be reduced or preserved (Must be recorded if the reading was on supplemental O2)
  - **Temperature:**
    - Fever (Infection)
- **Hands:**
  - Perfusion/CRT
  - Clubbing (Chronic Cyanosis)
  - Pale Nails (Anaemia)
  - Koilonychia (Iron Deficiency Anaemia)
  - Palmar Crease Pallor (Anaemia)
  - Tar Staining (Smoking)
  - Intrinsic Hand Muscle Wasting \*(Pancoast Tumour)
  - Hand Muscle Weakness \*(Pancoast Tumour)
  - Asterixis (CO2 Retention)
- **Arms:**
  - HPOA (Lung Tumour)
  - Pemberton's Sign \*(SVC Obstruction due to Pancoast Tumour)



Unattributable

- **Face:**
  - Plethora (Polycythaemia, SVC Obstruction)
  - Skin Lesions (SCC/BCCs)
  - Conjunctival Pallor (Anaemia); Subconjunctival Haemorrhages (Severe Cough)
  - Atrophic Glossitis/Angular Stomatitis (All Types of Anaemia)
  - Hydration
  - Central/Peripheral Cyanosis (All Lung Pathology)
  - Leucoplakia/Erythroplakia (Premalignancy from Smoking)
  - Tar-Stained Teeth (Smoking)
  - **HORNER'S Syndrome** (Sympathetic Nerve Palsy):
    - Unilateral Ptosis, Anhydrosis, Miosis (Pinpoint Pupil, Enophthalmos, Laryngeal Hoarseness)
- **Neck:**
  - Virchow's Node (Supraclavicular) (Malignancy)
  - Cervical Lymphadenopathy (Infection, Malignancy)
  - ↑JVP (SVC Obstruction – Pancoast Tumour)(or Pulmonary Hypertension)
  - Abdojugular Reflux (Negative if SVC Obstruction)
  - Tracheal Deviation (Atelectasis, Pneumothorax)
  - Tracheal Tug (Restrictive Lung Disease)

- **Chest – ONCE FROM BEHIND, ONCE FROM THE BACK:**
  - **Inspection:**
    - Chest Deformities
    - Scars
    - Visualise Chest Expansion FROM BEHIND
  - **Palpation:**
    - Chest Expansion >5cm = Normal (From Behind)
    - ↑Tactile Fremitus (Consolidation)
    - Hoover's Sign (Severe COPD, Emphysema, Asthma)
    - Chest Wall Tenderness (Malignancy)
  - **Percussion:**
    - Dullness = Consolidation
    - Stony Dullness = Pleural Effusion
    - Hyper resonant = COPD/Emphysema/Asthma/Pneumothorax
    - Lung Borders for Hyperinflation (6<sup>th</sup> rib anteriorly = normal) (COPD, Emphysema, Asthma)
  - **Auscultation:**
    - ↑Vocal Resonance
    - **Breath Sounds:**
      - Vesicular = Normal
      - Bronchial = Consolidation
      - Pleuritic Rub = Pleuritis/Mesothelioma
      - Muffled = Pleural Effusion
      - Inspiratory Crepitations/Crackles = Pulmonary Oedema/Pneumonia
      - Expiratory Wheezes = Bronchial Disease
  - **(+ Mention Cardiovascular Examination)**
- **Abdomen:**
  - **Percuss & Palpate for Liver Ptosis (due to Lung Hyperinflation)**
- **Legs:**
  - Oedema
  - Calf Tenderness & Erythema (DVT→PE)
  - Signs of Venous Stasis – Shiny Skin, Hair Loss, Venous Ulcers (DVT→PE)
- **Feet:**
  - Perfusion/CRT
  - Clubbing
  - Koilonychia (Iron Deficiency Anaemia)
  - Pale Nails (Anaemia)

### Acute Bronchitis:

- **TYPICAL Symptoms/Presentation:**
  - Fever
  - Productive, Wheezy Cough, +/- Dyspnoea
  - + URTI Symptoms (Sore-Throat, Runny Nose, Sneezing, Hoarseness)
  - + Non-Specific Viral (Fever, Malaise, Headache, Myalgia)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea
  - **Other:**
    - End-Expiratory Wheezes
  - **Signs of Causes:**
    - 2° to URTI

### Bronchial Asthma (Variable Obstructive):

- **TYPICAL Symptoms/Presentation:**
  - **Asymptomatic unless during an "Attack":**
    - Wheezy Dry Cough,
    - Dyspnoea
    - Anxiety
  - **If Severe:**
    - Exhaustion
    - Inability to Speak in Full Sentences
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia,
  - **Other:**
    - Peripheral Cyanosis, Wheals, Hives, Rhinitis
    - Accessory Muscle Usage
    - Inspiratory Wheezes, Marked Expiratory Wheezes, Prolonged Expiratory Phase,
    - ↑Chest Expansion (+/- Hyperinflated Lung Fields)
    - Reduced Breath Sounds (Silent if Severe)
    - NO signs of Consolidation (Normal ↑Fremitus, ↓Percussion & ↑Vocal Resonance)
    - (+/- Pulsus Paradoxus if Severe)
  - **Signs of Causes:**
    - Signs of Atopia (Wheals, Allergic Rhinitis, Hives)
    - Family History

### Bronchiectasis:

(Chronic Bronchial Thickening & Dilation + Mucus Accumulation due to ↓Mucociliary)

- **TYPICAL Symptoms/Presentation:**
  - Wheezy Productive Cough
  - Copious Purulent Sputum (Green/Yellow)
  - (+/- Haemoptysis)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia, (+/- Fever if Infection)
  - **Other:**
    - Clubbing, Cyanosis, Cachexia,
    - Foul-Smelling Sputum (Sometimes Haemoptysis)
    - Coarse Pan-Inspiratory Crackles, End-Expiratory Wheezes
    - (+/- Cor-Pulmonale if Severe)
  - **Signs of Causes:**
    - **\*\*Cystic Fibrosis (Clubbing, Peripheral Cyanosis, Sputum, Salty Frost, Wasting)**

### **Emphysema (Dry):**

- **TYPICAL Symptoms/Presentation:**
  - **Pink Puffers:**
    - Wheeze
    - Severe Dyspnoea
    - Weight Loss
    - (+/- Peripheral Oedema & Ascites in Cor-Pulmonale)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia
  - **Other:**
    - (Note: Does NOT cause Clubbing or Haemoptysis)
    - **Thin (No Oedema), Pink (No Cyanosis),**
    - Tachypnoea, Accessory Muscle Usage, Intercostal Recession, Tripoding (Stooping), Pursed-Lip Breathing, Barrel Chest,
    - ↓Chest Expansion, Hoover's Sign Positive, Tracheal Tug
    - Hyperinflated Lungs Fields, Hyper resonant Percussion (Gas Trapping)
    - ↓Breath Sounds, **Early Expiratory Crackles (Small Airway Disease)**, (+/- Wheeze).
    - (+/- ↑JVP, Ascites, Oedema if RVF Due to Cor-Pulmonale)
  - **Signs of Causes:**
    - Heavy Smoking (Tar Staining, Smoke Smell, Yellow Teeth, Leuko/Erythroplakia)
    - (If Young Age – Think Congenital  $\alpha$ 1-Antitrypsin Deficiency.)

### **Chronic Bronchitis (Wet): (+/- Emphysema)**

- **TYPICAL Symptoms/Presentation:**
  - **Blue Bloaters:**
    - Wet, Wheezy Productive Cough
    - Chronic Sputum Production (>3mths/year for >2years)
    - Severe Dyspnoea
    - Oedema
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea, Tachycardia
  - **Other:**
    - (Note: Does NOT cause Clubbing or Haemoptysis)
    - **Obese Patient, Gross Cyanosis (Central & Peripheral),**
    - **Cor-Pulmonale (Ascites, Oedema, Cyanosis, ↑JVP)**
    - Tachypnoea, Accessory Muscle Usage, Intercostal Recession, Tripoding (Stooping), Pursed-Lip Breathing, Barrel Chest
    - ↓Chest Expansion, Positive Hoover's Sign, Tracheal Tug
    - Hyperinflated Lung Fields, Hyper resonant Percussion
    - ↓Breath Sounds, **End-Expiratory Wheezes (Bronchial Disease)**
  - **Signs of Causes:**
    - Heavy Smoking (Tar Staining, Smoke Smell, Yellow Teeth, Leuko/Erythroplakia)



### Pneumonia (Consolidation):

- **TYPICAL Symptoms/Presentation:**
  - Acute High Fever + Chills + Rigors
  - Productive Cough (+/- Haemoptysis)
  - Pleuritic Chest Pain
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea
  - **Other:**
    - **Appears Ill:** P/C-Cyanosis, Exhaustion, Dyspnoeic, Respiratory Distress,
    - **Consolidation:** Area of ↓ Expansion, ↑ Tactile Fremitus, Dull Percussion, Bronchial Breathing, ↑ Vocal Resonance.
    - Bronchial Breath Sounds, Pleural Friction Rub, Pan-Inspiratory Crackles
  - **Signs of Causes:**
    - **Broncho Pneumonia → Diffuse, Patchy Consolidation**
    - **Lobar Pneumonia → Localised Consolidation**

### Melioidosis: (Burkholderia Pseudomallei)

- **TYPICAL Symptoms/Presentation:**
  - **Presents Similar to TB but with Pneumonia:**
    - **PUO** (Fever, Night Sweats, Chills, Rigors)
    - **Skin Lesions** (Abscesses/Ulcers)
    - **Pneumonia** (Dyspnoea, Cough, Sputum, Pleuritic Chest Pain)
  - (May → Sepsis)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea
  - **Other:**
    - Cyanosis, Dyspnoea, Skin Lesions (Melioidosis Abscesses/Ulcers), Lymphadenopathy,
    - **Consolidation:** Area of ↓ Expansion, ↑ Tactile Fremitus, Dull Percussion, Bronchial Breathing, ↑ Vocal Resonance.
    - Bronchial Breath Sounds, Pleural Friction Rub, Pan-Inspiratory Crackles
  - **Signs of Causes:**
    - Living in Tropical NQ

### Pulmonary Tuberculosis:

- **TYPICAL Symptoms/Presentation:**
  - Fever, Night Sweats,
  - Chronic Productive Cough (+/- Haemoptysis)
  - Weight Loss
  - +/- Pleuritic Chest Pain
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea
  - **Other:**
    - Cachexia, Dyspnoea, Lymphadenopathy, Hepatomegaly, Splenomegaly,
    - (Note: Cavitation Can → Pleural Effusion (Stony Dullness), Haemoptysis, Atelectasis)
  - **Signs of Causes:**
    - Immunocompromise (HIV), Immigrant, Overseas Travel,

### Pulmonary Embolism:

- **TYPICAL Symptoms/Presentation:**
  - Sudden, Severe Dyspnoea
  - Pleuritic Chest Pain
  - (+/- Haemoptysis)
  - (+/- Syncope)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea, Hypotension (LVF)
  - **Other:**
    - **RVF** → Cool Peripheries, ↑CRT, Peripheral Cyanosis, ↑JVP, RV-Heave, Tricuspid Regurgitation Murmur
    - **↓Respiratory Function** → Central Cyanosis
    - Pleural Friction Rub,
  - **Signs of Causes:**
    - **DVT** – Calf Pain, Calf Tenderness, Calf Swelling/Erythema, Pedal Oedema.
    - (B/G of Pregnancy, Air Travel, Recent Surgery, Clotting Disorders)

### Pulmonary Hypertension & Cor-Pulmonale: (LVF, COPD)

- **TYPICAL Symptoms/Presentation:**
  - **RVF Secondary to Pulmonary Hypertension:**
    - **COPD:** Dyspnoea, Cough, Wheeze
    - **Pulmonary HTN:** Cough/Dyspnoea/PND/Orthopnea
    - **RVF:** Swelling (Legs, Abdo), Chest Pain
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia (if LVF), Tachypnoea (if COPD/LVF), Hypotension (If LVF), Afebrile
  - **Other:**
    - **If LVF:** Cool Peripheries, ↑CRT, Peripheral & Central Cyanosis, Low Volume Pulse
    - **If COPD:** Clubbing, Tar Staining, Peripheral & Central Cyanosis
    - **RVF:** ↑JVP + a-Wave, RV-Heave, Medium Pan-Inspiratory Crackles in Lung Bases, Loud S2 (closure of Pulmonic Valve) Abdojugular Reflux Positive, Portal Hypertension (Tender Hepatomegaly), Ascites, Sacral/Pedal Oedema,
  - **Signs of Causes:**
    - LVF, Smoking, COPD, IPF

### Pneumothorax (Including Tension Pneumothorax):

(Air in Pleural Space)

- **TYPICAL Symptoms/Presentation:**
  - **Similar Presentation to Pulmonary Embolism:**
    - Sudden, Severe Dyspnoea
    - Pleuritic Chest Pain
    - (+/- Syncope)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachycardia, Tachypnoea, Hypotension (Mediastinal Compression),
  - **Other:**
    - Cyanosis, Dyspnoea, Anxiety, Respiratory Distress,
    - Peripheral Shutdown (↑CRT, Cool Pale Peripheries, Peripheral Cyanosis, Low-Vol Tachycardia)
    - **Tracheal Deviation AWAY from Affected Side (If Tension Pneumothorax),**
    - **@Site of Pneumothorax:** ↓Chest Expansion, ↓Tactile Fremitus, Hyper resonant Percussion, Absent Breath Sounds, Absent Vocal Resonance.
    - (IF KINKING OF GREAT VESSELS → Syncope, Negative Abdojugular Reflux)
  - **Signs of Causes:**
    - Signs of Trauma, Signs of Smoking (Emphysema), Mechanical Ventilation,

### **Atelectasis: (Collapsed Lung):**

(Due to Airway Obstruction by Foreign Object or Cancer)

#### **TYPICAL Symptoms/Presentation:**

- Dyspnoea
- Chest Pain
- (May Quickly → Pneumonia)

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Tachypnoea
- **Other:**
  - Dyspnoea,
  - **Tracheal Deviation TOWARDS the Affected Side.**
  - **@Site of Atelectasis:** ↓Chest Expansion, ↑Tactile Fremitus, Dull Percussion, Bronchial Breath Sounds, ↑Vocal Resonance
- **Signs of Causes:**
  - Mucous (Asthma, Cystic Fibrosis), Foreign Body Aspiration, Bronchial Ca (Cachexia)

### **Pleural Effusion:**

#### **TYPICAL Symptoms/Presentation:**

- Sudden Severe Pleuritic Chest Pain (Worse on Inspiration)
- Dyspnoea
- Dry Cough

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Fever, Tachypnoea, Tachycardia,
- **Other:**
  - **Tracheal Deviation AWAY from Effusion**
  - **Displaced Apex Beat AWAY from Effusion**
  - **@ Site of Effusion:** ↓Chest Expansion, ↓Tactile Fremitus, **STONY DULLNESS**, ↓Breath Sounds, ↓Vocal Resonance
- **Signs of Causes:**
  - Note: ALWAYS Suspect Mesothelioma
  - Portal Hypertension, Hypoalbuminaemia (CLD, Nephrotic Syndrome), Congestive L-Heart Failure, Lung Injury, Lung Infection.

### **Interstitial Lung Diseases: ("Pneumoconioses")**

(Inhaled Dusts: Anthracosis, Asbestosis, Silicosis)

#### **TYPICAL Symptoms/Presentation:**

- (On a Background of Occupational Exposure)
  - Chronic Cough (+/- Productive)
  - Dyspnoea (+/- Cyanosis)
- (Asbestosis is most severe and can → Mesothelioma → Pleural Effusions, Metastases)

#### **TYPICAL Clinical Signs:**

- **Vitals:**
  - Tachypnoea
- **Other:**
  - **Clubbing**, Dyspnoea, Cyanosis, Cachexia, Accessory Muscle Usage, Intercostal Recession, Tracheal Tug.
  - **Cough**
  - ↓Chest Expansion,
  - **Fine Pan-Inspiratory Crackles**
  - + Restrictive Pulmonary Function Tests.
- **Signs of Causes:**
  - Signs of Connective Tissue Diseases (Rheumatoid Arthritis, SLE, Scleroderma, Sjogren's)

### **Idiopathic Pulmonary Fibrosis:**

- **TYPICAL Symptoms/Presentation:**
  - Gradual Onset of Symptoms
    - Progressive Dyspnoea
    - Dry Cough
  - (Very Poor Prognosis – 3yrs – No Treatment)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea
  - **Other:**
    - Hypoxia/Cyanosis → Clubbing
    - Dry, Velcro-like inspiratory crackles
  - **Signs of Causes:**
    - Restrictive Pattern on Pulmonary Function Tests (↓VC, ↓TLC)

### **Sarcoidosis:** (Idiopathic Immune → Non-Caseating Granulomas)

- **TYPICAL Symptoms/Presentation:**
  - **Systemic Disease – (Lungs, Eyes, Skin, LNs, Liver & Spleen)**
    - **General:** Fatigue, Weight Loss,
    - **Lungs:** Dyspnoea, Dry Hacking Cough
    - **Eyes:** Uveitis
    - **Skin:** Erythema Nodosum (Nodules on Shins), Lupus Pernio (Red plaques), Hypertrophic Scars
    - **LNs:** Lymphadenopathy
    - **Liver/Spleen:** Organomegaly
    - **MSK:** Arthralgia, Finger Swelling
    - **Heart:** Heart Block, Syncope, Cor-Pulmonale
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Tachypnoea
  - **Other:**
    - Dyspnoea, Cyanosis, Cachexia, Accessory Muscle Usage, Intercostal Recession, Tracheal Tug.
    - **Cough**
    - ↓Chest Expansion,
    - **Fine Pan-Inspiratory Crackles**
    - + Restrictive Pulmonary Function Tests.

### **Wegener's Granulomatosis:** (Autoimmune)

- **TYPICAL Symptoms/Presentation:**
  - Many Non-Specific Symptoms (Arthralgia, Myalgia, Night Sweats, Weight Loss, Red Eyes, URTI, Chronic Ear Infections, Fever)
  - **BUT Relevant as it can → Pneumonia:**
    - Dyspnoea
    - Cough (+/- Haemoptysis)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachycardia, Tachypnoea,
  - **Other:**
    - Cachexia, Epistaxis, Nasal Sores, Various Skin Lesions, Haematuria, Conjunctivitis, Chest Pain, Cough (+/- Haemoptysis), Dyspnoea, Weakness, Wheezing.

**Mediastinal Compression:** (Lung Ca, Lymphoma, Retrosternal Goitre, Aortic Aneurysm)

- **TYPICAL Symptoms/Presentation:**
  - Facial Plethora
  - Supraclavicular Lymphadenopathy
  - Hoarseness
  - Horner's Syndrome
  - Dyspnoea
- **TYPICAL Clinical Signs:**
  - **Other:**
    - **SVC Obstruction:** Facial Cyanosis, Facial Plethora, Positive Pemberton's, Periorbital Oedema, ↑Non-Pulsatile JVP
    - **Mechanical Compression:** Stridor (Tracheal Compression), Tracheal Deviation, Hoarseness
    - **Nerve Compression:**
    - Horner's Syndrome (Unilateral Ptosis, Anhidrosis, Miosis, Enophthalmos, Laryngeal Hoarseness)
    - Unilateral Phrenic Nerve Paralysis → Unilateral Diaphragm Paralysis → Asymmetrical Chest Expansion.
  - **Signs of Causes:**
    - ↑Thyroid Gland (Retrosternal Goitre)
    - Virchow's Node (R-Supraclavicular Lymphadenopathy) for Lung Cancer.

**Lung Cancers:**

- **TYPICAL Symptoms/Presentation:**
  - **Often Asymptomatic..But may → :**
    - Fever, Night Sweats, Weight Loss, Fatigue
    - Dyspnoea
    - Cough (+/- Haemoptysis) (+/- Wheeze if Bronchial involvement)
  - **(Don't Forget +/- Paraneoplastic Syndrome)**
    - Hypercalcaemia (↑PTH)
    - Hyponatraemia (↑ADH)
    - Cushing's (↑ACTH)
    - Carcinoid Syndrome (↑Serotonin)
    - Gynaecomastia (↑Gonadotrophins)
- **TYPICAL Clinical Signs:**
  - **Vitals:**
    - Fever, Tachypnoea (late stages)
  - **Other:**
    - **General Signs:**
      - Anorexia, Clubbing, HPOA (Wrist Tenderness), Intrinsic Hand Muscle Wasting, Virchow's Node (R-Supraclavicular Lymphadenopathy), Axillary Lymphadenopathy, Liver & Bony Tenderness (Metastases)
    - **If Apical (Pancoast) Tumour →**
      - SVC obstruction → Pemberton's Sign (Facial Plethora)
      - Sympathetic Nerve Compression → Deficiency → Horner's Syndrome (Unilateral Ptosis, Anhidrosis, Miosis, Enophthalmos, Laryngeal Hoarseness)
      - C8/T1 Nerve Lesion → Intrinsic Hand Muscle Weakness & Wasting
    - **If Mesothelioma →**
      - Pleuritic Chest Pain
      - Pleural Effusion
      - Tender Ribs
    - **If Bronchocarcinoma →**
      - Wheezing (Partial Bronchial Involvement)
      - Atelectasis (Complete Bronchial Obstruction)
      - Haemoptysis
  - **Signs of Causes:**
    - Smoker, Miner, Sand-Blaster, Builder,

## **URTI:**

- **TYPICAL Symptoms/Presentation:**
  - Fever
  - Sore-Throat
  - Runny Nose, Sneezing, Hoarseness, Cough
  - Sinus Headache
  - +/- Non-Specific Viral (Fever, Malaise, Headache, Myalgia)
- **TYPICAL Clinical Signs:**
  - **Typical URTI Symptoms:**
    - Fever, Malaise, Headache, Cough
  - **+ Specific Symptoms:**
    - *Adolescent + Sore Throat + Lymphadenopathy* = **EBV (Glandular Fever)**
    - *Toddler + Drooling Saliva* = **Epiglottitis (Haemophilus Influenza)**
    - *Photophobia + Neck Stiffness* = **Meningitis (Eg: Neisseria Meningitidis)**
    - *Arthritis + New Cardiac Murmur after 2wks* = **GAB-Strep Pharyngitis → Rh-Fever/Rh-Heart Disease**
    - *Hx of Recurrent Pneumonia + SOB* = **Cystic Fibrosis/Immunocompromised/Smoker**
    - *SOB + Weight Loss* = **Tuberculosis (Mycobacterium Tuberculosis)**
    - *Community Outbreak + Travel History* = **SARS (SARS-Associated Coronavirus)**



## **THE RHEUMATOLOGICAL EXAMINATION**

## THE RHEUMATOLOGICAL EXAMINATION

### Full Rheumatological Exam:

- Introduction, Wash Hands, Consent,
- General Inspection:
  - Cushingoid Appearance (Steroids→ Osteoporosis)



- - Bird-like Facies (Scleroderma)



- - Weight Loss (Autoimmune, Scleroderma)
  - Butterfly Rash & Hair Loss (SLE)



- **Hands:**

○ **Rheumatoid Arthritis:**

- Symmetrical PIP Synovitis (Red + Swollen + Tender)
- Ulnar Deviation
- Subluxation of MCP Joints
- Z-Deformity of the Thumbs
- Swan Neck Deformity of the Fingers
- Vasculitic Splinters in the Fingers
- Small Muscle Wasting
- Palmar Tendon Crepitus
- Carpal Tunnel (Inverse Prayer Test)



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

○ **Reiter's Syndrome:**

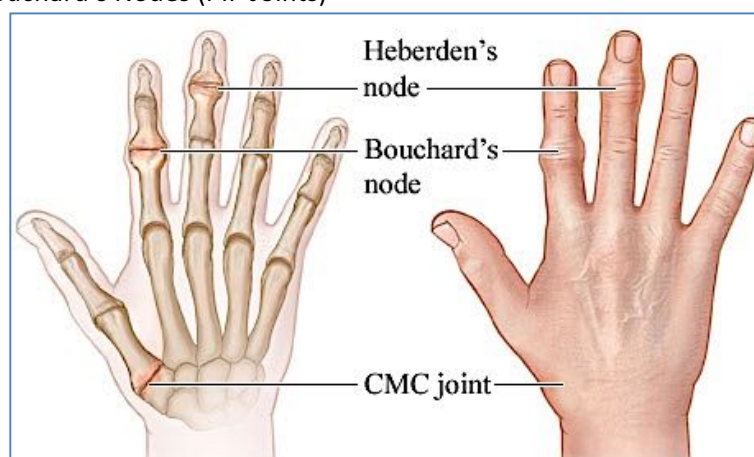
- Syphilitic-like Lesions on Soles & Palms.



CDC/ Dr. M. F. ReinTranswiki approved by: w:en>User:Dmcdevit, Public domain, via Wikimedia Commons

○ **Osteoarthritis:**

- Heberden's Nodes (DIP Joints)
- Bouchard's Nodes (PIP Joints)



<https://myhealth.alberta.ca/Health/pages/conditions.aspx?hwid=zm2488>

- **Gouty Arthritis:**

- Gouty Tophi in Hands



Arthritis Research UK Primary Care Centre, Primary Care Sciences, Keele University, Keele, UK.  
e.rodry@cphc.keele.ac.uk, CC BY 2.0 <<https://creativecommons.org/licenses/by/2.0>>, via Wikimedia Commons

- **Scleroderma (CREST):**

- Calcinosis (Calcific Nodules in Skin of Fingers)
- Raynaud's Phenomenon
- (+ Oesophageal dysmotility)
- Sclerodactyly
- Telangiectasias on the Fingers



<https://clinicalconnection.hopkinsmedicine.org/news/scleroderma-lung-disease-the-best-protocol-for-early-detection-and-treatment>

- **Arms:**

- Rheumatoid Nodules (Rheumatoid Arthritis)
- Gouty Tophi in Wrists & Elbows (Gout)
- Psoriatic Plaques on Extensor Surfaces (Psoriasis & Psoriatic Arthritis)



<https://www.health.harvard.edu/diseases-and-conditions/a-deeper-look-at-psoriasis>

- **Face:**
  - **Rheumatoid Arthritis:**
    - Dry Eyes (Sjogren's Syndrome)
    - Cataracts (Steroid Side Effect)
    - Conjunctival Pallor (Anaemia from NSAIDs → GI Bleeding)
    - Parotid Gland Enlargement (Sjogren's Syndrome)
    - Dryness of Mouth (Sjogren's Syndrome)
    - Mouth Ulcers & Gum Hypertrophy (Methotrexate Side Effects)
    - TMJ Joint Crepitus
  - **Ankylosing Spondylitis:**
    - Acute Iritis
  - **Reiter's Syndrome:**
    - Conjunctivitis
    - Iritis
  - **Gout:**
    - Gouty Tophi in the Ear
- **Neck:**
  - Cervical Lymphadenopathy (Rheumatoid, SLE)
- **Chest:**
  - **Rheumatoid Arthritis:**
    - Bony Tenderness (Particularly Spine)
    - Pleural Effusions
    - Pericardial Rub
    - Aortic Regurgitation Murmur
  - **Ankylosing Spondylitis:**
    - Lordosis + Kyphosis
    - Lumbar Tenderness
    - \*\*Markedly ↓ ROM of Spine
    - \*\*↓ Chest Expansion
    - Aortic Regurgitation Murmur
    - Sacroiliac Joint Tenderness
- **Abdomen:**
  - **Rheumatoid & SLE:**
    - \*\*Splenomegaly
    - \*\*Hepatomegaly
- **Genitals:**
  - **Reiter's Syndrome:**
    - Urethral Discharge
    - Balanitis
    - Prostatitis
- **Legs:**
  - **Rheumatoid Arthritis:**
    - Quadriceps Wasting
    - Knee Crepitus
    - Valgus (Bowing) Deformity of the Knee
    - Baker's Cysts (in Popliteal Fossa)
    - Peripheral Neuropathy (\*\*Including Foot Drop\*\*)
  - **Ankylosing Spondylitis:**
    - Achilles Tendonitis
    - Plantar Fasciitis
  - **Gouty Arthritis:**
    - Tophi in Big Toe (75% of cases)
    - Tophi in Achilles Tendon

- **Feet:**
  - **Rheumatoid:**
    - Foot Drop (Common Fibular Nerve Palsy)
    - MTP Joints Swelling & Subluxation
    - Achilles Tendon Nodules
    - Claw Toes
  - **Reiter's Syndrome:**
    - Sausage Toes
    - Plantar Fasciitis
    - Achilles Tendonitis
    - Syphilitic-like Lesions on Soles & Palms.

**Note: Limited Cutaneous Scleroderma = CREST Syndrome =**

- **Calcinosis (Calcific Skin Nodules in the Fingers)**
- **Raynaud's Phenomenon**
- **Oesophageal Dysmotility → Dysphagia**
- **Sclerodactyly of fingers**
- **Telangiectasia on fingers**



**THE PREOPERATIVE ASSESSMENT:**

## THE PREOPERATIVE ASSESSMENT:

- **Right Patient, Right Surgery, Right Side?**
  - "Time out" is done in the OR; X-References Pt info with Operation etc.
  - Ask Pt what surgery?
  - Mark the correct side.
- **Informed Consent:**
  - **General Risks:**
    - **Anaesthetic Risks:**
      - Atelectasis (Lung Collapse)
      - Pneumothorax (Lung Rupture)
      - Pneumonia
      - Aspiration
      - Mouth Injury
      - Sore Throat
      - Awareness
      - Coma
      - Malignant Hyperthermia (Rare → Rhabdomyolysis & Fever → **Usually Fatal**)
      - Anaphylaxis (Anaesthetic Reaction)
      - Cardiac Arrest
      - Death
    - **Surgical Risks:**
      - Bleeding
      - Transfusion (+/- Reaction)
      - Wound (Pain, Scars)
      - Infection
      - DVT +/- PE
      - Disability
      - Death
      - Nerve Palsy
  - **Specific Risks:**
    - Brain Damage (Neurosurgery)
    - Stroke (Carotid Endarterectomy)
    - Hypothyroidism (Thyroidectomy)
    - Hypoparathyroidism (Thyroidectomy)
    - CBD Dissection (Cholecystectomy)
    - Ileus (Abdominal Surgery)
    - Adhesions (Abdominal Surgery)
  - **Final Words:**
    - Consent is NOT a contract; It can be revoked at any time.
    - Risks are relative & depend on Pt Health & Procedure.
    - Anaesthetic Mortality  $\approx 1/200,000$  healthy pts.
    - Benefits should outweigh Risks
- **Anaesthetic Risk Assessment:**
  - **Cardiorespiratory Function**
  - **Co-Morbidities?**
    - IHD
    - Diabetes
    - Asthma
    - HTN
    - Epilepsy
    - Jaundice
    - Pregnant
  - **Allergies**
  - **Smoker**
  - **Previous Anaesthesia + any complications?**
  - **FamHx of Malignant Hyperthermia**

- **Drug Assessment:**
  - **Pt on Steroid Therapy?**
    - Pts on steroids have suppressed adrenals ∴ need extra cortisol to cope with the stress of surgery. Give **Hydrocortisone 50-100mg IV with premeds**, then **TDS for <3days**.
  - **Pt on Anticoagulants?**
    - Typically **Aspirin** is fine.
    - **Stop Warfarin >2-5days pre-op**. (Emergency Reversal with **Vitamin K +/- FFP**.)
    - **Stop Heparin 6hrs pre-op**. (Emergency Reversal with **Protamine**)
    - (Note: When re-warfarinizing, DO NOT stop Heparin until INR is Therapeutic, as Warfarin is *Pro-Thrombotic* in the early stages)
    - (Note: Avoid Epidural, Spinal & Regional Blocks)
  - **Pt on OCP/HRT?**
    - ↑Oestrogen = ↑ DVT Risk, ∴ **Stop OCP 4wks Pre-Op** (major/leg surgery); **Resume 2wks Post-Op**.
- **Pre-Operative Checklist:**
  - **Fast The Patient:**
    - **NBM <2hr Pre-Op**
    - **Clear Fluids >2 Pre-Op**
    - **No Solids >6hrs Pre-Op**.
  - **IV Cannula**
  - **Catheterise (if Necessary)**
  - **Group & Hold/Crossmatch**
    - **G&H for Moderate Surgery** (Eg: Mastectomy, Cholecystectomy)
    - **X-Match for Major Surgery** (Eg: Caesarean=2U, Gastrectomy=4U, AAA=6U)
  - **Usual Blood tests:**
    - **FBC** (Hb)
    - **U&E** (if Diabetic/on Diuretics/Burns Pt/Renal Dx/Liver Dx/Ileus/on TPN)
  - **Specific Blood Tests:**
    - **LFT** (if Jaundiced/Malignancy/ETOH Hx)
    - **Amylase** (if Acute Abdomen)
    - **Drug Levels** (Eg: Digoxin/Lithium)
    - **TFT** (if Thyroid Hx)
  - **CXR** (If Cardiac/Resp Hx, Possible Lung Mets, >65yrs)
  - **ECG** (If >55yrs/IHD/HTN/Other CVD)
  - **Book any imaging**
- **DVT Prophylaxis needed? (P580 oxford):**
  - **Graduated Compression Stockings (NOT FOR Vasculopathies!!!)**
  - **+ Heparin 5000Units SC 2hr Pre-Op (OR LMWH/Enoxaparin 20mg/d SC); then BD Post-Op until Walking**

- **Prophylactic Antibiotics:**

| <b><u>Procedure</u></b>                                                                                                                                                     | <b><u>Likely Pathogen/s</u></b>                                                                      | <b><u>Antibacterial Cover</u></b>                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>General Surgery:</b> <ul style="list-style-type: none"> <li>- Appendectomy (Non-Perforated)</li> <li>- Colorectal Surgery</li> <li>- Biliary/Duodenal Surgery</li> </ul> | Enteric G-Negs<br><br>Enteric G-Negs + G-Pos Enterococcus, Anaerobes<br>Enteric G-Negs + G-Pos Cocci | Cephalexin / Gentamicin<br><br>Cephalexin + Metronidazole<br>Cephalexin + Metronidazole |
| <b>Orthopaedic Surgery</b>                                                                                                                                                  | Staphs + Streps + G-Neg Bacilli + Anaerobes                                                          | Cephalexin / Gentamicin                                                                 |
| <b>Vascular Surgery</b>                                                                                                                                                     | Staphs + G-Neg Bacilli + G-Pos Enterococcus                                                          | Cephalexin / Gentamicin / Augmentin                                                     |
| <b>Urologic Surgery</b>                                                                                                                                                     | G-Neg Bacilli + G-Pos Enterococcus                                                                   | Cephalexin / Ciprofloxacin                                                              |
| <b>Gynaecologic Surgery:</b> <ul style="list-style-type: none"> <li>- C-Section</li> <li>- Hysterectomy</li> </ul>                                                          | Staphs + Strep + G-Pos Enterococcus<br>Enteric G-Negs + Group B Strep + G-Pos Enterococcus           | Cephalexin / Gentamicin<br>Cephalexin / Gentamicin / Ampicillin                         |

|                                       | <b><u>Egs:</u></b>                                                                                   | <b><u>Effective Antibiotics:</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>G. Positives ("-cocci")</u></b> | Enterococcus Spp.<br>Staphylococcus Spp.<br>Streptococcus Spp.                                       | <b>Penicillins</b> (Benz-Pen-G, Amoxicillin, Ampicillin, Flucloxacillin)<br>1. (Note: <b>Augmentin</b> for $\beta$ -lactamase resistant bacteria = Amoxil + Clavulanate)<br><b>Cephalosporins</b> (Ceftriaxone <sup>3</sup> , Cefipime <sup>4</sup> , Cephalexin <sup>4</sup> )<br><b>[Vancomycin (For resistant G-Pos/ if Penicillin Allergy)]</b>                                                                                                                 |
| <b><u>G. Negatives</u></b>            | E. Coli<br>Neisseria Spp.<br>Pseudomonas<br>Haemophilus Spp.<br>Klebsiella Spp.<br>Enterobacter Spp. | <b>Aminoglycosides</b> (Gentamicin, Tobramycin, Streptomycin)<br>2. (Note: Used with Penicillins/Cephalosporins for <i>Synergy</i> )<br><b>Tetracyclines</b> (Tetracycline, Doxycycline)<br><b>Macrolides</b> (Erythromycin, Azithromycin)<br><b>Quinolones</b> (Ciprofloxacin, Norfloxacin)<br><b>Cephalosporins</b> (Ceftriaxone <sup>3</sup> , Cefipime <sup>4</sup> , Cephalexin <sup>4</sup> )<br><b>[Benz-Pen-G (For Neisseria Gonorrhoeae/Meningitidis)]</b> |
| <b><u>Anaerobes</u></b>               | Bacteroides Spp.<br>Clostridium Spp.                                                                 | <b>[Metronidazole (For Bacteroides)]</b><br><b>[Vancomycin (For Clostridium Difficile)]</b>                                                                                                                                                                                                                                                                                                                                                                         |
| <b><u>Atypicals</u></b>               | Mycoplasma<br>Legionella                                                                             | <b>Tetracyclines</b> (Tetracycline, Doxycycline)<br><b>Macrolides</b> (Erythromycin, Azithromycin)                                                                                                                                                                                                                                                                                                                                                                  |

**Note: Triple Therapy: –Ampicillin, Gentamicin, Metronidazole- give great 'Broad Cover'. (Remember by AGM – Annual General Meeting...If you want to be around next year, take these 3)**

## **Cardiovascular Pre-Op Assessment**

- **Assess the Pump (Heart):**
  - **Power (Myocardium)**
  - **Valves (Stenosis/Regurgitation)**
    - Note: Stenosis is worse than Regurgitation because they cap the CO under load/stress.
  - **Piping (Vessels)**
  - **Control (Conduction)**
- **CVS Systems Review:**
  - **Cardiac Failure:**
    - Dyspnoea
    - PND
    - Orthopnoea
    - Peripheral Oedema/Ascites
  - **IHD:**
    - Angina (Stable/Unstable)
    - Prev. MI
    - FamHx of IHD
  - **Valve Disease:**
    - Hx of Rheumatic Fever
    - Old Age
  - **PVD:**
    - Claudication
    - Smoker?/Diabetic?
    - Dizziness/Blackouts/Prev. TIA
  - **Conduction Deficits:**
    - Palpitations
    - Arrhythmias
- **CVS Examination:**
  - **Signs of Failure:**
    - Basal Crepitations (LVF)
    - Peripheral Oedema/Ascites/Organomegaly (RVF)
  - **Signs of Structural Abnormality:**
    - Displaced Apex (Dextrocardia/Cardiomegaly)
    - Parasternal Heave
    - Murmurs/Thrills
    - Previous Surgery
  - **Signs of PVD:**
    - Peripheral Pulses
    - CRT
    - Ulcers (Arterial/Venous)
  - **Signs of Arrhythmias:**
    - Irregular Pulse
    - Rapid Pulse
    - ECG
- **Medications:**
  - **Diuretics** – because they ↓K<sup>+</sup> & ↑Mg<sup>+</sup>
  - **Antihypertensives**
  - **Antiarrhythmics**
  - **Anticoagulants** – esp. Dabigatran (Irreversible)
  - **Antibiotics**

**THE EAR HISTORY & HEARING TESTS:**



## THE EAR HISTORY & HEARING TESTS:

### The Interview – Ear Related History

#### **1. Presenting Symptoms**

- Pain,
- Discharge
- Itching,
- Vertigo
- Tinnitus
- Changing in Hearing

#### **2. History of Presenting Symptoms**

- Onset – weeks day, months
- Contributing factors – trauma, recent illness, exposure to sudden loud noise
- Which ear effected
- Does anything make it worse, does anything make it better.

#### **3. Medical History – Note specifically**

- Hypoxia at birth
- Low birth weight – less than 1500g
- Congenital abnormalities of the face or skull
- Maternal drug use
- Head trauma
- Ear trauma
- Genetic disease – Meniere's Disease
- Chronic and acute infections – ear, sinus, colds, flu

#### **4. Surgical History – Note specifically**

- Ear surgery – grommets etc
- Sinus surgery
- Head and facial surgery
- Neurosurgery

#### **5. Medications: Many medications can be ototoxic. Note specifically:**

- Large doses of Aspirin
- Gentamicin Sulphate
- Aminoglycosides

#### **6. Social History – Note specifically:**

- Exposure to loud noises at work, availability and use of PPE
- Exposure to Industrial noise
- Exposure to recreational noise

#### **7. Family History – Note Specifically**

- Family history of hearing loss
- Genetic disorders

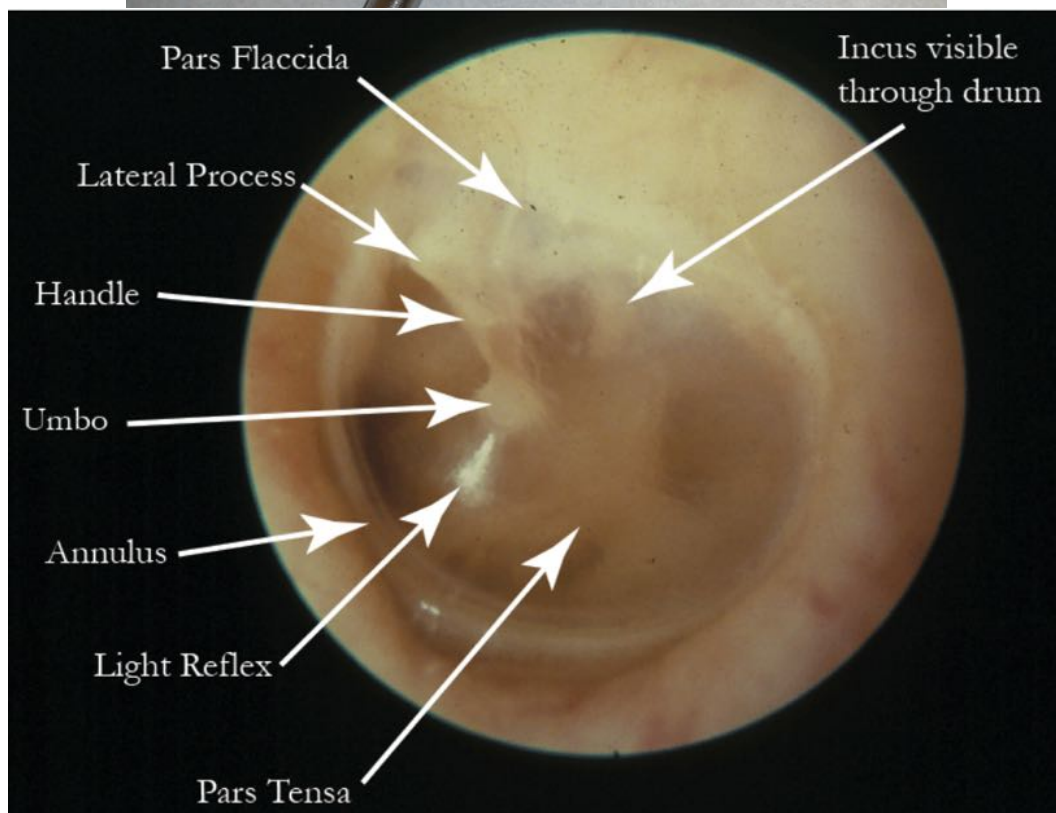
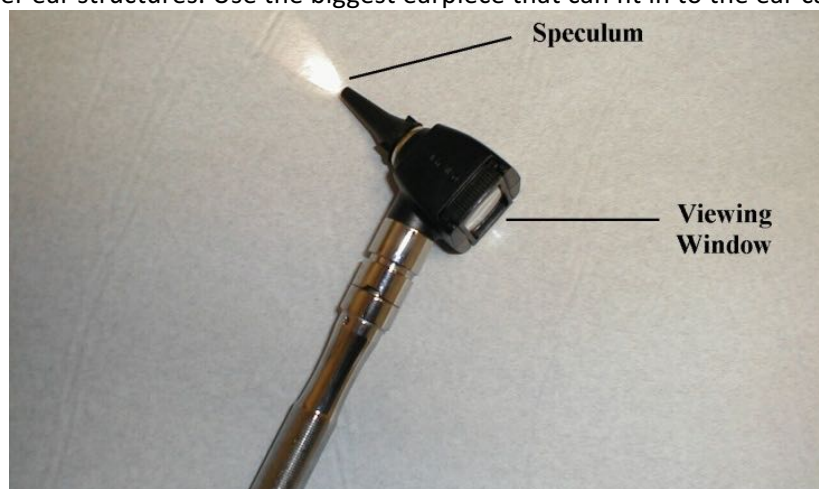
## EXAMINATION OF THE EAR

### External structures:

- Briefly examine the outer structures
  - Colour and texture of the skin
  - Any moles, cysts, nodes, or deformities
- Any skin changes suggestive of cancer (Eg: basal cell, melanoma,
  - Any signs of swelling and/or redness
  - Discharge – colour and odour
- Palpate the ear for pain, tenderness, skin texture and lesions.
  1. If pain increases – external ear infection likely
  2. No pain increase – middle ear infection likely
  3. Tenderness in mastoid area – possible mastoiditis

### External Auditory Canal and Tympanic Membrane:

- **The Otoscope:**
  - The otoscope allows you to examine the external canal,, as well as the tympanic membrane and a few inner ear structures. Use the biggest earpiece that can fit in to the ear canal.



<https://www.enteducationswansea.org/normal-ear-anatomy>

### Examination procedure:

#### **Always examine the unaffected ear first.**

1. Place the tip of the specula in the opening of the external canal.
2. Gently grasp the top of the left ear with your left hand and pull up and backwards. This straightens out the canal, allowing easier passage of the scope.
3. Look through the viewing window with either eye. Slowly advance the scope, heading a bit towards the patient's nose but without any up or down angle.
4. As you advance, pay attention to the appearance of the external canal. In the setting of infection, called otitis externa, the walls become red, swollen and may not accommodate the speculum. In the normal state there should be plenty of room.

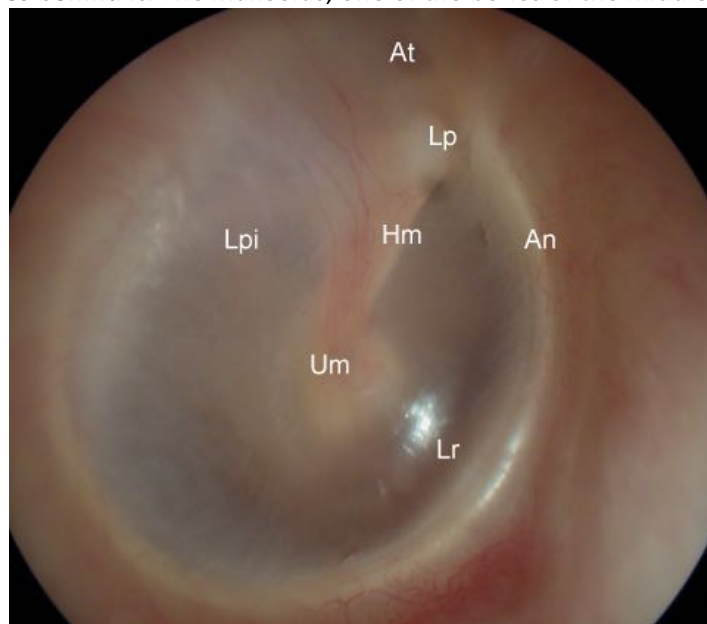


### The Tympanic Membrane:

- **The tympanic membrane (a.k.a. ear drum) should be visible.**

#### **Pay particular attention to:**

- a. **The colour:** When healthy, it has a greyish, translucent appearance.
- b. **The structures behind it:** The malleolus, one of the bones of the middle ear, touches the drum.



An *annulus fibrosis*

Lpi *long process of incus* - sometimes visible through a healthy translucent drum

Um *umbo* - the end of the malleus handle and the centre of the drum

Lr *light reflex* - antero-inferiorly

Lp *Lateral process of the malleus*

At *Attic* also known as *pars flaccida*

Hm *handle of the malleus*

- a. In the setting of infection within the middle ear, the drum becomes diffusely red and the light reflex is lost.
- b. fluid collecting behind the drum. This is called a middle ear effusion and can cause the drum to bulge outwards.

### Detecting Conductive v. Sensorineural Deficits:

1. **Conduction:** The passage of sound from outside to the level of the 8th cranial nerve. This includes transmission of sound through the external canal and middle ears.
2. **Sensorineural:** The transmission of sound through the 8th nerve to the brain.

Hearing loss can occur at either level. To determine which is affected, the following tests are performed:

### Weber Test:

- 1) Grasp the tuning fork by its stem and place the stem towards the back of the patient's head equidistant from either ear. The bones of the skull will transmit this sound to the 8th nerve, which should then be appreciated in both ears equally.
  - a. If there is a conductive deficit (Eg: wax in the external canal), the sound will be heard better in that ear. This is because impaired conduction has prevented any competing sounds from entering the ear via the normal route.
  - b. In a sensorineural abnormality (Eg: an acoustic neuroma, a tumor arising from the 8th CN), the sound will be best heard in the normal ear.
- 2) If sound is heard better in one ear it is described as lateralizing to that side. Otherwise, the Weber test is said to be mid-line.



DrSaltMD, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

### Rinne Test:

- 1) Place the stem on the mastoid bone and instruct the patient to let you know as soon as they can no longer hear the sound.
- 2) Then place the tines of the still vibrating fork right next to, but not touching, the external canal. They should again be able to hear the sound. This is because transmission of sound through air is always better than through bone.
- 3) This will not be the case if there is a conductive hearing loss which causes bone conduction to be greater than or equal to air.
- 4) If there is a sensorineural abnormality (Eg: medication induced toxicity to the 8th CN), air conduction should still be better than bone as they will both be equally affected by the deficit.



<https://www.mitchmedical.us/physical-diagnosis/the-rinne-test.html>

## **Ear, Nose & Throat Examination:**

- **Introduction, Wash Hands, Consent,**
- **General Inspection:**
  - Hair distribution (PCOS, Hypothyroidism)
  - Hair Infestations (nits, lice)
  - Oily, sweaty hair (Hyperthyroidism, Acromegaly)
  - Thin, Dry Hair (Hypothyroidism)
  - Scars
  - Skull Deformities (Injuries, Congenital, Cleft Lip)
  - High Arch Palate (Marfan's)
  - Skin lesions (SCCs, BCCs)
  - Skin Pigmentation (Addison's, Haemochromatosis, Acanthosis Nigricans in Diabetes/Cushings/PCOS/Acromegaly)
  - Skin Scaling (Psoriasis, Autoimmune Disorders)
  - Abnormal Facies – (Cushing's, Acromegaly, Addison's, Graves Hyperthyroidism, Hashimoto Hypothyroidism, Downs Syndrome, Turner's Syndrome(Female).)
  - Facial Symmetry (Facial Nerve Palsies)
  - Facial Muscle Fasciculations
  - Eye Inspection (Conjunctival Pallor, Scleral icterus, Conjunctivitis, Discharge, Periorbital Oedema)
- **Vital Signs:**
  - **Pulse:**
    - Tachycardia (Infection); Bradycardia (if ↑ ICP)
  - **Blood Pressure:**
    - Hypertension (if ↑ ICP)
  - **Respiratory Rate:**
    - Bradypnoea (if ↑ ICP)
  - **Temperature:**
    - Fever (Infection)
- **Ears:**
  - **External Inspection:**
    - Redness (Infection)
    - Swelling (Gouty Tophi)
    - Scars
    - Lesions (SCC/BCC)
    - Discharge
  - **Palpation:**
    - Tug Test (for External Ear Infection)
    - Cervical Lymphadenopathy (Infection, Malignancy)
  - **Otoscopy:**
    - Explain to the patient that it may feel uncomfortable
    - Pull Pinna Back and Upward to straighten the canal
    - Examine the Left Ear with the Left hand holding the Otoscope (and vice versa)
    - **Look for:**
      - Inflammation of External Ear Canal (Otitis Externa)
      - Discharge, Blood, Pus, CSF
      - Light Reflex of Tympanic Membrane (Eardrum)
      - Inflammation of Eardrum (Otitis Media)
      - Bulging of Eardrum (Otitis Media)
      - Visible Malleus (Normal)
      - Identify Borders (Pars Flaccida, Pars Tensa, Umbo)
      - Sclerosis (Chronic Otitis Media)
      - Tympanic Membrane Rupture
      - Impaction by Cerumen (earwax)
  - **Hearing:**
    - **Whisper Test** – “69 & 100” – Whilst distracting the other ear with your hand.
    - **Weber's Test** – Lateralising? (Conductive Deafness)
    - **Rinne's Test** – Normally Air Conduction is better than bone conduction.

- **Nose:**

○ **Visual Inspection:**

- Redness (Rosacea, SLE, Acne)
  - Enlargement (Rosacea, Acromegaly)
  - Skin Lesions (Comedones, BCCs, SCCs)
  - Scars, Deformities (Broken Nose)
  - Discharge, Pus, Blood (Epistaxis)
  - Nasal Deviation
  - Nasal Polyps (inflammatory from chronic URTI)
  - Septal deformities
  - Enlarged, Inflamed Turbinates (URT)
- **Palpation:**
- Tenderness
  - Swelling
  - Deformities
  - Block Each nostril & Assess for Obstruction
  - **Sinuses:**
    - Press/Percuss over Frontal Sinuses (Tenderness = Sinusitis/Congestion)
    - Press/Percuss over Maxillary Sinuses (Tenderness = Sinusitis/Congestion)
    - Transilluminate Maxillary Sinuses (Shine otoscope through Maxillae & observe illumination in the mouth)

- **Throat:**

○ **Visual Inspection:**

- Hydration
- Central or Peripheral Cyanosis
- Mouth Ulcers (UC/Cohn's/SCC/BCC)
- Leucoplakia/Erythroplakia (Smoking – Precancerous)
- Buccal Petechiae (Leukaemias, Bleeding Disorders, DIC, Haemorrhagic Infections)
- Buccal Pigmentation (Addison's Disease)
- Angular Stomatitis (Microcytic OR Macrocytic Anaemias)
- Atrophic Glossitis (Microcytic OR Macrocytic Anaemias)
- Gum Hyperplasia (AML, Methotrexate)
- Bleeding Gums (AML)
- Dentition & Dead Teeth (CVS Risk Factor)
- High Arched Palate (Marfan's Syndrome)
- Inflamed, Enlarged Tonsils
- Tonsillar Exudate
- Inflamed Pharynx (URT)
- "Say Ah" – (Glossopharyngeal & Vagus Nerve Lesions)
- Note any Fotor/Bad Breath (Uraemic, Hepatic, Acetone)

○ **Palpation:**

- Bimanual Palpation for enlarged Parotid Gland (Alcoholism, Sialolithiasis)
- Feeling Tongue & Buccal Mucosa for Lumps/Tenderness/Ulcers
- **+ Cervical Lymph Nodes**

- **Thank patient "that concludes my examination", I'll go organise further tests.**

**THE EYE EXAMINATION:**



## THE EYE EXAMINATION:

**History:** – (at least 80% of the diagnosis comes from the history)

- **Take a meaningful, focussed history.**
- **Presenting complaint**
  - **Nature of Presenting complaint:**
    - **1: Abnormal Vision?** (Eg: Flashing lights/blind spot/etc)
    - **2: Abnormal Sensation?** (Eg: Itching/throbbing/pain)
    - **3: Altered Appearance?**
  - **Time course**
    - Sudden onset/Rapid Onset?
    - Duration
    - Exacerbating (What makes it worse)
    - Relieving (what makes it better)
    - Associated (What is it associated with)
- **Past History**
  - Meds / allergies
  - Eg: Are they diabetic
  - Eg: What environmental insults (Eg: Welder)
- **Family history**
- **Social history**

### Possible Presenting Complaints:

#### 1: Abnormal vision

- **A) Reduced vision: (Note: If Sudden → *Probably* Vascular in Origin)**
  - **Central Loss:**
    - **Far/Near/Both**
  - **Peripheral Loss:**
    - **Partial/Total loss**
    - **Eg: Scotoma:**
      - An area of degenerated visual acuity in one's field of vision, which is surrounded by a field of normal vision.



- **Hemianopia (bilateral):**
  - Type of partial [blindness](#) where vision is missing in the outer half of both the right and left visual field.
  - Usually associated with Optic Chiasm lesions.



- **Impaired Night vision (Ie: “Night Blindness”):**
  - (a *symptom* of several eye diseases) – Eg: From Vit-A Deficiency.
  - A condition making it difficult or impossible to see in relatively low light.
- **Colour Blindness:**
  - 8%males 0.5% females
  - (X-Linked Recessive)

- **B) Floaters:**

- Haemorrhage
- Inflammation
- Vitreous degeneration
- Muscae volitantes (Small Spots)



- **C) Flashers:**

- Due to Irritative stimulation of retinal or visual pathway (no sensation)
  - **Unilateral (retinal)**
  - **Bilateral (Visual Pathway)(Eg: From migraine/basilar artery insufficiency)**



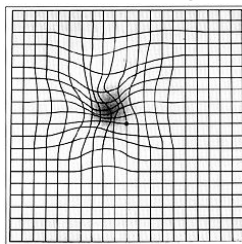
- **D) Haloes**

- **Rainbow coloured rings around lights**
  - Due to Diffraction
  - Corneal oedema
  - Can suggest Increased intraocular pressure



- **E) Metamorphopsia / Micropsia:**

- **Metamorphopsia** = Apparent distortion of straight lines



- **Micropsia** = Objects are perceived to be smaller than they actually are
- **Possible Causes:**
  - **Retinal oedema** → Require urgent referral
  - **Macular degeneration**

- **F) Diplopia (Double Vision):**

○ **Binocular Diplopia:**

- When axes of both eyes are not directed to the same object
- **Causes:**
  - Muscle Weakness (Eg: Myasthenia)
  - Nerve – Eg: CN-III with ptosis mydriasis
    - Eg: CN-VI with head injury raised IC pressure

○ **Monocular Diplopia:**

- Patient sees double when viewing with only one eye.
- **Possible Causes:**
  - Corneal Surface Deformity
  - Structural Defect in Eye
  - Lesion in Visual Cortex
  - Sub-Luxation of Lens



**2: Abnormal Sensation:**

- **A) (Stinging/scratching pain) Foreign body sensation:**

- **Causes include:**
  - Entropion & Trichiasis (Eyelid folds inward → Eyelashes touching cornea)
  - Conjunctivitis
  - Xerophthalmia (dry eye)
  - Can be minimal in IOFB (Intra-Ocular Foreign Body)
- **Local anaesthetic relieves it.**

- **B) (Achy Pain):**

- Eg: Photophobia (excessive sensitivity to light and the aversion to sunlight)
- Eg: Iritis (Inflammation of Iris)
- Eg: Keratitis (Corneal Inflammation)

- **C) (Severe Deep Pain):**

- Eg: Acute Closed Angle Glaucoma → requires Pupil Constriction to open the Canal of Schlemm.
- Eg: Herpes Zoster infection of the Eye.

- **D) Asthenopia (eye strain):**

- **After intensive use of eyes**
- **Due to:**
  - **Inadequately corrected refractive error**
  - **Heterophoria** (motion of the eyes are not parallel to each other)

- **E) Watery eyes / discharge**

- **Overproduction of tears (ocular irritation/ FB)**
- **Faulty drainage of tears**
- **Instability of tears (Tears run down face instead of staying in the eyes → Dry Eyes)**

### **3: Altered Appearance**

#### **- Proptosis (Eye Dislocation)**

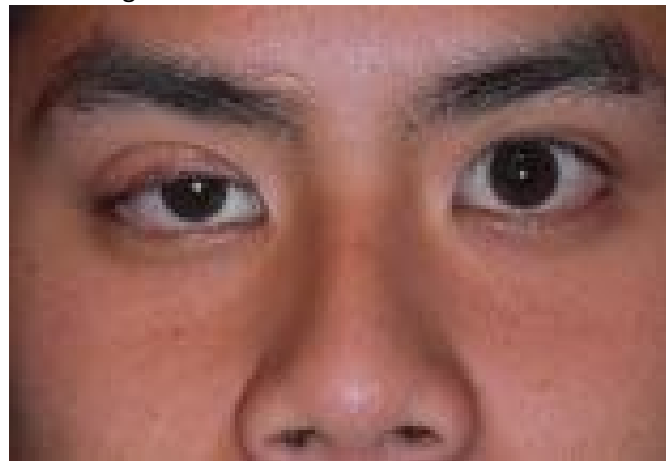
- From Orbital Infection, trauma, mass
- Eg thyroid



<http://kellogg.umich.edu/theeyeshaveit/common/proptosis.html>

#### **- Ptosis ('small' eye)**

- CN-III nerve lesions
- Levator abnormalities
- Local lid abnormalities eg infections



Andrewya, Public domain, via Wikimedia Commons

#### **- Lid retraction**

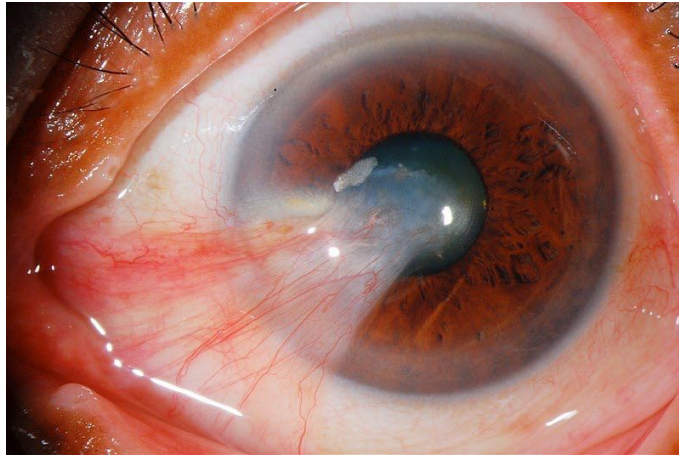
- Where the lid retracts or moves away from the surface of the eye.



Jonathan Trobe, M.D. - University of Michigan Kellogg Eye Center, CC BY 3.0  
<<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons

- **Lesions**

- On lids
- On globe eg **pterygium**



Jmvaras, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- **Diffuse redness**



- **Ciliary injection**



- **Corneal/Iris Opacities:**





## Eye Examination:

- **Introduction, Wash Hands, Consent,**
- **General Inspection of Patient:**
  - Guide Dog
  - Walking Aids
  - Visual Aids/Glasses/Eye Patch
- **Vital Signs:**
  - **Pulse:**
    - Particularly Irregularity (AF can → TIAs → Transient Visual Loss **" Amaurosis Fugax "**)
  - **Blood Pressure:**
    - Particularly Hypertension (Hypertensive Retinopathy)
  - **Respiratory Rate:**
  - **Temperature:**
    - Fever in Infection
- **Eye Examination:**
  - **Visual Acuity:**
    - **Snellen's Charts**
    - Unilateral Corrected + Bilateral Corrected
    - If Poor Visual Acuity – Use a Pinhole to see if it is a REFRACTIVE ERROR?
    - Blind Spot (Enlarged in Macular Degeneration or Papilloedema)
  - **Colour Vision:**
    - **Ichihara Charts** (Colour Blindness)
  - **Visual Inspection:**
    - **Eyelids:**
      - Lid Retraction (Graves Hyperthyroid)
      - Lid lag (Graves Hyperthyroid)
      - Ptosis (Facial Nerve Palsy, Horner's Syndrome)
      - Periorbital Oedema (Nephrotic Syndrome, Hashimoto Hypothyroidism, Allergies)
      - Xanthelasma (Liver Disease, Diabetes, CVD, Cushing's, Acromegaly, Hypothyroidism)
      - Skin Lesions (SCCs, BCCs)
      - Eyelashes (Trachoma)
    - **Eyeball:**
      - Exophthalmos/Proptosis (Grave's Hyperthyroid, Leukaemias, Head Injury, Cushing's)
      - Enophthalmos (Horner's Syndrome, Dehydration)
    - **Conjunctiva:**
      - Conjunctival Pallor (Anaemia)
      - Scleral Icterus (Liver Disease)
      - Subconjunctival Haemorrhages (Extreme Coughing)
      - Conjunctivitis (Inflammation + Pus)
    - **Iris, Cornea & Lens:**
      - Arcus Senilis (Sign of CVD Risk factors)
      - Copper Ring around the Iris (Wilson's Disease)
      - Horner's Syndrome (Unilateral Miosis [pinpoint pupils])
      - Band Keratopathy (Hypercalcaemia, Hyperparathyroidism, Renal Failure)
      - Cataracts (Diabetes, Hypertension, Iodine Deficiency ∴ Hypothyroidism)
      - Corneal Ulceration (Herpes, Trachoma)
      - Pus/Blood in Anterior Chamber
  - **Pupil Reactivity To Light:**
    - **Pupil Size** (normally 2.5mm-3.5mm)
    - **Direct Response** (If absent = Afferent *OR* Efferent Nerve Lesion in THAT Eye)
    - **Consensual Response** (If absent = Efferent Nerve Lesion in THAT Eye)
    - **Marcus-Gunn Pupil** (Swinging Torch Test – both pupils should stay the same after a few oscillations due to the Consensual Response) (If one pupil dilates whilst the torch is on the other eye, then that is a Marcus-Gunn Pupil)

- **Visual Fields (Remove Glasses & in Confrontation Position + Patient covers one eye):**
  - Bitemporal Hemianopsia = Optic chiasm compression (Pituitary Adenoma)
  - Homonymous Hemianopsia = Optic Radiation Lesion
  - Unilateral Hemianopsia = Optic Nerve Lesion
  - + **Blind Spot** (If enlarged – May = Macular Degeneration, or Papilloedema)
- **Extraocular Movements:**
  - Oculomotor = Superior Rectus, Medial Rectus, Inferior Rectus, Inferior Oblique
  - Trochlear = Superior Oblique (Down & Out) - ∴ In III-Nerve Palsy, eye faces Down & Out.
  - Abducens = Lateral Rectus
  - + **Nystagmus** = Points to the side of a Cerebellar Lesion.
- **Fundoscopy:**
  - Papilloedema
  - Macular Degeneration
  - Diabetic Retinopathy
  - Hypertensive Retinopathy
  - Roth's Spots in Infective Endocarditis.

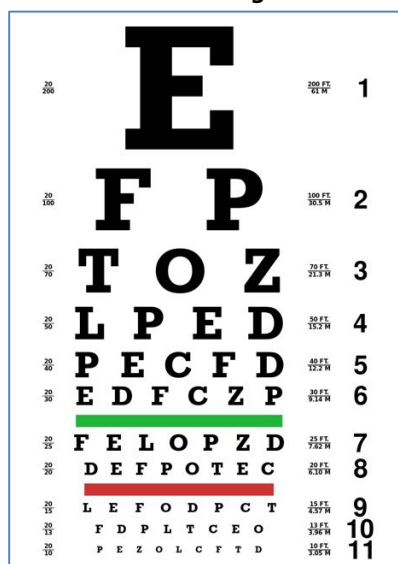
**“Thankyou, that concludes my Examination”**



## TESTING THE VISUAL SYSTEM:

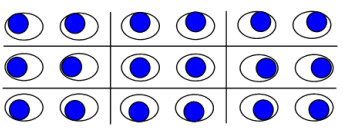
### 1: Visual Acuity Tests (Snellen's Chart):

- All Australian charts are read at 6m (Or are standardised to 6m)
- Tests 'Corrected' visual acuity @ a distance. (Those with 'corrected' vision use their glasses)
- Each eye is tested Separately; Followed by vision with both eyes open.
  - Record results for:
    - Right Eye
    - Left Eye
    - Both Eyes
  - The Results are Fraction:
    - Numerator = the standardised distance that letter should be read at
    - Denominator = The distance that it should be visible from with perfect vision
- Try using a Pinhole to Improve Visual Acuity (A pinhole eliminates the need for a lens)
  - If acuity improves with pinhole, it is a refractive error;
  - If acuity doesn't improve, it is a retinal problem.
- IF they can't read the Snellen's chart (even from up close), you need to ask them whether they can see:
  - CF (count Fingers)
  - HM (Hand movements)
  - PL, NPL (Perception of Light/nil perception of light)(Shine a light in their eye)
- IF testing for Presbyopia – use a 'Near-Reading Chart'.



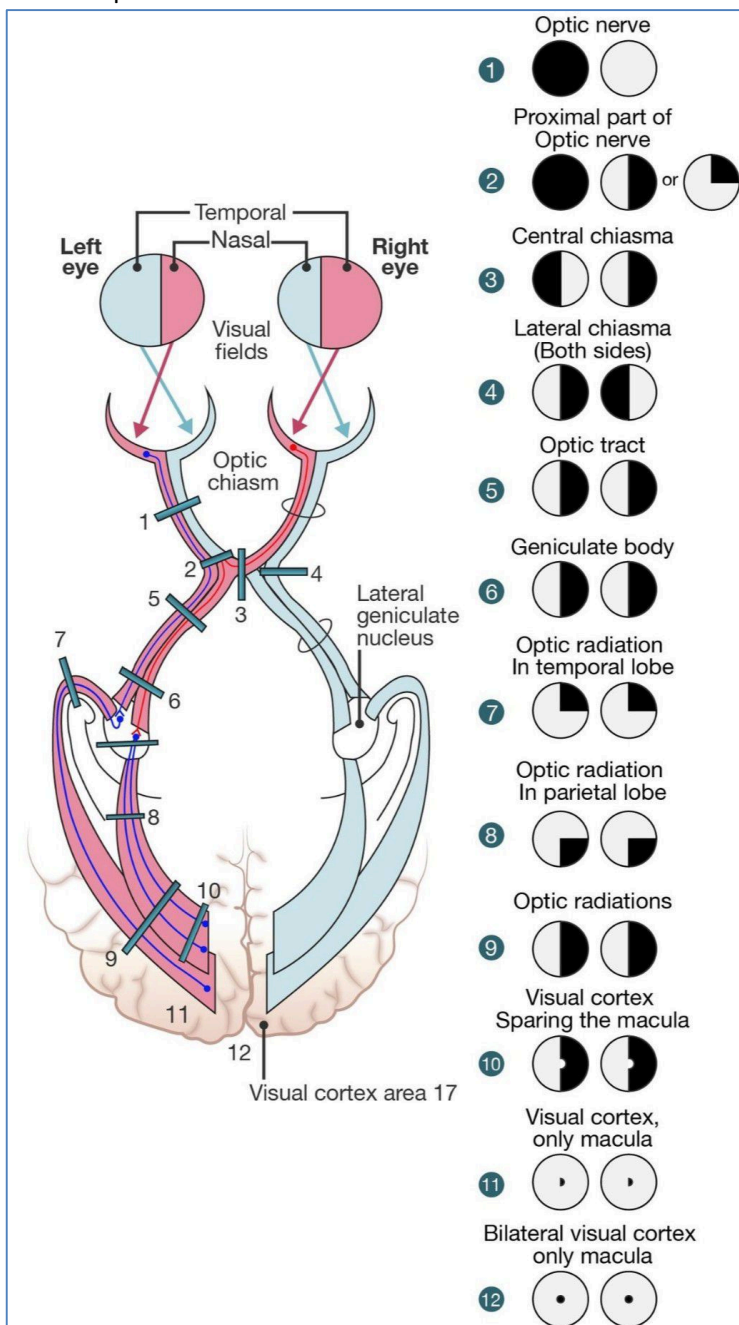
### 2: Testing Extraocular Muscles (Eye Movement):

- Start with the target 6inches in front of your partner; Then move your target slowly and smoothly in an wide "H" pattern.
  - Look for **Conjugate Movement**:
    - Movement of both eyes in a coordinate manner
  - Look for **Nystagmus**:
    - Rhythmic, involuntary oscillation of the eyes.
- Is movement Comitant/Incomitant?
  - (To the eyes move parallel to each other?)
  - (If there's strabismus, is the same angle of misalignment maintained in all directions?)
- Note: Deviation greatest when move towards field of action of involved muscle.
- Test Convergence for near Vision.

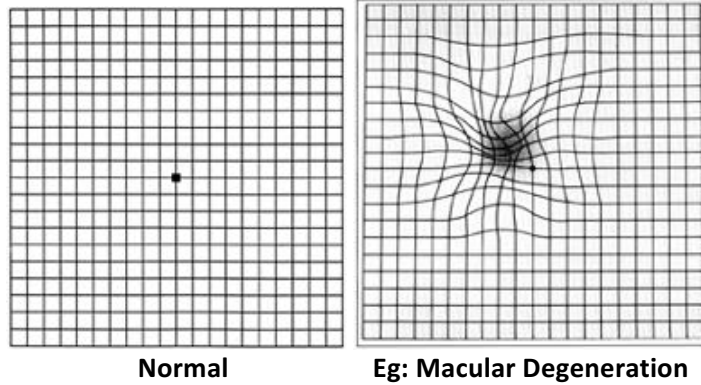
| Name             | Action                               | Controlling cranial nerve |  |         |           |
|------------------|--------------------------------------|---------------------------|---------------------------------------------------------------------------------------|---------|-----------|
| Lateral rectus   | Moves eye laterally                  | VI (abducens)             | right up                                                                              | up      | left up   |
| Medial rectus    | Moves eye medially                   | III (oculomotor)          | right                                                                                 | primary | left      |
| Superior rectus  | Elevates eye and turns it medially   | III (oculomotor)          | right down                                                                            | down    | left down |
| Inferior rectus  | Depresses eye and turns it medially  | III (oculomotor)          |                                                                                       |         |           |
| Inferior oblique | Elevates eye and turns it laterally  | III (oculomotor)          |                                                                                       |         |           |
| Superior oblique | Depresses eye and turns it laterally | IV (trochlear)            |                                                                                       |         |           |

### - 3: Visual field Test:

- Pt stares at your nose.
- **A) Testing Peripheral Visual Field:** Use your fingers to test vision in peripheries of each quadrant.  
(Best to use a red-topped bottle/pin)
  - Sit in front of Pt with eyes at the same level, about an arm's length away.
  - Cover your left eye.
  - Ask your Pt to cover their Right eye and to stare into your open eye.
  - Compare your partner's peripheral visual field to your own:
    - Hold up your finger midway between yourself and the Pt (so it is just out of your field of vision).
    - Move your finger slowly towards the centre asking the patient when they first see it.
    - Test all four quadrants of the visual field (upper right, upper left, lower right, lower left).
- Repeat for the right eye.
- **Remember that the visual field and the retina have an inverted and reversed relationship:**
  - Upper field on the inferior retina
  - Lower field on the superior retina
  - Nasal field on temporal retina
  - Temporal field on nasal retina.

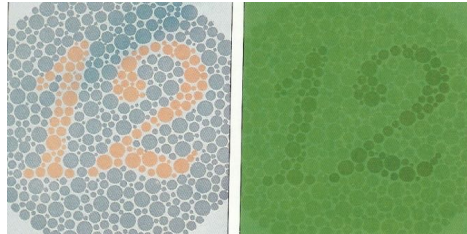


- **B) Testing Central Visual Field:** Use an Amsler Grid to test Patient's *Central* Visual Field. (Pts with macular disease may see wavy lines/missing lines)



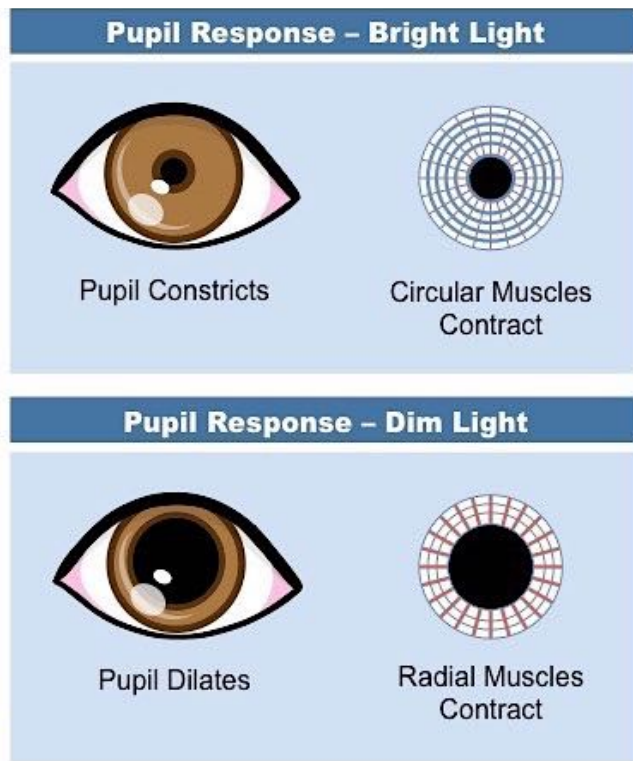
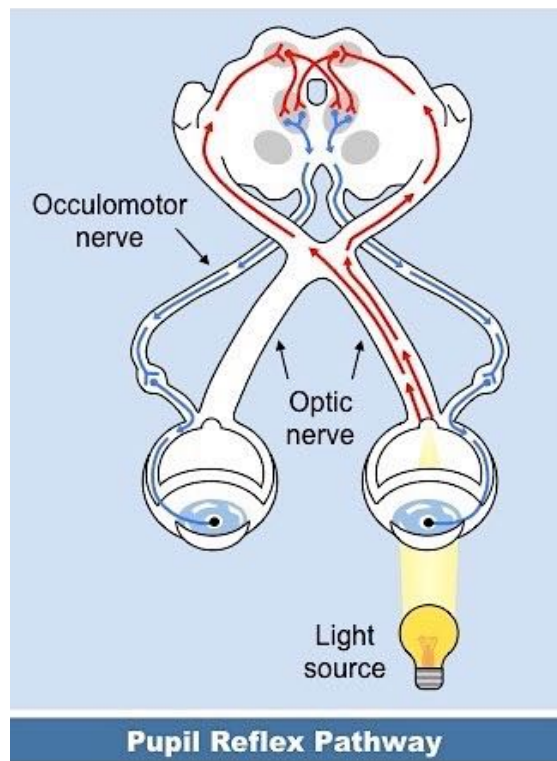
- **4: Colour vision (Ishihara Tests):**

- Tests for colour blindness
- ∴ Tests Cone Receptors.
- Note: Men are Most Affected by Colour Blindness:
  - Because it is *Sex-Linked Recessive* (on the X-Chromosome), and therefore acts dominantly when inherited in males (who only have 1x X-Chromosome).
  - Red-Green photoreceptor disorders are most common.
- People with normal colour vision can see numbers in the pics below:



- **5: Pupillary Reflexes:**

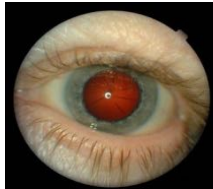
- (Note: The *constriction* reflexes are Parasympathetic-Mediated)
- **PERLA is Pupils Equal Round Reactive to Light and Accommodation.**
- A bright source of light is used to test pupillary reflexes by swinging the light in front of right eye 2 times, while observing 1<sup>st</sup> right pupil and then the left. Test is then repeated for left eye.
  1. Dim lights.
  2. Ask Pt to stare at a target in the distance.
  3. Turn on the light and look at the **right pupil** as you swing the light over the **right eye**. Watch as the pupil constricts.
    - **This is the direct response.**
  4. Repeat the same procedure again but this time look at the **left pupil** as you swing the light over the **right eye**. Does it also constrict?
    - **This is the consensual response.**
  5. Now repeat the procedure with the **left eye**. First looks at the left pupil as you shine the light on the left eye, then look at the right pupil.
- **Direct:**
  - Eg: Right Pupil contracts when Right Eye exposed to Light.
- **Consensual:**
  - Eg: Left Pupil Contracts when Right Eye exposed to Light.
- **Testing Accommodation-Mediated Pupillary Constriction:**
  - (Pupil Constriction Associated with the Accommodation)
  - 1. Ask your Pt to hold their finger about 8 inches from their nose.
  - 2. Ask your Pt to look in the distance and then look at their finger.
  - 3. Watch for convergence (crossed eyes) and pupillary constriction.
- **Note: Corneal Opacities/Cataracts can Affect the Pupillary Response:**
  - → Decreased pupillary response in both the affected eye (Direct response) and the other eye (Consensual response) because less light falls on the retina of the affected eye.



Unattributable

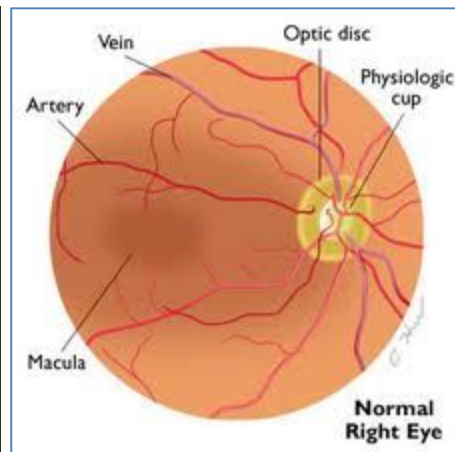
### **Ophthalmoscopy Technique:**

- Ask pt to look straight ahead both eyes open
- Set to zero
- Use right hand and right eye for patient's right eye, and left ditto for left eye
- Hold sight hole as close as possible to your eye, steady it against your nose
- Gently raise lid with your thumb
- **Find red reflex (the reddish-orange reflection from the eye's retina when using an ophthalmoscope)**
  - o Stand at arm's length and focus on the pupillary area with ophthalmoscope
  - o Should be able to see an uniform orange glow
  - o This is due to light being reflected from the choroidal vessels
  - o Look for any opacity in the red reflex
  - o (Abnormal or absent reflex especially in children could mean sight threatening condition. Need to be referred and treated quickly otherwise vision fails to develop)



- Follow it in at about 15 degrees temporal to line of vision
- Focus
- Find a blood vessel
- Follow it to the disc
- **Systematically examine**
  - o **Vessels / Macula / Vessels**
  - o **Optic Disc**
    - Colour (normal pink)
    - Margins – Well defined
    - Cup:Disc Ratio. Normal is less than half of the disc diameter. (If ratio is increased → Probably Glaucoma)
  - o **Retina:**
    - Look for:
      - Haemorrhages/Exudates/New Blood Vessels/Arteries/Veins (Bigger & Darker)
    - Common causes of retinal changes:
      - Diabetic Retinopathy (leading cause of blindness in <60yrs)
      - Age Related macular degeneration (Leading cause of blindness in >60yrs)
      - Hypertensive retinopathy
      - Retinal Artery Occlusion
      - Retinal Vein occlusion
  - o **Macula:**
    - Darker than rest of retina (Due to pigment and thick ganglion cell layer)
    - Temporal to the disc
    - Centre of macula is the fovea.
      - No blood vessels overlying the fovea

**Normal Fundus**

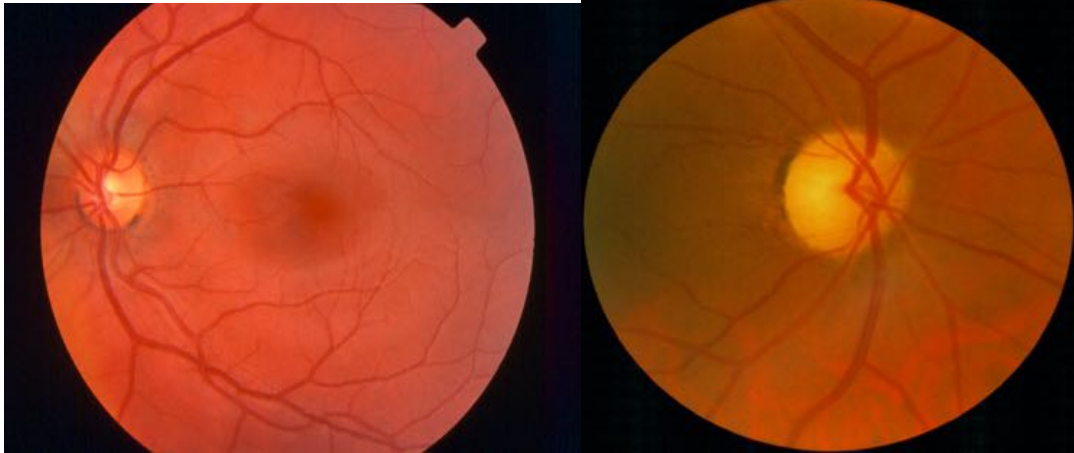




## ABNORMAL OPHTHALMOSCOPY FINDINGS

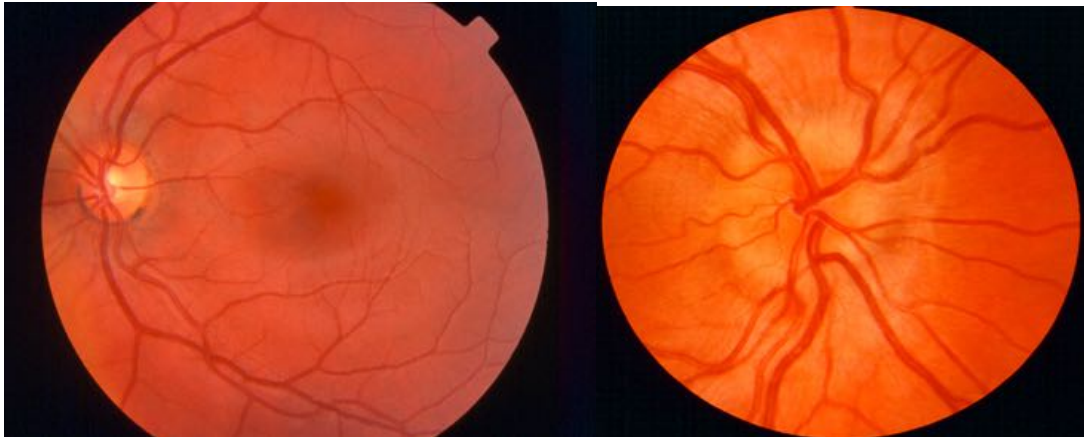
### OPTIC DISC ABNORMALITIES (NORMAL VS. ABNORMAL)

#### Optic Atrophy



#### Swollen Disc

(Notice disc margin is not well defined)



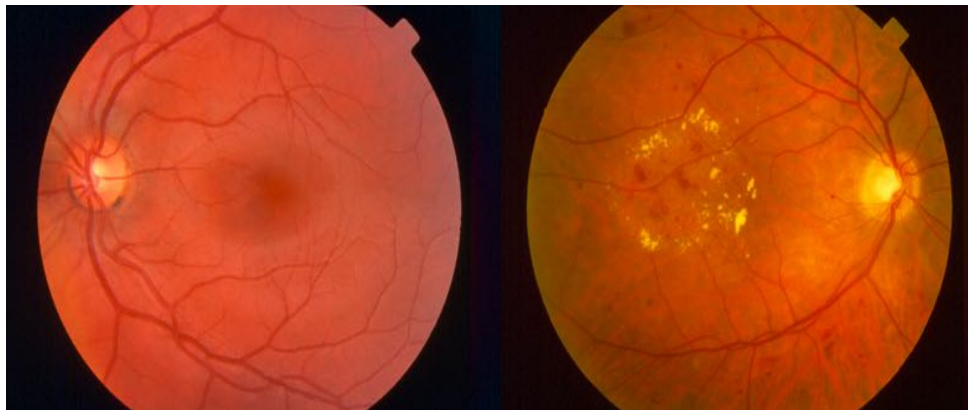
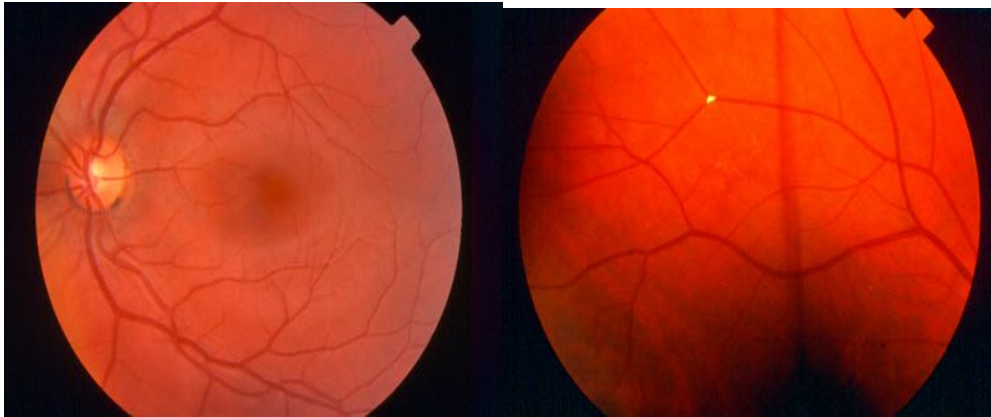
#### "Cupping of the Disc"



This ratio is approx. 0.6

**RETINAL ABNORMALITIES**  
**(NORMAL VS. ABNORMAL)**

**Exudates:**



**These are hard exudates because areas of leakage → protein & calcium deposits.**

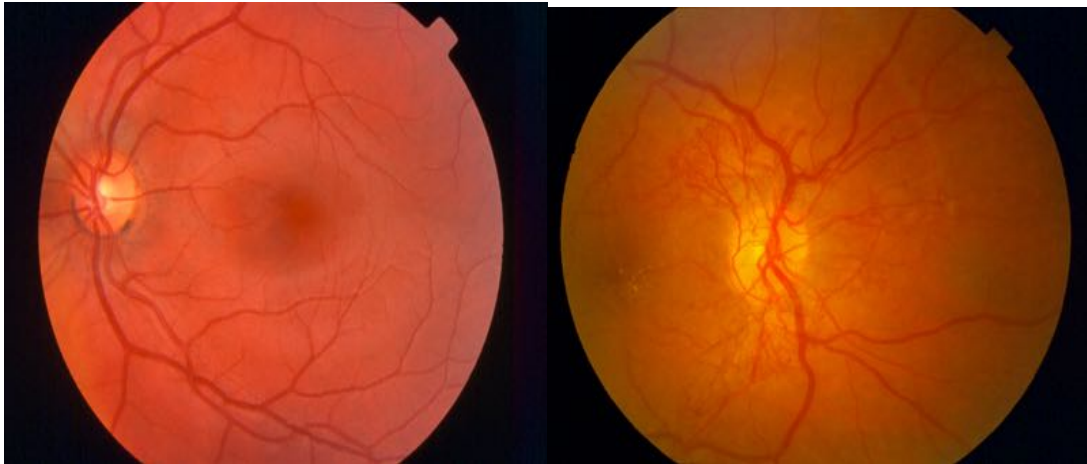


**These are soft exudates (Aka: "Cotton wool Spots") due to ischaemia**



**BLOOD VESSEL ABNORMALITIES**  
**(NORMAL VS. ABNORMAL)**

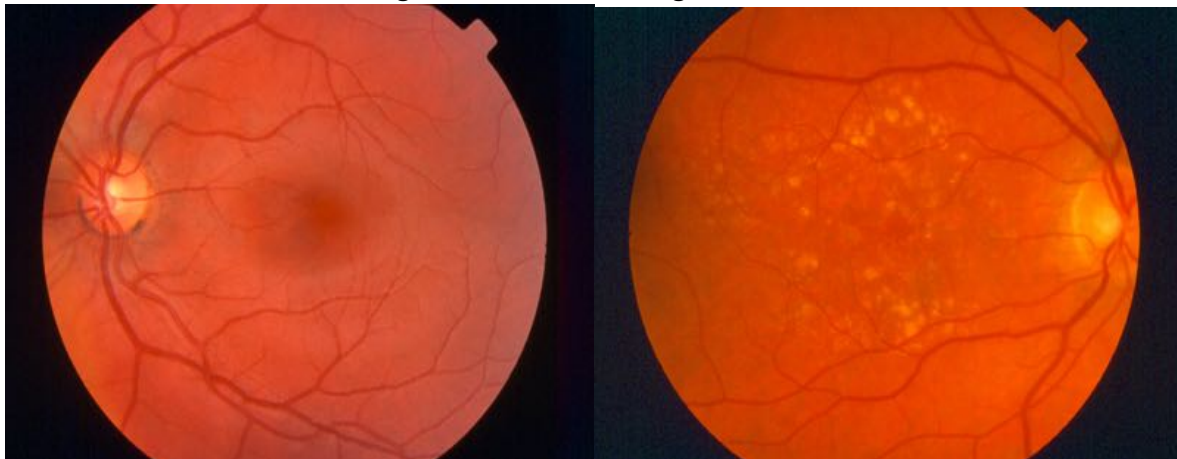
**New Vessels (Proliferative Diabetic Retinopathy)**



New Vessels (abnormal) - Proliferative diabetic retinopathy (Most common fundal abnormality that you'll see)  
These are dangerous because they are very fragile and can bleed very easily. → Impairs Vision

**MACULAR ABNORMALITIES**  
**(NORMAL Vs. ABNORMAL)**

**Age Related Macular Degeneration**



- Loss of Vision in the Centre of the Visual Field (the macula) due to Damage to the Retina.
- →Can make it Difficult/Impossible to Read or Recognize Faces, although enough peripheral vision remains to allow other activities of daily life.
- Most common form of blindness in over 65s

**THE WELL-WOMAN'S HEALTH CHECK**

## THE WELL-WOMAN'S HEALTH CHECK

### Introduction

**Consent** – Can I ask you some questions regarding your reproductive and sexual health?

### Menstrual History

- Age of Menarche (& Menopause if relevant)?
- LMP – Last menstrual period? (& Was the last period '*normal*'?)
- Regularity of periods? (N ≈ Predictable timing of menses)
  - o Duration of cycle? (N ≈ 28 days)
  - o Duration of menstruation? (N ≈ 5 days)
  - o (Note: Irregular can = PCOS/Stress/Anorexia/PID/Fibroids/etc)
- Quantity of bleeding? (Amenorrhoea, Menorrhagia)
- Intermenstrual Bleeding – i.e: bleeding between periods?
- Dysmenorrhoea – i.e: Painful periods?
- Dyspareunia – Painful Intercourse (Endometriosis)
- Associated Symptoms: Abdominal pain, Fever, Vaginal Discharge

### Sexual History:

- Are you sexually active?
- Do you currently have one or more sexual partners?
- Heterosexual/Homosexual
- How many sexual partners have you had in the past 6 months?
- Have you ever been diagnosed with any STIs?
  - o Which ones?
  - o Treated?
- Last STI screen?
- Do you practice safe sex?

### Contraceptive use:

- Any current contraceptives?
  - o Which one/s?
  - o For how long?
  - o Compliance?
  - o Understanding of how to use it effectively?
  - o Any side effects? (weight gain, mood swings)
- Any past contraceptives?
- Barrier protection

### Past Gynaecological History?

- Up to date with Pap-Smears?
- Gardasil Vaccinations? – All 3?
- Results of last Pap-Smear?
- Any Previous Smear Abnormalities?
- Previous Colposcopy?

### Past Gynaecological Surgeries?

- LLETZ Procedures? (For CIN1/3)
- Cone Biopsy? (For CIN1/3)
- Hysterectomy?
- Oophorectomy?
- Tubal Ligation? (For Contraception)

### (Other History):

- Obstetric History
- PMH
- PSH
- Current Medications
- Allergies
- SocHx
- Systems Review

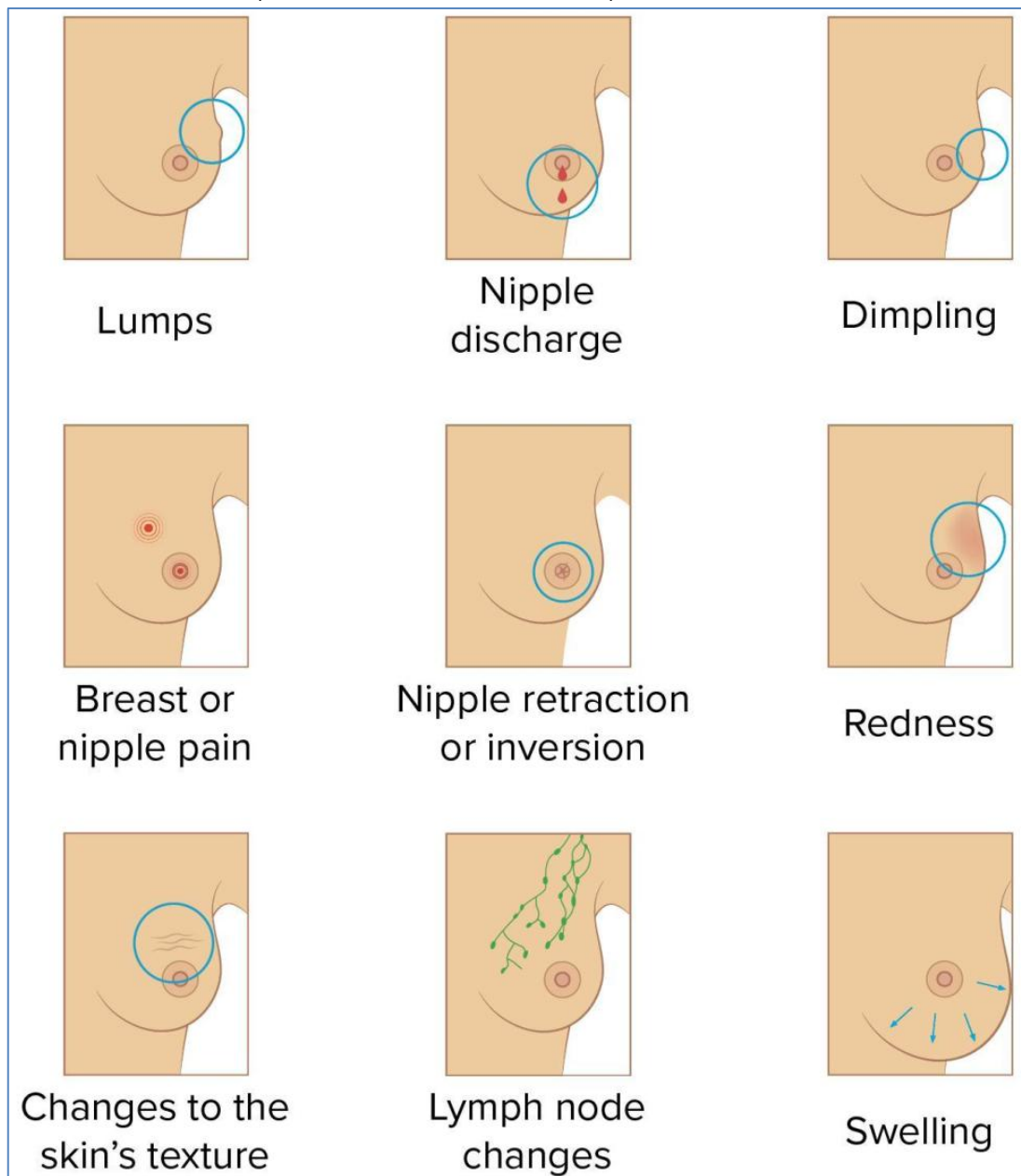
### Examination

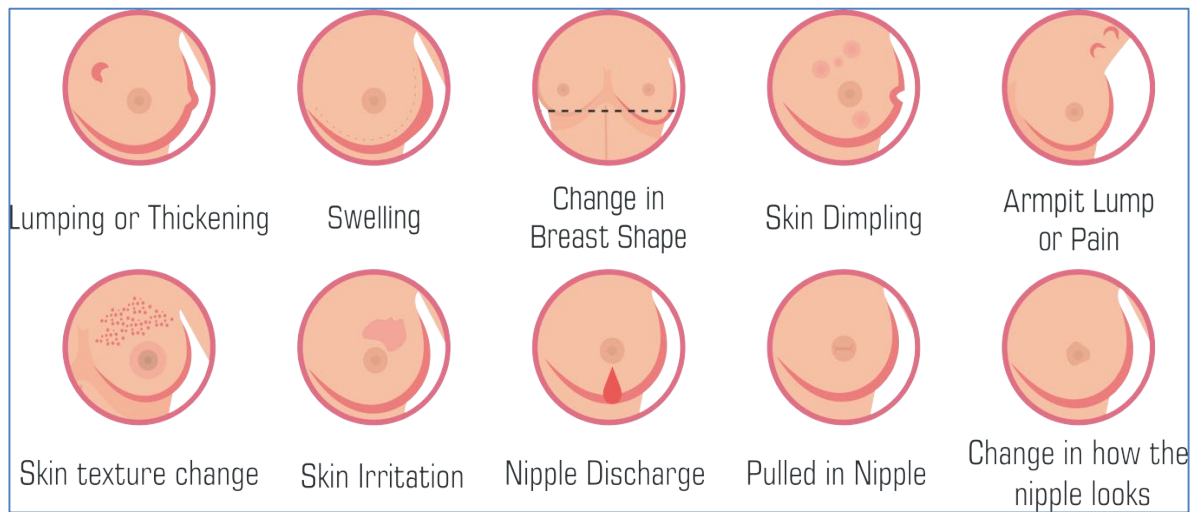
### The “Well-Woman’s Health Check” - (Breast, Pap Smear, Pelvic Exam):

- Wash hands
- Ask about the well woman’s check.
  - o Are they sexually active? When was their last pap smear? (**Pap smears recommended every 2 years starting 1-2 years after first sexual activity until death**)
  - o Have they ever been screened for breast cancer? (**Mammogram Screening is recommended every 2 years starting at age 50-70**)
  - o Do they wish to be screened for STI’s? Gonorrhoea, Chlamydia, HIV, Syphilis? (**Recommended For patients < 25 years old for High Vaginal Swab Chlamydia +/- Urine Sample for Gonorrhoea**)
- Explain the examination & Get consent + ASK FOR A CHAPERONE.
- Explain that the exam may be discontinued at their discretion
- Ask patient to undress and drape themselves in privacy and let you know when they’re ready
- Ask: Any areas of concern?

### The Breast Examination:

- **General Inspection (Peu’d’Orange, Skin Puckering, Nipple Retraction, Lumps/Masses, Discharge, Tethering):**
  - o Sitting up with hands on hips
  - o Roll Shoulders forward
  - o Roll shoulders back
  - o Hands above head (do twice and look at both sides)



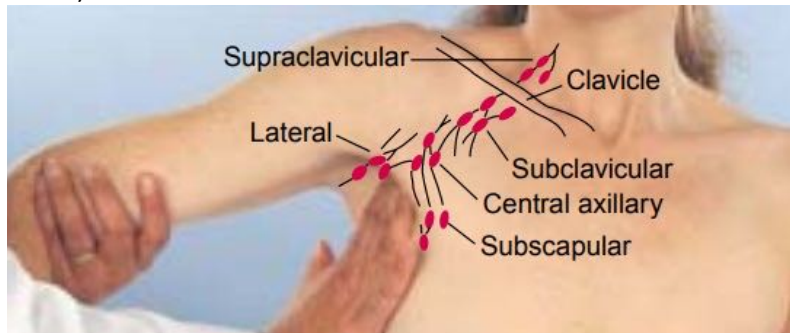


Source: <https://www.pinkribbon.org.pk/>

- **Palpation:**

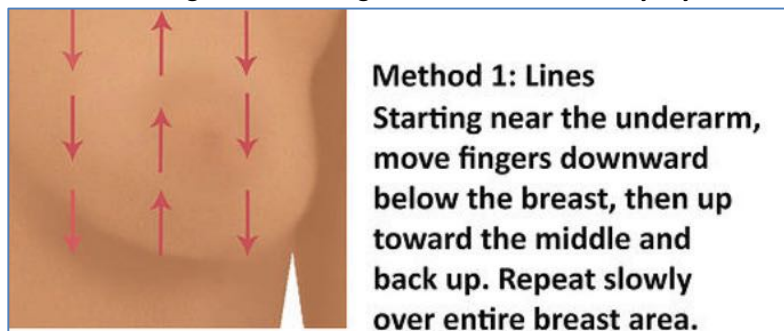
○ **Sitting Up (Lymphadenopathy):**

- Lymph Nodes (Supraclavicular, Infraclavicular, Parasternal, Pectoral, Subscapular, Central, Lateral)



○ **Lying Down with Arm Behind head (Firm Masses distinct from normal nodularity of breasts):**

- Zig Zag in a North-South Fashion starting at Upper Sternal Border, all the way across the breast, and finish up in the tail (overlying the shoulder).
  - Start with light pressure, then deep pressure
  - Do not lift hand when moving to next area
- Assess for Subareolar Masses
- Assess for Nipple Discharge
- **Note: Whilst doing this, Educate Patient about Breast Awareness:**
  - Do you do your own breast self-examination? (Most lumps are found by themselves)
  - You don't need a FamHx to get breast cancer (> 90% breast cancer have no FamHx)
  - Women with a FamHx of breast/ovarian cancer are at ↑ risk of developing breast cancer
  - Mammogram Screening is recommended **every 2 years starting at age 50-70**



○ **REMEMBER TO DO THE OTHER SIDE AS WELL!!**

- **Conclusion:**

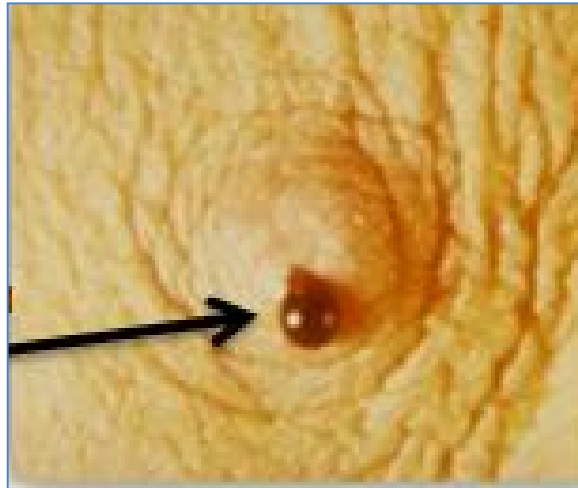
- "Everything looks healthy and normal"
- "Ok we are now going to move onto the pap smear and pelvic examination. Are you okay with that?"

### **Giant Fibroadenoma**



Giant breast fibroadenomas in adolescents: Diagnostic and therapeutic procedures; Beatriz Corredor Andrés, María Márquez Rivera; DOI: [10.1016/j.anpede.2018.01.013](https://doi.org/10.1016/j.anpede.2018.01.013)

### **Bloody nipple discharge**



Source: Unattributable

### **Purulent nipple discharge**



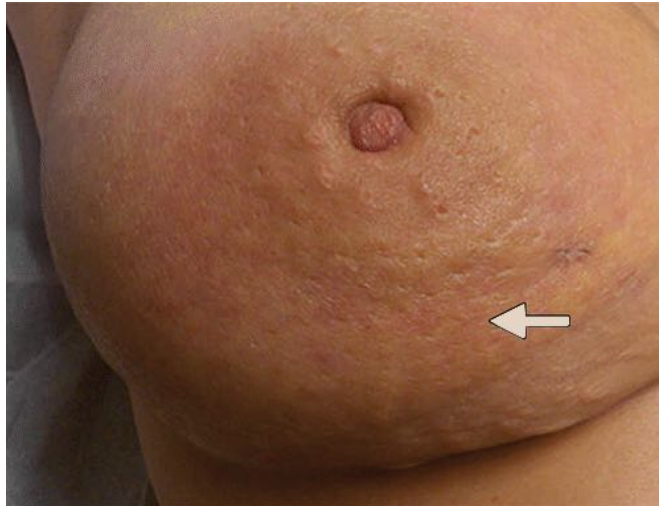
<http://www.meddean.luc.edu/lumen/meded/medicine/pulmonar/pd/step30b.htm>

### **Mastitis**





JayneLut, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0>>, via Wikimedia Commons  
**Peu'de'Orange & Nipple Retraction**



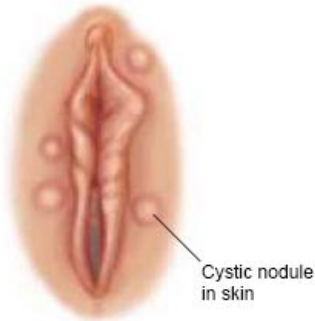
Eren D. Yeh, Heather A. Jacene, Jennifer R. Bellon; <https://doi.org/10.1148/rg.337135503>

## The Pelvic Examination

### General Inspection & Palpation:

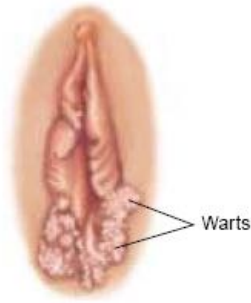
- First I am going to **inspect** the external genitalia, you will feel my fingers palpating you, please put soles of feet together and relax knees to side.
  - Touch inner thigh before touching genitals
  - Open labia majora
  - Inspect labia minora, clitoral hood, urethral orifice, ask pt to bear down and inspect for cyst/rectocele, urethra/uterine prolapse, inspect Bartholin's glands, fourchette, perineum, anus. Also inspect skin, mons pubis, intertriginous areas and hair distribution

TABLE 11-1 ■ Lesions of the Vulva



#### Epidermoid Cyst

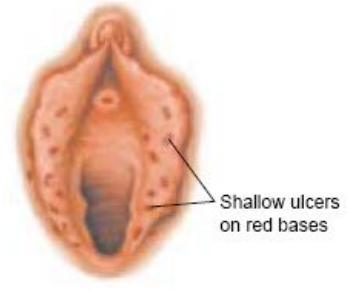
Small, firm, round cystic nodules in the labia suggest epidermoid cysts. They are sometimes yellowish in color. Look for the dark punctum marking the blocked opening of the gland.



#### Venereal Wart

(*Condyloma Acuminatum*)

Warty lesions on the labia and within the vestibule suggest condylomata acuminata. They are due to infection with human papillomavirus.



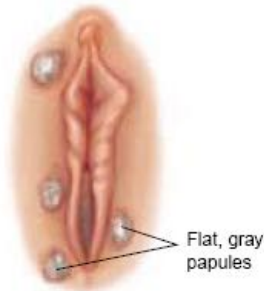
#### Genital Herpes

Shallow, small, painful ulcers on red bases suggest a herpes infection. Initial infection may be extensive, as illustrated here. Recurrent infections are usually confined to a small local patch.



#### Syphilitic Chancre

A firm, painless ulcer suggests the chancre of primary syphilis. Since most chancres in women develop internally, they often go undetected.



#### Secondary Syphilis

(*Condyloma Latum*)

Slightly raised, flat, round or oval papules covered by a gray exudate suggest condylomata lata. These constitute one manifestation of secondary syphilis and are contagious.



#### Carcinoma of the Vulva

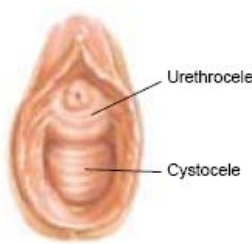
An ulcerated or raised red vulvar lesion in an elderly woman may indicate vulvar carcinoma.

**TABLE 11-2 ■ Bulges and Swelling of the Vulva, Vagina, and Urethra**



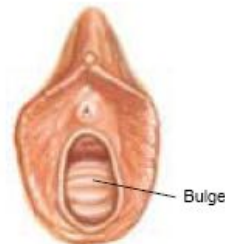
### Cystocele

A cystocele is a bulge of the anterior vaginal wall, together with the bladder above it, that results from weakened supporting tissues. The upper two thirds of the vaginal wall are involved.



### Cystourethrocele

When the entire anterior vaginal wall, together with the bladder and the urethra, is involved in the bulge, a cystourethrocele is present. A groove sometimes defines the border between urethrocele and cystocele, but is not always present.



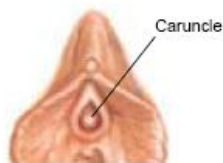
### Rectocele

A rectocele is a herniation of the rectum into the posterior wall of the vagina, resulting from a weakness or defect in the endopelvic fascia.



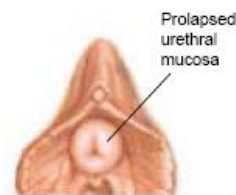
### Bartholin's Gland Infection

Causes of a Bartholin's gland infection include gonococci, *Chlamydia trachomatis*, and other organisms. Acutely, it appears as a tense, hot, very tender abscess. Look for pus coming out of the duct or erythema around the duct opening. Chronically, a nontender cyst is felt. It may be large or small.



### Urethral Caruncle

A urethral caruncle is a small, red, benign tumor visible at the posterior part of the urethral meatus. It occurs chiefly in postmenopausal women, and usually causes no symptoms. Occasionally, a carcinoma of the urethra is mistaken for a caruncle. To check for this, palpate the urethra through the vagina for thickening, nodularity, or tenderness, and feel for inguinal lymphadenopathy.



### Prolapse of the Urethral Mucosa

Prolapsed urethral mucosa forms a swollen red ring around the urethral meatus. It usually occurs before menarche or after menopause. Identify the urethral meatus at the center of the swelling to make this diagnosis.

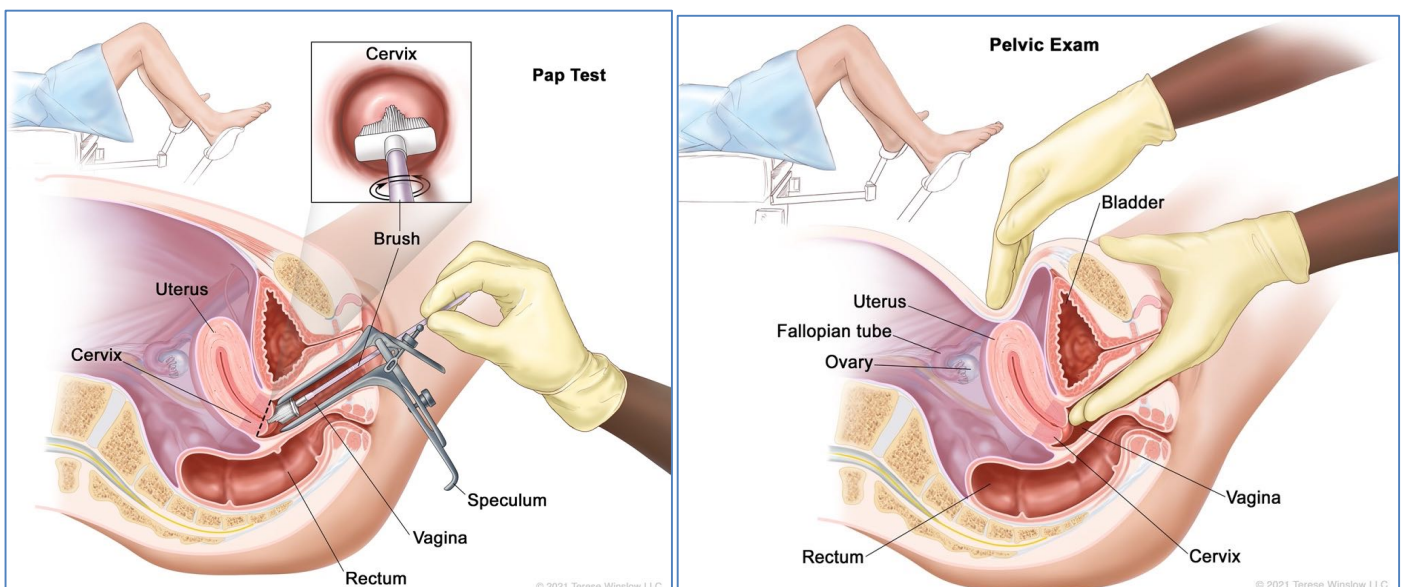
Unattributable

## Speculum Examination & Pap Smear:

- I will now insert the speculum so I can see your cervix and obtain a pap smear (and cultures)
- Pick up speculum in left hand. Apply lubricant to back 2/3 of speculum with right index finger. Switch speculum to right hand
- Touch the inner thigh before touching genitalia
- Place bottom bill of speculum against the fourchette and introduce speculum into vagina with downward pressure until it is fully inserted
- Open speculum by pressing on lever, open and adjust until cervix comes into view, tighten screw to hold speculum in open position while accessing pap/culture supplies
- **Obtain pap/cultures – DON'T FORGET TO SPRAY THE SAMPLES WITH FIXATIVE IMMEDIATELY!!**
- Note position of cervix – normal/prolapsed
- Note whether cervix is parous or nulliparous
- Note any discharge from cervix (mucoid, clear, yellow, bloody) or in vaginal vault
- Note if cervix is friable or has any lesions
- Note whether vagina is ruggae or is atrophic
- Release screw while keeping pressure on level (to avoid bills closing on cervix) remove speculum approximately 1 cm and then release lever so speculum is closed as it is removed from the vagina
- Everything appears to be healthy and normal OR discuss abnormal findings after exam is completed and patient is dressed

### Bimanual Pelvic Examination:

- I am now going to do the bimanual **exam** – this consists of inserting 2 gloved fingers into the vagina and palpating the uterus and ovaries:
  - Touch inner thigh before touching genitals
  - Insert gloved index and middle finger of right hand into vagina with downward pressure on fourchette, may say relax this muscle here
  - Determine location of cervix, move fingers behind cervix and move cervix upward to test for cervical motion tenderness, observe patient's face for reaction
  - Place fingers on cervix while placing left hand on patient's abdomen and assess size, contour and mobility of uterus
  - Remove fingers a few cm and place along left side of cervix in fornix. Fingers point upward as left hand palpates along left side of uterus to assess adnexa. Remove fingers a few cm and repeat on right side
  - Remove fingers and then gloves and say to patient 'everything appears healthy and normal' 'I'll leave so you may get dressed' 'I will be back to answer any questions you may have'
  - Hand patient box of tissues
  - Rectal exam as indicated



<https://www.cancer.gov/publications/dictionaries/cancer-terms/def/pelvic-exam>

### Clinical Investigations

- Pap, GC/Chlamydia, other STI, urine pregnancy test, urinalysis, mammogram, breast/pelvic US.

### Post Exam Education - explain

- Any questions or concerns?
- **Explain Pap Smear:**
  - On visual inspection everything appeared "Healthy and normal", but we will await the results of the pap smear to see if there are any microscopic abnormalities.
  - In the meantime, you may experience some **mild spotting** – If this is severe go straight to the ED.
  - Results are expected in one week. Arrange for call back.
  - Would you like to be placed on the Pap Smear Register (free register which will inform you of when you should come in for your next pap smear)
  - Remind the patient that Screening pap smear is recommended every 2 years starting 1-2 years after first sexual activity, and for as long as sexually active.
- **Thank the patient and conclude the examination.**

### Gynaecological History (Isolated):

#### **OBGYN History**

##### **Intro + Consent**

##### **Presenting Complaint:**

- Any current symptoms? Unusual vaginal d/c or itching? Painful IC? Urinary complaints? Menstrual problems?

##### **Date of Last Period –**

- Was it normal?

##### **Regularity? –**

- Are your periods generally monthly (10-12 periods a year)?

##### **Symptoms related to Periods? –**

- Do you have excessive bleeding or cramping with menses?

##### **Age of Menarche –**

- How old were you when you started having periods?

##### **Has Menopause Occurred? –**

- How old were you when you stopped having periods? (menopause = 1 year since last menses)
- Any vaginal bleeding since menopause?

##### **Possibility of Pregnancy? –**

- Are you using contraception? Sexually active? New partner?
- Any chance you might be pregnant now?

##### **Maternal History? –**

- Gravida - # pregnancies
- Para - # deliveries
- Live births – full term or premature
- Still births
- Multiple births
- C-Sections?
- Spontaneous abortions
- Therapeutic abortions
- Living children

##### **“Well-Woman Screening” History:**

- When was last pap smear? Was it normal?
- When was your last breast exam? Do you examine yourself regularly?
- Any Family History of Female Cancers?

##### **PMH/PSH:**

- Past STIs?
- Any OB/GYN surgeries?

##### **Follow up with Sexual History**



**Sexual History (Isolated):**

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sexual History</b> | <p><b>Intro + Consent</b></p> <p><b>Presenting Complaint:</b></p> <ul style="list-style-type: none"><li>- Any current symptoms? Unusual vaginal d/c or itching? Painful IC? Urinary complaints? Menstrual problems?</li></ul> <p><b>Relationship Status, Sexual Orientation:</b></p> <ul style="list-style-type: none"><li>- Are you in a relationship?</li><li>- DO you have a regular partner? Male or female?</li><li>- DO you partake in sexual activities with males, females or both?</li></ul> <p><b>How many partners?</b></p> <ul style="list-style-type: none"><li>- How many partners have you had in the last 12 months?</li><li>- Have you had sex with anyone different in the last 3 months</li><li>- DO you know your partners? – friends?, randoms?, what country are they from?</li></ul> <p><b>Monogamous/Promiscuous?</b></p> <p><b>Safe Sex</b></p> <ul style="list-style-type: none"><li>- Do you use condoms? Effectively?</li><li>- When was your last unprotected sex?</li><li>- What type of sexual activities do you partake in?</li><li>- Sex with Prostitutes</li></ul> <p><b>History of STIs</b></p> <ul style="list-style-type: none"><li>- Have you had any STD's before?</li></ul> <p><b>When was your last Pap Smear?</b></p> <p><b>Is there anything else you would like to add?</b></p> <p><b>Follow up with OBGYN History</b></p> |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



**MALE GENITOURINARY EXAM**

## MALE GENITOURINARY EXAM

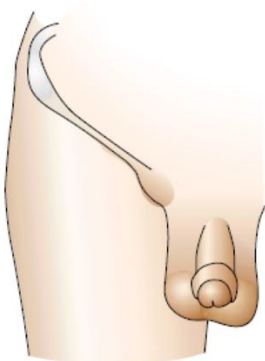
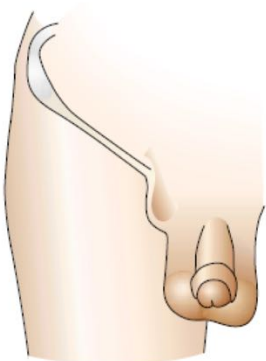
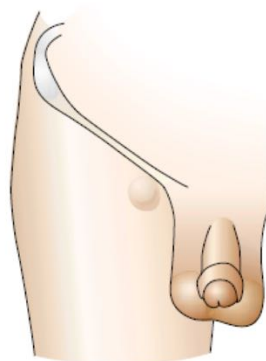
1. Wash hands
2. Introduce your self
3. Ask the patient's name
4. Ask the patients age
5. Explain the procedure
  - a. Involves inspection and palpation of the penis, testes and examination for hernia's.
6. Ask the patient to gown or drape.
7. Ask the patient to expose their groin and genitalia
8. Patient can be lying or standing – standing preferable (hernia's are more easily seen this way)
9. General appearance
  - a. Anxious
  - b. Comfortable
  - c. In distress
  - d. Unable to stand
10. Penis
  - a. GLANS TO SHAFT
    - i. Glans
      1. Comment on circumcision
      2. Urethra – location (hypospadias), discharge (patient may need to milk to penis)
        - a. Press end to open urethra
      3. Retract prepuce – smegma, phimosis, paraphimosis
    - ii. Shaft
      1. Inspect - vesicles, verruca (condyloma acuminatum), ulcerations, nodules, scars, inflammation (balanitis)
      2. Palpate for tenderness, induration, deformity,
    - iii. Skin at base of shaft
      1. Excoriation, inflammation, burrows, nits, lice
    - iv. Hair of pubic region
  - b. Scrotum
    - i. Inspect
      1. Rashes, epidermoid cyst, other lesions
      2. View above, below, lift penis
    - ii. Palpate
      1. Testis + epididymis
        - a. Size, shape, consistency, nodules
      2. Spermatic cord
      3. Transilluminate mass
11. Hernia Evaluation
  - a. Inguinal
    - i. Inspect
      1. Ask patient to bear down, cough
    - ii. Palpate
      1. Right index finger for right inguinal
      2. Ask patient to cough
      3. A hernia will touch your finger
    - iii. Auscultate
      1. Bowel sounds
  - b. Femoral
    - i. Inspect
    - ii. Palpate
      1. Femoral artery – hernia will be around there somewhere
      2. Ask patient to cough

## RECOMMEND

- Self-examination of the testicles every month
- Clean the inside of the penis – if uncircumcised

**TABLE 10-4 ■ Differentiation of Hernias in the Groin**

Differentiation among these hernias is not always clinically possible. Understanding their features, however, improves your observation.

|                                                                           | Inguinal                                                                          |                                                                                        | Femoral                                                                                                                     |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                                                           | Indirect                                                                          | Direct                                                                                 |                                                                                                                             |
|                                                                           |  |      |                                          |
| <b>Frequency</b>                                                          | Most common, all ages, both sexes                                                 | Less common                                                                            | Least common                                                                                                                |
| <b>Age and Sex</b>                                                        | Often in children, may be in adults                                               | Usually in men over age 40, rare in women                                              | More common in women than in men                                                                                            |
| <b>Point of Origin</b>                                                    | Above inguinal ligament, near its midpoint (the internal inguinal ring)           | Above inguinal ligament, close to the pubic tubercle (near the external inguinal ring) | Below the inguinal ligament; appears more lateral than an inguinal hernia and may be hard to differentiate from lymph nodes |
| <b>Course</b>                                                             | Often into the scrotum                                                            | Rarely into the scrotum                                                                | Never into the scrotum                                                                                                      |
| With the examining finger in the inguinal canal during straining or cough | The hernia comes down the inguinal canal and touches the fingertip.               | The hernia bulges anteriorly and pushes the side of the finger forward.                | The inguinal canal is empty.                                                                                                |

**TABLE 10-1 ■ Abnormalities of the Penis**



**Venereal Wart**  
(*Condyloma acuminatum*)

Venereal warts are rapidly growing excrescences that are moist and often malodorous. They result from infection by human papillomavirus.



**Genital Herpes**

A cluster of small vesicles, followed by shallow, painful, nonindurated ulcers on red bases, suggests a herpes simplex infection. The lesions may occur anywhere on the penis. Usually there are fewer lesions when the infection recurs.



**Syphilitic Chancre**

A syphilitic chancre usually appears as an oval or round, dark red, painless erosion or ulcer with an indurated base. Nontender enlarged inguinal lymph nodes are typically associated. Chancres may be multiple, and when secondarily infected may be painful. They may then be mistaken for the lesions of herpes. Chancres are infectious.



**Hypospadias**

Hypospadias is a congenital displacement of the urethral meatus to the inferior surface of the penis. A groove extends from the actual urethral meatus to its normal location on the tip of the glans.



**Peyronie's Disease**

In Peyronie's disease, there are palpable nontender hard plaques just beneath the skin, usually along the dorsum of the penis. The patient complains of crooked, painful erections.



**Carcinoma of the Penis**

Carcinoma may appear as an indurated nodule or ulcer that is usually nontender. Limited almost completely to men who are not circumcised in childhood, it may be masked by the prepuce. Any persistent penile sore must be considered suspicious.

TABLE 10-2 ■ Abnormalities of the Male Genitalia



Fingers can get above mass

#### Hydrocele

A hydrocele is a nontender, fluid-filled mass within the tunica vaginalis. It transilluminates and the examining fingers can get above the mass within the scrotum.



Fingers cannot get above mass

#### Scrotal Hernia

A hernia within the scrotum is usually an *indirect inguinal hernia*. It comes through the external inguinal ring, so the examining fingers cannot get above it in the scrotum.



#### Scrotal Edema

Pitting edema may make the scrotal skin taut. This may accompany the generalized edema of congestive heart failure or nephrotic syndrome.



#### Cryptorchidism

In cryptorchidism, the testis is atrophied and may lie in the inguinal canal or the abdomen, resulting in an undeveloped scrotum as above. There is no palpable left testis or epididymis. Cryptorchidism markedly raises the risk of testicular cancer.



#### Acute Orchitis

The testis is acutely inflamed, painful, tender, and swollen. It may be difficult to distinguish from the epididymis. The scrotum may be reddened. Seen in mumps and other viral infections; usually unilateral.



#### Small Testis

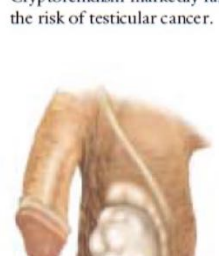
In adults, the length is usually  $\leq 3.5$  cm. Small firm testes in *Klinefelter's syndrome*, usually  $\leq 2$  cm. Small soft testes suggesting atrophy seen in *cirrhosis*, *myotonic dystrophy*, use of estrogens, *hypopituitarism*; may also follow orchitis.



Early

#### Tumor of the Testis

Usually appears as a painless nodule. Any nodule within the testis warrants investigation for malignancy.



Late

As a testicular neoplasm grows and spreads, it may seem to replace the entire organ. The testicle characteristically feels heavier than normal.

TABLE 10-2 ■ Abnormalities of the Male Genitalia (Continued)



#### Acute Epididymitis

An acutely inflamed epididymis is tender and swollen and may be difficult to distinguish from the testis. The scrotum may be reddened, and the vas deferens inflamed. It occurs chiefly in adults. Coexisting urinary tract infection or prostatitis supports the diagnosis.



#### Spermatocele and Cyst of the Epididymis

A painless, movable cystic mass just above the testis suggests a spermatocele or an epididymal cyst. Both transilluminate. The former contains sperm and the latter does not, but they are clinically indistinguishable.



#### Tuberculous Epididymitis

The chronic inflammation of tuberculosis produces a firm enlargement of the epididymis, which is sometimes tender, with thickening or beading of the vas deferens.



#### Varicocele

Varicocele refers to varicose veins of the spermatic cord, usually found on the left. It feels like a soft "bag of worms" separate from the testis, and slowly collapses when the scrotum is elevated in the supine patient. Infertility may be associated.



#### Torsion of the Spermatic Cord

Torsion, or twisting, of the testicle on its spermatic cord produces an acutely painful, tender, and swollen organ that is retracted upward in the scrotum. The scrotum becomes red and edematous. There is no associated urinary infection. Torsion, most common in adolescents, is a surgical emergency because of obstructed circulation.



#### Epidermoid Cysts

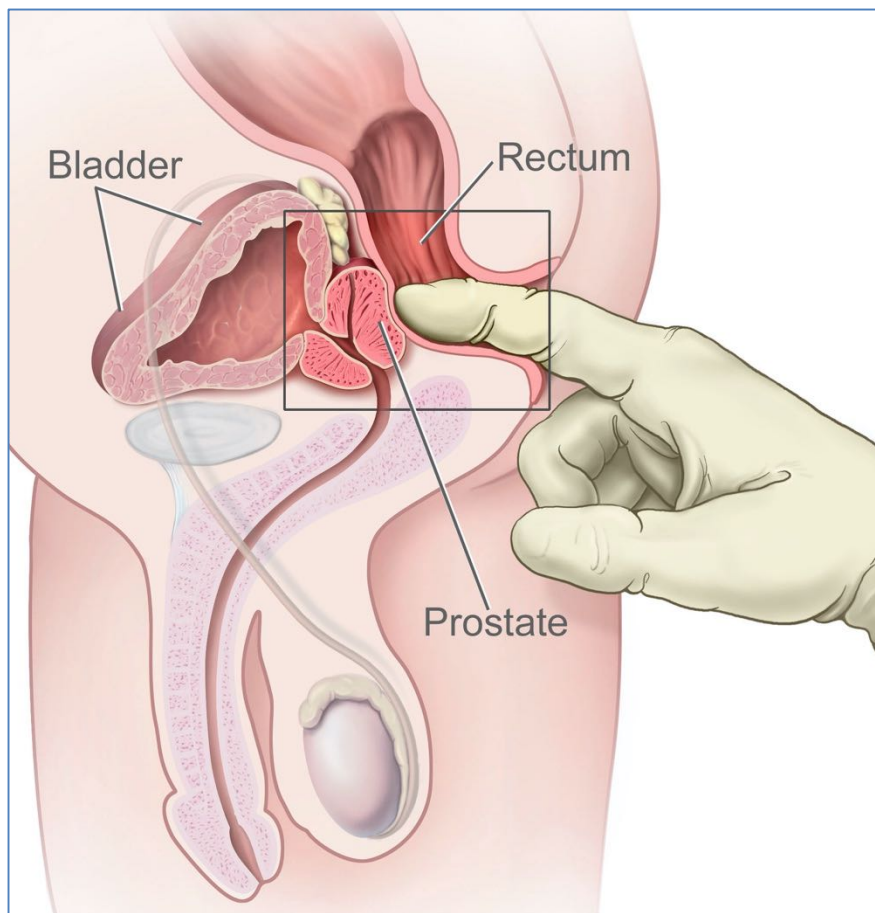
These are firm, yellowish, nontender, cutaneous cysts up to about 1 cm in diameter. They are common and frequently multiple.

Unattributable



**Male Digital Rectal Examination & Prostate Examination:**

12. Patient in Left Lateral Position – Get them to shuffle towards the edge of the bed until they touch your hip
13. General Inspection of Perineum (Scars, Fissures, Haemorrhoids, Skin Tags, Infection, Discharge)
14. “Ok we’re going to begin the rectal examination now”
15. Lubricate Index Finger and place on the anus, tell the patient to exhale or bear down, and as they do, curl your finger inside the anus.
16. Assess Sphincter tone & power by asking them to tighten their anus
17. Rotate finger a full 360° noting:
  - a. rectal masses, tenderness, faeces
  - b. Prostate: Normal rubbery & smooth with median sulcus? Nodular, hard and enlarged? Loss of median sulcus? Tenderness (Prostatitis)
18. Advise patient that you’re removing your finger.
19. Wipe finger onto white towel and inspect for any blood, pus or mucus.
20. Wipe patient’s anus with towel
21. Advise patient to get dressed



Unknown author, Public domain, via Wikimedia Commons



MEDSTUDENTNOTES.COM

## Best Wishes From Our Team!

On behalf of the MedStudentNotes team, we sincerely hope you found these study notes valuable. Our mission is to help aspiring medicos like yourself to become the best clinicians that you can possibly be. We wish you all success in your ongoing studies and career ahead!!

If you have any comments, suggestions, or feedback, please feel free to email us at [admin@medstudentnotes.com](mailto:admin@medstudentnotes.com)

If you've found these notes valuable and you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit the link below for instructions:

>>> <https://www.medstudentnotes.com/pages/bundle-upgrade-process> <<<

Just like you, we have big plans for the future, so be sure to keep an eye on your email inbox for updates.

